

Continual Learning and Model Editing for Foundation Models: A Comprehensive Survey of Adaptation, Correction, and Lifelong Intelligence

1 Foundations, Taxonomies, and the Paradigm Shift to Dynamic AI

1.1 Core Definitions and the Evolution from Static to Dynamic Paradigms

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The advent of Foundation Models (FMs) has fundamentally reshaped the landscape of artificial intelligence, establishing a new baseline for capability across natural language processing, computer vision, and multimodal tasks. Traditionally, the development of these models has been characterized by a static paradigm: a massive, one-time pre-training phase on vast corpora followed by immutable deployment. While this approach has yielded models with remarkable zero-shot capabilities, it inherently assumes a stationary world where the data distribution encountered during deployment mirrors that of the training set. However, real-world environments are intrinsically non-stationary, subject to concept drift, evolving user preferences, and emerging knowledge domains. This discrepancy necessitates a paradigm shift from static, frozen intelligence to dynamic, lifelong adaptation, giving rise to the critical fields of Continual Learning (CL) and Model Editing (ME). This transition sets the stage for understanding the rigorous mathematical constraints—specifically the stability-plasticity dilemma and catastrophic forgetting—that govern these dynamic processes, as explored in the following section.

In the context of FMs, Continual Learning is defined as the capability of a model to incrementally acquire new knowledge from a non-stationary stream of data while preserving previously learned information. Unlike traditional CL settings that often involve training from scratch on small datasets, CL for FMs leverages rich, generic features extracted during large-scale pre-training to facilitate rapid adaptation. The primary challenge here is mitigating “catastrophic forgetting,” where the acquisition of new tasks leads to a severe degradation in performance on prior tasks. Recent research highlights that the integration of pre-training into CL is not merely beneficial but essential, yet it introduces multifaceted complexities regarding how different CL algorithms interact with pre-trained weights [1]. For instance, methods that rely heavily on regularization may behave differently when initialized with robust pre-trained representations compared to random initialization, suggesting that the definition of effective CL must be re-evaluated in light of modern architectural scales.

Parallel to CL is the emerging domain of Model Editing (ME), which focuses on the precise, localized modification of a model’s knowledge base without the need for comprehensive retraining. While CL addresses the sequential acquisition of tasks or domains, ME targets the correction of specific facts, the removal of biases, or the injection of time-sensitive information. This distinction is crucial; whereas CL often deals with distributional shifts across entire tasks, ME operates at the granularity of individual knowledge triplets or conceptual nodes within the model’s latent space. The necessity for such precision is driven by the prohibitive cost and environmental impact of retraining massive models for every minor update. As noted in broader discussions of the field, foundation models are increasingly viewed as modular components that must be adaptable to downstream tasks with minimal computational overhead, a requirement that static pre-training fails to meet [2].

The evolution from static to dynamic paradigms is also marked by a transition in how we view the lifecycle of an AI system. The traditional “train-once, deploy-forever” model is becoming obsolete

in favor of systems capable of “test-time adaptation” and continuous refinement. This shift is evident in scenarios involving non-stationary time series data, where models must adapt to evolving distributions such as macroeconomic cycles or pandemic-induced shifts to maintain forecasting accuracy [3]. In these dynamic settings, the ability to select relevant historical data or adjust internal representations on the fly becomes a defining feature of intelligent systems. Approaches like adaptive sampling strategies have been developed to specifically address these distribution shifts, allowing models to focus on relevant time steps rather than treating all historical data equally [4].

Furthermore, the dynamic paradigm extends beyond single-device learning to distributed and federated settings, where data heterogeneity and privacy constraints complicate the adaptation process. The integration of FMs into Federated Learning (FL) frameworks, such as Federated Domain-Adaptive Pre-Training (FDAPT), demonstrates how dynamic adaptation can occur across decentralized clients without centralizing sensitive data [5]. These frameworks illustrate that the future of AI lies not in monolithic, static artifacts but in ecosystems of models that continuously evolve through interaction with diverse, changing environments. The emergence of “super-model” paradigms further theorizes this evolution as a diffusion process where model parameters are transported from a general pre-trained distribution to specific, domain-adapted steady states, mathematically grounding the shift towards adaptable intelligence [6].

Despite these advancements, significant challenges remain in defining the boundaries of dynamic adaptation. For example, in continual test-time adaptation, many sophisticated methods have been shown to collapse over long timescales, performing worse than simple resetting strategies, which indicates that our current understanding of stable dynamic paradigms is still incomplete [7]. Additionally, the robustness of foundation models to provenance-related distribution shifts—where data sources differ in language use or class distributions—requires deliberate adjustment mechanisms rather than relying solely on out-of-the-box generalization [8]. As the field progresses, the definitions of CL and ME must continue to expand, encompassing not just the avoidance of forgetting, but the active management of model plasticity, the efficiency of parameter updates, and the ethical implications of automated knowledge modification. The transition to dynamic AI is thus not merely a technical upgrade but a fundamental reimagining of how intelligent systems learn, remember, and evolve, leading directly to the formal analysis of the stability-plasticity trade-off that underpins these capabilities.

1.2 Formalizing the Stability-Plasticity Dilemma and Catastrophic Forgetting

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Building upon the paradigm shift from static to dynamic AI outlined in the previous section, the operational realization of lifelong adaptation hinges on resolving the stability-plasticity dilemma. This fundamental tension articulates the competing requirements of stabilizing previously acquired knowledge against the necessity of maintaining sufficient plasticity to assimilate new information. In the context of Foundation Models (FMs), this is not merely a biological analogy but a rigorous mathematical constraint governing the optimization landscape of high-dimensional parameter spaces. Catastrophic forgetting (CF) manifests as the primary symptom of this dilemma: gradient updates required to minimize the loss function of a new task inadvertently overwrite the critical weight configurations established during prior tasks, leading to an abrupt and often total degradation of performance on historical data. This phenomenon represents a fatal shortcoming in gradient-based neural networks, where the acquisition of novel knowledge results in destructive interference with learned representations [9].

Mathematically, the stability-plasticity trade-off can be formalized through the geometry of the loss landscape. In a sequential learning setting involving tasks $T_1, T_2, \dots, T_k$, the objective is to identify a parameter set θ that minimizes cumulative loss without deviating significantly from the minima of previous tasks. However, standard stochastic gradient descent (SGD) effectively treats the parameter space as a *Tabula Rasa* for each new iteration, ignoring the curvature of the loss surfaces associated with prior tasks. Theoretical analyses reveal that in overparameterized linear models, the extent of forgetting is intrinsically linked to the alignment of task subspaces. Specifically, when tasks are presented sequentially, the model’s projection onto the subspace of earlier tasks decays; this forgetting can be bounded by factors related to the number of tasks and the dimensionality of the input space [10]. This demonstrates that forgetting is not random noise but a structured consequence of the orthogonal projections inherent in high-dimensional optimization.

The severity of catastrophic forgetting is further exacerbated by the scale of the model, challenging the assumption that size alone confers robustness. Empirical studies on Large Language Models (LLMs) indicate that as the parameter count increases, the intensity of forgetting can paradoxically intensify during continual instruction tuning. Investigations into models ranging from 1 billion to 7 billion parameters have demonstrated that larger architectures do not inherently possess immunity to CF; rather, the complex interactions within deep decoder-only structures can lead to significant erosion of domain knowledge and reasoning capabilities when exposed to non-stationary data distributions [11]. This counters the intuition that massive overparameterization provides sufficient capacity to store disjoint tasks without interference. Instead, the high-dimensional nature of these models implies that the “flatness” of the minima becomes crucial. Methods that explicitly seek to create and maintain flat regions in the loss landscape around the current task’s minimum have shown that preserving the geometry of the solution space is essential for accommodating future tasks without erasing past ones [12].

Furthermore, the mechanism of forgetting is deeply rooted in the dynamics of hidden representations. Research into the anatomy of CF suggests that deeper layers of neural networks are disproportionately responsible for the loss of information, as these layers encode high-level semantic features most susceptible to shift when task distributions change [13]. When a model learns a new task, the internal representational geometry shifts, causing a misalignment between the frozen readout layers (classifiers) and the evolving hidden states. This “readout misalignment” has been identified as a primary driver of accuracy loss, often accounting for a larger portion of performance degradation than the actual erasure of information within the weights themselves [14]. Consequently, formalizing CF requires distinguishing between the loss of representational capacity and the failure of the decoding mechanism to interpret those representations correctly.

To address these challenges, various mathematical frameworks have been proposed to constrain the optimization trajectory. One approach involves modifying gradient updates to ensure they lie in a subspace orthogonal to the gradients of previous tasks, effectively projecting the learning signal away from directions that would increase the loss on old data [15]. Another perspective utilizes the Fisher Information Matrix to estimate the importance of parameters, penalizing changes to weights that are critical for prior tasks, although this often requires reparameterization techniques like network rotation to approximate the diagonal assumption of the Fisher matrix accurately [16]. Additionally, the concept of “trust regions” has been introduced to dynamically select related old tasks and reuse their frozen weights via scaling matrices, facilitating forward knowledge transfer while strictly bounding interference [17].

Theoretical bounds also extend to the limits of parameter isolation. While some methods aim to provide provable guarantees against forgetting by isolating parameters for specific tasks, the

geometry of the loss landscape often dictates that such isolation is only possible if the tasks are sufficiently dissimilar or if the network capacity is expanded dynamically [18]. In scenarios where tasks share significant semantic overlap, the intersection of low-error parameter subspaces shrinks, making it increasingly difficult to find a single set of weights that satisfies all tasks simultaneously without cumulative error growth [19]. Thus, formalizing the stability-plasticity dilemma in Foundation Models requires a nuanced understanding of how task similarity, overparameterization, and optimization dynamics interact to either preserve or destroy the intricate knowledge structures embedded within massive neural networks. While these mechanisms rigorously define the protection of valid prior knowledge, they set the stage for a complementary challenge discussed in the following section: distinguishing this preservation of valid memory from the necessary updating of information that has become temporally obsolete.

1.3 Knowledge Staleness versus Catastrophic Forgetting

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While the previous section formalized the stability-plasticity dilemma through the lens of catastrophic forgetting (CF)—where new learning destructively interferes with valid prior knowledge—the deployment of Foundation Models (FMs) in dynamic, real-world environments introduces a complementary yet distinct challenge: knowledge staleness. In static machine learning, the objective is maximizing performance on a fixed, stationary distribution. However, lifelong intelligence requires navigating non-stationary data streams where the ground truth itself evolves. Within this context, distinguishing between CF and knowledge staleness is critical. Although both phenomena manifest as performance degradation over time, their underlying mechanisms, temporal characteristics, and requisite solutions differ fundamentally. Addressing this distinction is essential for designing adaptation strategies that move beyond simple retention to include the active updating of obsolete information, a necessity that bridges the theoretical constraints of Section 1.2 with the practical taxonomies explored in Section 1.4.

Catastrophic forgetting, as detailed previously, arises from the interference between tasks during optimization. When a model acquires new patterns, the representational space shifts, inadvertently overwriting the delicate weight configurations required for prior tasks [20]. Here, the “old” knowledge remains factually correct and valuable; the failure is one of capacity and interference management. The primary goal in mitigating CF is stability: ensuring the model serves as a competent repository of all previously encountered information through regularization, replay buffers, and architectural isolation [21]. The paradigm assumes that the past should be preserved exactly as learned.

In sharp contrast, knowledge staleness refers to the temporal obsolescence of facts stored within the model. Unlike CF, where valid knowledge is unintentionally lost, staleness involves the persistence of information that is no longer accurate due to changes in the external world. Foundation models are typically pre-trained on massive corpora collected at specific snapshots in time, effectively freezing their world knowledge to that epoch. As time progresses, entities change roles, scientific consensus shifts, and geopolitical landscapes alter, rendering a significant portion of the model’s internal knowledge base outdated [22]. This creates a state of “temporal misalignment,” where the model confidently recites facts that were true during pre-training but are false in the present [23]. The core problem here is not the loss of memory, but rather an excess of stability: the model retains its “memory” too well, clinging to deprecated facts that conflict with current reality.

The distinction between these two failure modes dictates divergent resolution mechanisms. Address-

ing catastrophic forgetting typically involves protecting specific parameters or rehearsing old data to prevent weight drift. However, applying these same protective mechanisms to address knowledge staleness can be counterproductive. If a model is heavily regularized to prevent forgetting, it may become resistant to updating outdated facts, thereby exacerbating staleness. For instance, in domains involving time-evolving objects or dynamic networks, the lack of a time dimension in knowledge encoding leads to frequent model updates and poor decision-making if staleness is not explicitly managed [24]. The solution to staleness requires mechanisms capable of identifying and selectively “forgetting” or overwriting specific time-sensitive facts while retaining invariant knowledge—a capability that standard continual learning frameworks, designed primarily for stability, often lack.

Recent literature highlights the unique challenges posed by Continual Knowledge Learning (CKL), where the objective shifts from merely accumulating tasks to maintaining an ever-changing knowledge base. Studies indicate that parameter expansion is often necessary to reliably retain time-invariant knowledge while simultaneously acquiring and integrating new, conflicting information that replaces outdated facts [22]. Furthermore, the concept of “temporal knowledge editing” has emerged to address the need for precise updates. Traditional editing methods often conflate correcting intrinsic errors with updating outdated knowledge, leading to the catastrophic forgetting of historical context. Effective systems must distinguish between these edit types, preserving the recollection of historical knowledge while integrating new facts, a balance that requires optimizing for both temporal accuracy and historical consistency [25].

Moreover, the dynamics of staleness extend beyond simple fact replacement to structural representation decay. In complex systems like Temporal Knowledge Graphs, the absence of events for extended periods can lead to “memory staleness,” where node embeddings become obsolete due to temporal deactivation. Overcoming this requires aggregating information from neighbors or incorporating time-aware attributes to refresh representations without full retraining [26]. Similarly, in language modeling, facts possess varying durations of validity; some are permanent, while others are volatile. Mechanisms that predict fact duration can help models decide whether to rely on parametric memory or seek external, up-to-date sources, effectively discarding volatile facts that are prone to rapid change [23].

Ultimately, the transition from static to dynamic AI demands a nuanced understanding of these two failure modes. While catastrophic forgetting represents a failure of stability in the face of new learning, knowledge staleness represents a failure of plasticity in the face of a changing world. Addressing staleness requires moving beyond the binary of “remember vs. forget” to a more sophisticated framework of “refresh vs. retain.” This involves developing algorithms capable of detecting temporal conflicts, predicting the shelf-life of knowledge, and executing targeted updates that modify specific factual representations without destabilizing the model’s general reasoning capabilities [27]. Only by disentangling these concepts can we build Foundation Models that are not only life-long learners but also timely and reliable sources of information. This duality sets the stage for the unified taxonomy in the following section, which categorizes strategies—ranging from regularization to direct editing—capable of navigating both the preservation of valid history and the correction of temporal obsolescence.

1.4 A Unified Taxonomy of Adaptation and Editing Strategies

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Building on the critical distinction between catastrophic forgetting and knowledge staleness established in Section 1.3, this section proposes a unified taxonomy that transcends the traditional boundaries between Continual Learning (CL) and Model Editing (ME). While the previous discussion highlighted that mitigating forgetting requires stability mechanisms while addressing staleness demands targeted plasticity, existing literature often treats these domains as orthogonal silos. To bridge this gap, we categorize adaptation strategies into five interrelated paradigms: regularization-based, memory-replay, parameter-efficient, architectural expansion, and direct editing methods. This framework not only organizes existing literature but also elucidates the synergistic potential of hybrid approaches capable of navigating the dual challenges of preserving valid history and correcting temporal obsolescence, setting the stage for the diverse learning scenarios explored in Section 1.5.

The first category, **regularization-based strategies**, focuses on constraining parameter updates to protect important weights associated with previously learned tasks, directly addressing the stability side of the dilemma. These methods implicitly encode past knowledge by adding penalty terms to the loss function, thereby preventing drastic shifts in the model’s decision boundaries. Early works established that architectural priors and regularizers are crucial for tackling distributional shifts, yet designing flexible solutions that balance efficiency and effectiveness remains a challenge [28]. A systematic examination reveals that regularization techniques can be further subdivided based on whether they affect data, network architectures, error terms, or optimization procedures, revealing fundamental similarities between seemingly disparate methods [29]. However, standard regularization often assumes uniform task difficulty, which can lead to significant forgetting when new tasks diverge sharply from previous ones or unnecessary parameter overhead for simple tasks [30]. To address class-dependent biases where global regularization sacrifices performance on specific categories, recent studies emphasize the need for more nuanced, class-aware regularization schemes [31]. Furthermore, specialized regularizers, such as those leveraging binary concepts in latent spaces, have been shown to improve representation quality and mitigate forgetting in generative replay settings [32]. While effective for stability, these methods must be carefully tuned to avoid the “over-stability” that exacerbates knowledge staleness, a tension that necessitates integration with more dynamic approaches.

The second pillar, **memory-replay mechanisms**, explicitly retains knowledge by storing and rehearsing past experiences, serving as a buffer against both interference and obsolescence. This approach ranges from simple experience replay buffers to sophisticated generative synthesis. While vanilla experience replay remains highly competitive in large-scale settings, its efficacy is often limited by small buffer sizes and distributional drift in intermediate representations [33]. To overcome these limitations, advanced strategies now incorporate compressed activation replay to regularize intermediate layers without prohibitive memory costs [33]. In scenarios where storing raw data is infeasible due to privacy or capacity constraints, memory-free generative replay synthesizes pseudo-data from old classifiers, though this requires careful diversity regularization to avoid sample homogenization [34]. The utility of stored examples can also be enhanced through gradient-based editing, which modifies stored instances to create more challenging samples for future replay, thereby improving robustness in task-free continual learning [35]. Cognitive-inspired approaches further refine this paradigm by adaptively allocating replay quantities based on forgetting rates and selecting high-quality representative data, mimicking human review processes [36]. By curating what is replayed,

these methods begin to address the selection problem inherent in distinguishing between facts to retain and those to update.

The third category encompasses **parameter-efficient and architectural expansion methods**, which aim to adapt models by modifying only a subset of parameters or expanding the network capacity. These strategies are critical for scaling adaptation to massive FMs where full fine-tuning is computationally prohibitive and where isolating new knowledge from old is essential to prevent interference. Parameter-efficient tuning, such as prompt-tuning, has demonstrated remarkable scalability for personalization tasks, allowing efficient customization with minimal trainable parameters [37]. Architectural approaches often involve dynamic resource allocation or structural re-parameterization to optimize the trade-off between model size and accuracy [38]. For instance, automated search strategies for re-parameterization operations can identify superior architectures that outperform manually designed counterparts [39]. Hybrid frameworks like ARR effectively combine architectural growth with regularization to achieve state-of-the-art results across diverse data streams [28]. Moreover, unified frameworks have been proposed to reconcile these methodologies under a single optimization objective, introducing concepts like “refresh learning” to unlearn and relearn data dynamically [40]. Such expansion provides the necessary capacity to host updated facts without overwriting the parameters encoding stale information.

Finally, **direct editing methods** represent a paradigm shift towards precise, localized updates that correct specific facts or behaviors without global retraining, offering a potent solution specifically for knowledge staleness. This field distinguishes itself from continual learning by targeting specific knowledge triplets rather than general task adaptation. However, the reliability of direct editing is under scrutiny, with debates arising regarding its suitability compared to retrieval-based or attribution methods for ensuring factuality [41]. Despite these concerns, novel architectures like neuron-indexed dynamic LoRA (MELO) have emerged to support consecutive and batch-supportive editing with high efficiency, satisfying multiple editing properties simultaneously [42]. The challenge of maintaining consistency across sequential edits has led to the development of memory-friendly approaches that utilize hook layers to manage weight updates without expanding external memory indefinitely [43]. Comprehensive evaluations reveal a dual nature in editing strategies: parameter-modifying methods may degrade general capabilities after sequential edits, whereas parameter-preserving methods maintain foundational skills but struggle with recall in varied formats [44]. These methods provide the surgical precision required to replace outdated facts while leaving the rest of the model’s knowledge base intact.

In synthesis, this unified taxonomy demonstrates that the future of dynamic AI lies not in isolating these strategies but in integrating them to address the full spectrum of the stability-plasticity dilemma. Frameworks that harmonize regularization constraints with generative replay, or those that couple parameter-efficient adapters with direct editing hooks, offer the most promising pathway toward lifelong intelligence. By recognizing the interconnectedness of these approaches, researchers can design systems that are both plastic enough to acquire new knowledge and stable enough to retain the vast corpus of pre-trained information inherent in modern Foundation Models. However, the efficacy of these mechanistic strategies is heavily contingent upon the context in which they are deployed. As we move beyond idealized experimental settings, the following section explores how these adaptation and editing paradigms must be reconfigured to operate in complex, real-world learning scenarios beyond standard supervision.

1.5 Learning Scenarios Beyond Standard Supervision

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While the unified taxonomy presented in Section 1.4 categorizes adaptation strategies by their mechanistic approach, the efficacy of these methods is heavily contingent upon the learning scenario in which they are deployed. The traditional paradigm of Continual Learning (CL) has predominantly relied on idealized, task-incremental settings where data arrives in clearly delineated blocks, task identities are explicitly provided during training, and supervised labels are abundant for every sample. While these assumptions facilitate controlled experimentation, they create a significant gap between theoretical models and the messy, non-stationary realities of real-world deployment. To bridge this divide, the field is rapidly expanding into more rigorous and practical scenarios that abandon standard supervision, known task boundaries, and static data distributions. This shift necessitates a fundamental rethinking of the regularization, replay, and architectural strategies discussed previously, adapting them to operate effectively in task-free, online, unsupervised, and reinforcement learning environments where explicit guidance is scarce or entirely absent.

A primary departure from standard supervision is the transition to **task-free continual learning**, where the learner must adapt to non-stationary data streams without knowledge of task boundaries or identities. In realistic applications, data distributions shift gradually or abruptly without any external signal indicating a “new task” has begun, rendering static regularization or fixed-capacity architectures insufficient. Early approaches to this challenge involved adapting regularization methods, such as Memory Aware Synapses, to an online protocol that dynamically decides when to update importance weights and which data to utilize for consolidation [45]. However, reliance on fixed architectures often proves insufficient for handling complex distribution shifts in a task-agnostic manner. Consequently, expansion-based methods have gained prominence, utilizing mixture models that dynamically grow network capacity to accommodate new data distributions. For instance, evolved mixture models leverage metrics like the Hilbert-Schmidt Independence Criterion to evaluate the distance between stored knowledge and current inputs, triggering architectural expansion only when necessary [46]. Similarly, Bayesian non-parametric frameworks, such as the Continual Neural Dirichlet Process Mixture, allow for the principled creation of new neural experts to handle unseen data subsets, effectively eliminating catastrophic forgetting by isolating distinct data regimes [47].

Closely related to the task-free setting is the challenge of **online streaming with unknown boundaries**, often characterized by “blurry” task transitions and the requirement for anytime inference. In these scenarios, the model must process data in a single pass, making immediate predictions before updating its parameters, all while facing inputs that may belong to overlapping or evolving categories. This imposes strict constraints on memory-replay mechanisms, which must operate under severe buffer limitations. Research has introduced setups like “i-Blurry,” which formalizes online, class-incremental learning with blurry task boundaries, demanding robust memory management and novel learning techniques to maintain performance under continuous pressure [48]. To address the ambiguity in such streams, some frameworks employ discrepancy distance learning to theoretically bound generalization errors and selectively store representative samples based on their divergence from accumulated knowledge [49]. Furthermore, proactive strategies like Doubly Perturbed Continual Learning inject adversarial perturbations into both input data and decision-making processes to anticipate future distribution shifts, thereby enhancing plasticity and robustness in the face of unknown future samples [50].

Beyond the absence of task boundaries, many real-world environments lack **supervised labels**

entirely, necessitating **unsupervised continual learning (UCL)**. In UCL, the agent must extract and memorize meaningful feature representations from an unlabelled, continuous stream of data, challenging the assumption that direct editing or parameter-efficient tuning can rely on gradient signals from labeled losses. Surprisingly, empirical studies suggest that unsupervised visual representations can be more robust to catastrophic forgetting than their supervised counterparts, often learning smoother loss landscapes that generalize better to out-of-distribution tasks [51]. To capitalize on this, methods like SCALE (Self-Supervised ContrAstive Lifelong LEarning) have been developed to operate without prior knowledge of class boundaries, utilizing pseudo-supervised contrastive losses and self-supervised forgetting losses to extract representations on the fly [52]. The complexity increases in “Open-Set Semi-Supervised Continual Learning,” where the model must distinguish between in-distribution and out-of-distribution unlabeled data while learning from a sparse set of labeled examples [53]. Additionally, frameworks for Generalized Continual Category Discovery (GCCD) have emerged to unify continual learning with category discovery, enabling agents to identify novel classes within unlabeled streams while retaining knowledge of known categories [54].

Finally, the scope of dynamic learning extends into **reinforcement learning (RL)** and **open-world settings**, where agents must learn policies through interaction rather than static datasets. In these domains, the assumption of independently and identically distributed tasks is frequently violated, and the feedback loop is delayed and sparse, complicating the application of standard editing or replay techniques. Task-agnostic online RL approaches, such as those utilizing infinite mixtures of Gaussian Processes, enable agents to solve unseen physical tasks by maintaining a mixture of experts that model different dynamics and update sequentially via variational inference [55]. Similarly, architectures like Unicorn demonstrate strong continual learning capabilities in complex 3D domains with sparse rewards by jointly representing multiple policies in a parallel off-policy setup, effectively handling implicit task sequences without explicit curricula [56]. These advancements collectively illustrate a paradigm shift toward lifelong intelligence systems capable of thriving in unstructured, label-scarce, and boundary-less environments. However, as these scenarios push the limits of algorithmic design, they also expose fundamental constraints regarding memory and computation, leading naturally to the theoretical analysis of information flow and complexity bounds discussed in the following section.

1.6 Theoretical Bounds and Information Flow Limits

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While the preceding section highlighted the expansion of continual learning into unstructured, label-scarce, and boundary-less environments, these practical advancements inevitably collide with fundamental theoretical constraints. The pursuit of lifelong intelligence is not merely an algorithmic challenge but is strictly bounded by limits regarding memory capacity, computational complexity, and the dynamics of information propagation within neural architectures. Although empirical heuristics have proliferated, a rigorous theoretical framework is essential to delineate the feasible boundaries of stability-plasticity trade-offs and to assess the viability of zero-forgetting scenarios in scaling foundation models.

A foundational theoretical insight establishes that optimal continual learning—defined as mastering a sequence of tasks without any catastrophic forgetting—is computationally intractable under general conditions. Research within the Probably Approximately Correct (PAC) learning framework demonstrates that a learner capable of solving a sequence of k distinct tasks while retaining

proficiency in earlier ones requires memory resources that grow linearly with k [57]. This finding implies that the problem is inherently intractable for large-scale applications unless specific relaxations are introduced, such as allowing logarithmic passes over data or employing improper learning strategies. Furthermore, from a complexity-theoretic perspective, it has been proven that an optimal CL algorithm possessing perfect memory to avoid all forgetting generally solves an NP-hard problem [58]. This theoretical barrier provides a formal justification for the empirical success of experience replay, episodic memory, and core set methods; unlike purely regularization-based approaches, these methods effectively approximate the “perfect memory” required to navigate this computational hardness.

Beyond global memory complexity, the internal flow of information through network layers during sequential training imposes strict constraints on performance. The degradation observed in continual learning is not solely a function of parameter interference but is deeply rooted in how information is preserved or lost as it traverses network depth. A probabilistic framework analyzing information flow reveals that catastrophic forgetting correlates strongly with the disruption of task-specific knowledge passing between layers [59]. Consequently, optimal CL strategies must explicitly optimize for information preservation between layers, ensuring that mutual information relevant to previous tasks is not overwritten by the gradient updates of new tasks. This perspective necessitates a shift from global parameter regularization to the localized management of information channels within the network architecture.

The geometry of the loss landscape and the properties of the Neural Tangent Kernel (NTK) further illuminate the mechanisms driving these limits. Theoretical analysis indicates that the severity of catastrophic forgetting increases as the alignment between tasks grows, a phenomenon quantified by the NTK overlap matrix [15]. When tasks share similar feature representations, the gradient directions for new tasks are more likely to interfere with the optimal manifolds of previous tasks. Consequently, projected gradient algorithms that constrain updates to orthogonal subspaces are theoretically justified as they minimize this overlap. Additionally, studies on overparameterized linear models have derived explicit forms for expected forgetting and generalization errors, revealing that task ordering and similarity are critical determinants of performance [60]. These insights suggest that in high-dimensional foundation models, the sheer scale of parameters does not automatically guarantee immunity to forgetting; rather, the structural relationship between task gradients dictates the extent of interference.

Architectural choices also play a pivotal role in defining these theoretical limits, challenging the assumption that algorithms alone can resolve continual learning issues. The intrinsic width and depth of a network significantly influence forgetting dynamics; for instance, wider neural networks tend to forget less catastrophically, a phenomenon attributed to increased gradient orthogonality and the tendency of wide networks to operate in a “lazy training” regime where parameters change less drastically [61]. Similarly, the design of network components, such as skip connections and pooling layers, can be optimized to create CL-friendly architectures that naturally resist forgetting [62]. This implies that the theoretical bounds of continual learning are not fixed constants but are contingent upon the architectural inductive biases embedded in the model.

Furthermore, the concept of “zero forgetting” faces significant challenges when viewed through the lens of information theory and bit-level parameter dynamics. Forgetting can be conceptualized as the loss of information gain on model parameters; thus, preserving knowledge requires maintaining the specific bit-level configurations that encode previous tasks [63]. However, achieving this without unbounded memory growth is theoretically constrained. Recent work proposes that by reformulating CL as a continual contraction of the parameter space—where each task occu-

pies a hyperrectangle contained within previous ones—it is possible to provide guarantees against forgetting while reducing the problem complexity to polynomial time [64]. Yet, such guarantees often come at the cost of reduced plasticity, highlighting the inherent trade-off enforced by these theoretical limits.

Finally, the distinction between feature forgetting and output-level forgetting adds nuance to these bounds. While some theories suggest that representations may be robust, empirical and theoretical investigations reveal that feature forgetting occurs concurrently with knowledge accumulation, potentially slowing down the overall learning trajectory [65]. This duality suggests that theoretical limits must account for both the retention of raw features and the accumulation of abstract knowledge. In summary, the theoretical landscape dictates that achieving zero forgetting in scaling architectures is bounded by NP-hard complexity, linear memory requirements, and the geometric alignment of task gradients. These constraints necessitate a shift towards architectures and algorithms that explicitly manage information flow and parameter space contraction, setting the stage for the subsequent discussion on how model editing techniques can offer precise, theoretically grounded interventions to navigate these limits.

2 Architectural Innovations and Parameter-Efficient Adaptation

2.1 Advanced Low-Rank Adaptation Variants and Optimization Dynamics

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The advent of Large Language Models (LLMs) has precipitated a paradigm shift in natural language processing, where the sheer scale of parameters renders full fine-tuning computationally prohibitive for many practitioners. While Low-Rank Adaptation (LoRA) emerged as a foundational technique to mitigate these costs by injecting trainable rank decomposition matrices into frozen weights, standard implementations often exhibit a performance gap compared to full parameter fine-tuning. This discrepancy has spurred a wave of innovations aimed at refining the optimization dynamics, architectural decomposition, and learning capacity of low-rank adapters. These advanced variants seek to bridge the divide between parameter efficiency and model quality by addressing inherent limitations in rank allocation, scaling factors, and update mechanisms, laying the necessary groundwork for the more complex modular architectures discussed in subsequent sections.

A primary avenue of research focuses on decomposing the weight update process to better mimic the learning dynamics of full fine-tuning. Standard LoRA treats the weight update as a single low-rank matrix product, which may fail to capture the nuanced magnitude and directional changes occurring during full optimization. To address this, Weight-Decomposed Low-Rank Adaptation (DoRA) introduces a novel decomposition strategy that separates pre-trained weights into magnitude and direction components [66]. By specifically employing low-rank updates for the directional component while preserving the magnitude learning capacity, DoRA enhances both training stability and final performance. Empirical results indicate that this approach consistently outperforms standard LoRA across diverse tasks, including commonsense reasoning and visual instruction tuning, effectively closing the accuracy gap without incurring additional inference overhead [66].

Another critical optimization challenge lies in the scaling of low-rank adapters. Traditional LoRA implementations scale the adapter matrices by a factor inversely proportional to the rank, a heuristic that inadvertently stifles learning when higher ranks are employed. This phenomenon forces practitioners to rely on very low ranks, limiting the expressive power of the adapters. Addressing this, the concept of Rank Stabilization Scaling Factor for Fine-Tuning with LoRA proposes a theo-

retical correction, proving that adapters should be scaled by the square root of the rank rather than the rank itself [67]. This modification, termed rsLoRA, enables the utilization of higher ranks to trade increased computational resources during training for superior performance, all while maintaining the zero inference cost characteristic of PEFT methods [67].

Beyond static decomposition and scaling, dynamic strategies have been developed to overcome the rigid low-rank bottleneck inherent in fixed architectures, serving as a conceptual bridge toward the adaptive routing mechanisms found in Mixture-of-Experts systems. PeriodicLoRA introduces a multi-stage training framework that accumulates low-rank updates over time to achieve a higher effective update rank [68]. By periodically unloading learned LoRA weights into the backbone parameters and reinitializing the adapter states, this method significantly amplifies learning capacity—up to 1.8 times that of standard LoRA—without increasing memory usage [68]. Complementing this, Chain of LoRA employs an iterative optimization framework inspired by the Frank-Wolfe algorithm, utilizing a residual learning procedure where learned modules are merged and new ones initialized sequentially [69]. This approach provides theoretical convergence guarantees and demonstrates consistent performance gains over standard LoRA across various model scales [69].

Furthermore, the optimization landscape of LoRA has been scrutinized through the lens of learning rate dynamics and gradient flow. It has been observed that updating the constituent matrices of LoRA with identical learning rates leads to suboptimal feature learning, particularly in large-width networks. LoRA+ rectifies this by assigning distinct learning rates to the projection matrices based on scaling arguments, resulting in improved performance and accelerated fine-tuning speeds [70]. Similarly, investigations into memory efficiency have led to LoRA-FA, which freezes the projection-down weight to eliminate the need for storing full-rank input activations, thereby reducing overall memory costs by up to 1.4 times compared to standard LoRA while retaining competitive accuracy [71].

Theoretical underpinnings have also been strengthened to explain the efficacy of these methods. For instance, analysis in the Neural Tangent Kernel (NTK) regime suggests that LoRA with sufficient rank can eliminate spurious local minima, allowing gradient descent to find solutions that generalize well [72]. Additionally, the expressive power of LoRA has been quantified, showing that specific rank thresholds are sufficient to adapt fully connected networks to target models with high fidelity [73].

Finally, innovative approaches continue to emerge that blend these concepts to further push the boundaries of adaptation. Delta-LoRA propagates learning to pre-trained weights via the delta of low-rank matrix products, addressing the inadequacy of incremental updates for complex representations [74]. Meanwhile, methods like ResLoRA integrate residual paths during training to expedite convergence, merging them away before inference to ensure no latency penalty [75]. Collectively, these advanced variants represent a sophisticated evolution of parameter-efficient fine-tuning, moving beyond simple rank reduction to encompass complex optimization dynamics that rival full fine-tuning. While these methods refine the monolithic adapter structure, the following section explores how decomposing these adapters into specialized, dynamically routed experts can further mitigate interference and enhance multi-task capabilities.

2.2 Mixture-of-Experts and Modular Adapter Architectures

2.2 Mixture-of-Experts and Modular Adapter Architectures

While the advanced LoRA variants discussed in Section 2.1 refine the optimization dynamics of monolithic adapters, they still largely rely on a static, global application of low-rank updates. This uniform approach can lead to knowledge interference when handling diverse or conflicting tasks, as a single set of parameters struggles to accommodate heterogeneous feature distributions. To address this limitation, the integration of Mixture-of-Experts (MoE) principles into Parameter-Efficient Fine-Tuning (PEFT) represents a significant architectural evolution. By replacing static adapters with a dynamic ensemble of lightweight experts governed by a routing mechanism, modular architectures introduce sparsity and specialization. This shift allows the model to selectively activate specific parameter subsets for a given input token or task context, thereby enhancing multi-task capabilities and mitigating the catastrophic forgetting often observed in dense adaptation strategies.

A critical challenge in deploying MoE-based adapters is ensuring that individual experts learn distinct and complementary representations rather than collapsing into redundant functions. Recent advancements have addressed this through the application of contrastive learning objectives during the fine-tuning process. For instance, [76] proposes a novel framework that treats LoRA modules as a mixture of experts and employs contrastive learning to enforce diversity among them. By explicitly encouraging experts to learn distinct features, this method prevents the “random routing” phenomenon where the gate fails to differentiate between experts, thereby significantly outperforming standard LoRA on complex reasoning benchmarks. Similarly, [77] draws inspiration from cognitive neuroscience to design rank-1 experts that mimic human intuition, utilizing semantic clustering to guide the router. This architecture demonstrates that implicit guidance based on instance clustering can optimize feature allocation across experts, leading to superior efficiency and accuracy in multitask settings.

Beyond diversification, the granularity of routing and the allocation of experts across layers are pivotal for balancing computational cost and performance. While token-level routing offers fine-grained adaptability, it can introduce significant communication overhead and training instability. To address this, researchers have explored task-level and sequence-level routing strategies. [78] introduces shared dynamic task-based adapters that incorporate task information at different granularities, proving that task-aware routing allows models to generalize efficiently to new tasks while preventing interference between languages and domains. Furthermore, [79] demonstrates that routing at the task level enables the extraction of smaller, ready-to-deploy sub-networks from large sparse models, preserving performance gains that are often lost during distillation. This suggests that modular adapter architectures can be designed to operate at varying levels of abstraction, from individual tokens to entire sequences, depending on the specific requirements of the deployment environment.

The architectural design of these modular systems also extends to how experts are distributed across the network depth, moving beyond uniform allocation to heterogeneous structures that align computational resources with layer-specific complexity. [80] highlights that dynamic expert selection based on input difficulty allows the model to activate more experts for complex reasoning tasks while conserving resources for simpler inputs. This dynamic allocation is particularly effective when combined with layer-wise heterogeneity, as different transformer layers may require different degrees of specialization. Additionally, [81] leverages Neural Architecture Search to design heterogeneous MoE structures under computational constraints, automatically determining the optimal number and placement of experts to maximize efficiency without sacrificing accuracy.

To further enhance the stability and effectiveness of these modular architectures, recent works have focused on improving the routing mechanisms themselves to ensure balanced load and comprehensive feature coverage. [82] proposes an “expert choice” method where experts select the top-k tokens rather than tokens selecting experts, which helps balance the load and prevents certain experts from being under-trained. This approach contrasts with traditional top-k routing and has been shown to improve training convergence speeds significantly. Moreover, [83] introduces a multi-head mechanism that splits tokens into sub-tokens, allowing diverse experts to process different semantic aspects of the same token in parallel. This deepens context understanding and alleviates overfitting by enhancing expert activation rates. In the realm of vision-language models, [84] demonstrates that MoE layers can naturally partition modalities, with specific experts emerging organically to handle image-specific or text-specific features, thereby improving zero-shot performance in multimodal tasks.

Finally, the scalability of these modular adapter architectures is supported by advances in system-level optimizations that reduce memory footprints and communication costs, making them viable for widespread deployment. [85] shows that by combining MoE architectures with lightweight experts, it is possible to update less than 1% of a model’s parameters while achieving performance on par with full fine-tuning. This extreme parameter efficiency is crucial for deploying large models on resource-constrained devices. Additionally, techniques such as [86] allow for the progressive dropping of non-professional experts during fine-tuning, effectively reducing a sparse MoE model into a single-expert dense model for specific downstream tasks without significant performance degradation. These developments collectively underscore the potential of MoE-based modular adapters to provide a flexible, scalable, and robust solution for the lifelong adaptation of foundation models. While this section establishes the benefits of dynamic routing among distinct experts, the following section explores a complementary perspective: optimizing the internal structure of the trainable parameters themselves through dynamic resource allocation and sparse tuning mechanisms.

2.3 Dynamic Resource Allocation and Sparse Tuning Mechanisms

2.3 Dynamic Resource Allocation and Sparse Tuning Mechanisms

While the modular adapter architectures discussed in Section 2.2 introduce sparsity through dynamic routing among distinct experts, a complementary line of research focuses on optimizing the internal structure of the trainable parameters themselves. In the context of Continual Learning (CL) and Model Editing (ME), where computational resources are constrained and data distributions are non-stationary, static parameter allocation strategies—such as assigning uniform ranks or fixed sparsity patterns across all layers—prove increasingly inefficient. These static approaches fail to account for the heterogeneous importance of different network components during adaptation. Consequently, recent advancements have shifted toward dynamic resource allocation and sparse tuning mechanisms that adaptively manage model capacity from within. These techniques aim to optimize the trade-off between representation power and computational cost by allocating varying ranks across layers, employing sparse low-rank updates, and integrating advanced pruning strategies directly into the fine-tuning process, serving as a crucial bridge between the expert-based modularity of the previous section and the structural disentanglement paradigms explored subsequently.

A primary avenue for this internal dynamic allocation involves the adaptive adjustment of low-rank decomposition parameters, a core concept in Parameter-Efficient Fine-Tuning (PEFT). Standard Low-Rank Adaptation (LoRA) typically employs a fixed rank hyperparameter across all target

modules, which may lead to under-parameterization in critical layers and redundancy in others. To address this, **PRILoRA** introduces a mechanism that linearly allocates different ranks for each layer in an increasing manner while simultaneously performing pruning based on the temporary magnitude of weights and accumulated input statistics [87]. This approach acknowledges that deeper layers or specific functional modules may require higher capacity to capture complex task-specific features, whereas others can operate with minimal overhead. Similarly, **IncreLoRA** proposes an incremental parameter allocation method that adaptively adds trainable parameters during training based on module importance scores [88]. Unlike static pruning methods limited by preset rank upper bounds, IncreLoRA allows the model to grow its effective capacity dynamically, proving particularly effective in low-resource settings where efficient parameter utilization is paramount. Furthermore, **SoRA** (Sparse Low-Rank Adaptation) extends this flexibility by enabling dynamic adjustments to the intrinsic rank during adaptation through a gate unit optimized via proximal gradient methods [89]. By initializing with a higher rank and subsequently pruning zeroed-out ranks during inference, SoRA strengthens representation power while maintaining a concise final architecture.

Beyond rank adaptation, the integration of sparsity into the tuning process offers a robust mechanism for managing resources by leveraging the strengths of both compression paradigms. The interplay between low-rank approximation and sparse pruning has been explored to create hybrid models that are both compact and expressive. **LoSparse** proposes a novel technique that approximates weight matrices as the sum of a low-rank matrix and a sparse matrix [90]. This hybrid approach compresses coherent, expressive parts of the neurons via low-rank factors while removing incoherent, non-expressive parts through pruning. The synergy between these two methods prevents the loss of diverse features that might occur with pruning alone and avoids the expressiveness bottlenecks of pure low-rank methods. In a similar vein, **Compresso** utilizes a collaborative prompting paradigm where the model itself assists in learning optimal pruning decisions during instruction tuning [91]. By incorporating LoRA into L_0 regularization, Compresso achieves structured pruning that maintains or even surpasses the performance of the original dense model, demonstrating that dynamic sparse tuning can effectively replace static dense fine-tuning.

Dynamic resource allocation also extends to the management of sparsity patterns across the network topology, moving away from static criteria determined before or after training, which are suboptimal for continual learning scenarios where data distributions shift over time. **MAP** (Magnitude Attention-based Dynamic Pruning) addresses this by applying weight importance metrics throughout both forward and backward passes, allowing the model to dynamically explore and exploit sparse structures [92]. This method transitions from exploration to exploitation during training, ensuring that updates are focused on crucial weights, thereby achieving performance comparable to dense models with significantly reduced parameters. For scenarios requiring strict adherence to resource budgets, **DSA** (Differentiable Sparsity Allocation) offers an end-to-end solution that allocates sparsity in a continuous space using gradient-based optimization [93]. This differentiable approach eliminates the need for discrete, inefficient searches for layer-wise pruning ratios and supports pruning-from-scratch, making it highly adaptable to evolving task requirements.

Moreover, the challenge of maintaining model robustness and stability during dynamic sparse tuning cannot be overstated, as aggressive pruning or rank reduction can lead to ill-conditioned weight matrices, adversely affecting convergence and generalization. **Robust low-rank training via approximate orthonormal constraints** mitigates this by enforcing orthonormal constraints on low-rank weight matrices, preventing the explosion of singular values and ensuring better adversarial robustness without compromising accuracy [94]. Additionally, **EsaCL** (Efficient Continual Learning

of Sparse Models) introduces a directional pruning strategy informed by the sharpness of the loss landscape [95]. This method automatically prunes redundant parameters without the need for expensive retraining, a critical feature for continual learning systems that must adapt to new tasks rapidly. EsaCL also incorporates an intelligent data selection strategy to identify critical instances for estimating the loss landscape, further enhancing data efficiency in sparse settings.

The application of these dynamic mechanisms is not limited to language models but extends to multimodal and recommendation systems as well, underscoring the universality of adaptive sparsity. **ECoFLaP** presents a coarse-to-fine layer-wise pruning approach for Vision-Language Models, utilizing global importance scores to determine sparsity ratios before performing local unstructured pruning [96]. This two-stage process balances global perspective with local efficiency, achieving significant compression in high-sparsity regimes. In the domain of recommendation systems, where data is inherently non-stationary, **Adaptive Dense-to-Sparse Paradigm** proposes an algorithm that automatically learns sparsity across layers to adapt to shifting data distributions, avoiding the accuracy loss associated with applying static pruning to dynamic environments [97].

In summary, dynamic resource allocation and sparse tuning mechanisms represent a paradigm shift from static, one-size-fits-all compression to adaptive, context-aware optimization of the model’s internal weights. By leveraging techniques such as rank-increasing allocation [87], incremental parameter growth [88], hybrid low-rank and sparse decomposition [90], and differentiable sparsity control [93], researchers are developing Foundation Models that are not only smaller and faster but also more capable of lifelong adaptation. These methods collectively address the stability-plasticity dilemma by ensuring that model capacity is allocated precisely where and when it is needed. While this section focused on optimizing the density and rank of parameters within the adaptation modules, the following section explores structural paradigms that go a step further by completely decoupling task-specific parameters from the backbone through side-tuning, hypernetworks, and prompt decomposition.

2.4 Structural Paradigms: Side-Tuning, Hypernetworks, and Prompt Decomposition

2.4 Structural Paradigms: Side-Tuning, Hypernetworks, and Prompt Decomposition

While Section 2.3 focused on optimizing the internal density and rank of parameters within adaptation modules, a more radical structural shift involves completely decoupling task-specific parameters from the frozen backbone. Moving beyond the dynamic allocation of weights discussed previously, this section explores architectural paradigms that disentangle adaptation mechanisms entirely, enabling efficient multi-task deployment without the overhead of storing distinct model copies or modifying the core infrastructure. These approaches—side-tuning, hypernetwork-based conditional generation, and advanced prompt decomposition—prioritize structural separation to balance memory efficiency, sample efficiency, and inference latency, thereby laying the groundwork for the compositional fusion strategies examined in Section 2.5.

Side-tuning represents a fundamental departure from adapter-based methods that embed trainable modules directly into transformer layers. Instead, it constructs a separate, lightweight network that runs in parallel to the frozen backbone, interacting only through shortcut connections. A prime example is Ladder Side-Tuning (LST), which introduces a small side network receiving intermediate activations from the backbone via “ladders” [98]. This design offers a distinct advantage in training memory consumption: because the backbone remains entirely frozen and gradients do not backpropagate through its massive parameter set, LST can reduce memory costs by up to 69% compared to full fine-tuning, outperforming other parameter-efficient methods that still require

gradient storage for the backbone [98]. The concept of unbinding tuners from the backbone has been further generalized into flexible frameworks like Res-Tuning, which theoretically demonstrates that various tuning approaches can be viewed as equivalent counterparts under an unbinding formulation [99]. By detaching the bypass formed by tuners from the main branch, such methods allow for one-time backbone forwarding during multi-task inference, effectively decoupling tuner design from the specific architecture of the base model [99]. Recent advancements have augmented these structures with lightweight attention mechanisms, as seen in Res-Attn, which utilizes parallel multi-head attention modules with low-rank projections to streamline adaptation while strictly adhering to the side-tuning philosophy [100].

Parallel to side-tuning, hypernetworks have emerged as a powerful tool for sample-efficient multi-task adaptation by dynamically generating task-specific parameters rather than learning fixed weights for each task. In this paradigm, a hypernetwork learns a mapping from task descriptors to the parameters of the target model, facilitating rapid adaptation. The Prototype-based HyperAdapter (PHA) exemplifies this by integrating an instance-dense retriever with a prototypical hypernetwork to generate conditional modules [101]. This framework is particularly effective in low-data regimes, leveraging retrieved prototypes to inform adapter generation and striking a superior trade-off between trainable parameters and accuracy on streaming tasks [101]. Similarly, HyperPrompt extends this concept to the self-attention mechanism of transformers, using a hypernetwork to generate “hyper-prompts” that serve as task-global memories for queries to attend to [102]. This allows for flexible information sharing among tasks while maintaining high parameter efficiency, achieving competitive performance with merely 0.14% of additional task-conditioning parameters [102]. These hypernetwork-based architectures effectively address the limitations of independent task adaptation by explicitly modeling knowledge transfer between tasks.

Finally, prompt decomposition strategies have been developed to mitigate the sequence length overhead inherent in standard prompt tuning. While prompt tuning is highly parameter-efficient, the addition of soft prompt tokens increases input sequence length, leading to quadratic growth in computational complexity and memory usage during inference—a bottleneck for large language models facing heavy query loads. Decomposed Prompt Tuning (DePT) addresses this by reparameterizing the soft prompt into a shorter component and a pair of low-rank matrices [103]. This decomposition allows the model to achieve better performance while substantially saving memory and time costs without increasing the total number of trainable parameters [103]. The efficacy of this approach is supported by observations that soft prompts often exhibit low intrinsic rank characteristics, suggesting that decomposed structures are naturally aligned with the underlying geometry of the adaptation space [104]. Moreover, variations of decomposed tuning have been designed to resolve the Base-New Tradeoff dilemma, where improving generalization to base tasks typically degrades performance on new tasks. By decoupling base-specific knowledge into an isolated feature space, these frameworks preserve task-shared knowledge, thereby enhancing zero-shot generalization capabilities [105]. Collectively, these structural paradigms—side-tuning, hypernetworks, and prompt decomposition—provide a diverse toolkit for adapting foundation models. By achieving a complete disentanglement of task-specific parameters from the backbone, they not only enable scalable and robust lifelong learning but also set the stage for the next evolutionary step: orchestrating these isolated modules for cross-task generalization and compositional reuse, as explored in the following section.

2.5 Cross-Task Generalization and Compositional Adapter Fusion

2.5 Cross-Task Generalization and Compositional Adapter Fusion

Building upon the structural paradigms introduced in Section 2.4—which effectively disentangle task-specific parameters from the frozen backbone through side-tuning, hypernetworks, and prompt decomposition—the field has evolved from isolated adaptation toward cross-task generalization and compositional reuse. While the previous subsections established *how* to efficiently insert trainable modules, this section addresses *how* to orchestrate multiple such modules to solve novel problems. The core hypothesis driving this shift is that distinct tasks often share underlying semantic or structural skills; explicitly modeling the composition of these skills can yield superior generalization compared to training isolated adapters. This paradigm moves beyond simple modular insertion toward dynamic, learned fusion mechanisms that aggregate knowledge from multiple source domains, setting the stage for the scalable serving challenges discussed in Section 2.6.

A foundational approach in this domain is the concept of non-destructive task composition, exemplified by frameworks that decouple knowledge extraction from knowledge composition. By first learning task-specific adapters and subsequently combining them through a dedicated composition layer, models can exploit representations learned from diverse tasks without suffering from catastrophic forgetting. This two-stage process allows the classifier to effectively synthesize information from multiple sources, outperforming traditional multi-task learning strategies where gradient conflicts often degrade performance [106]. Building on this, recent advancements have introduced mechanisms to dynamically route input instances to specific subsets of adapter parameters. For instance, multi-head adapter routing strategies have been developed to select variable-size subsets of adapters for each task, providing finer-grained expressivity than static fusion methods. Research indicates that such routing mechanisms are particularly beneficial during the multi-task pre-training phase, where they facilitate better gradient alignment across tasks, although simplified variants that average pre-trained adapters can also achieve competitive zero-shot transfer performance with extreme parameter efficiency [107].

The challenge of composing adapters becomes even more critical when addressing the “combinatorial explosion” of parameters in multi-task and multilingual settings. Traditional approaches that train separate adapters for every task-language pair become computationally prohibitive as the number of domains grows. To address this, the hypernetwork architectures mentioned in Section 2.4 are repurposed here as a powerful solution for generating adapter weights conditionally. These systems utilize shared hypernetworks that take task and language embeddings as input to produce the parameters for lightweight adapter modules. This approach not only unifies multi-task and multilingual learning but also enables positive transfer by learning to encode relationships between different languages and tasks within the hypernetwork’s latent space [108]. Similarly, frameworks employing shared hypernetworks to generate adapter parameters for all layers and tasks have demonstrated the ability to share knowledge across tasks while maintaining task-specific adaptability, achieving state-of-the-art results with minimal parameter overhead per task [109].

Beyond hypernetworks, novel mixture-of-experts styles of adaptation have been proposed to handle diverse dataset mixtures without negative interference. Specifically, featurized low-rank mixtures associate unique dataset features, such as language or task type, with their own low-rank weight update parameters. By composing these feature-specific parameters dynamically for each input, these methods can accommodate highly heterogeneous data distributions and generalize effectively to unseen datasets, overcoming the capacity limitations of standard low-rank adaptation techniques [110]. This concept of dynamic construction extends to multimodal domains as well, where condi-

tional mixture-of-LoRA approaches dynamically construct low-rank adaptation matrices tailored to the unique demands of each input instance, thereby mitigating task interference in complex multimodal instruction tuning scenarios [111].

Furthermore, the composition of adapters has proven highly effective for few-shot learning and cross-domain transfer where data is scarce. Methods that merge pre-trained adapters through model fusion, rather than adding complex composition layers, have shown substantial improvements over both single adapters and heavier fusion baselines. By merging adapters from similar pre-training tracks, these approaches can surpass full fine-tuning performance while maintaining a minimal computational footprint [112]. In scenarios involving low-resource languages or domains, ensembling multiple existing adapters has also been shown to significantly improve robustness. Techniques that optimize ensemble weights by minimizing prediction entropy allow models to leverage related language adapters for uncovered varieties, effectively transferring knowledge without requiring new training data [113]. Additionally, specialized strategies like “target language-ready” adapters address the training-inference discrepancy in cross-lingual transfer by exposing task adapters to target language adapters during training, ensuring compatibility and enhancing modularity [114].

The scalability of these compositional approaches is further enhanced by hierarchical and recurrent designs. Hierarchical recurrent adapters, for example, utilize a shared controller network with multiple task-level heads, reusing adapter parameters across different layers of the pre-trained model. This architecture significantly reduces per-task parameter overhead while maintaining high performance in large-scale multi-task adaptation settings, such as automatic speech recognition [115]. Similarly, systems designed for small language models employ mixture-of-task-adapters to capture intra-task and inter-task knowledge simultaneously, enabling efficient multi-task learning on resource-constrained devices [116]. While these compositional strategies demonstrate that the future of PEFT lies in intelligently orchestrating modular knowledge units, their practical realization introduces significant system-level complexities. As the number of composable experts grows, the management of concurrent adapters, memory fragmentation, and communication overhead becomes paramount, necessitating the advanced serving, compression, and federated deployment solutions explored in the following section.

2.6 Scalable Serving, Compression, and Federated Deployment of PEFT

2.6 Scalable Serving, Compression, and Federated Deployment of PEFT

While the compositional strategies detailed in Section 2.5 demonstrate that the future of Parameter-Efficient Fine-Tuning (PEFT) lies in intelligently orchestrating modular knowledge units, their practical realization introduces significant system-level complexities. As the number of composable experts grows, the management of concurrent adapters, memory fragmentation, and communication overhead becomes paramount. Consequently, the field must shift focus from architectural design to the infrastructural mechanisms required for scalable serving, compression, and federated deployment. This transition sets the stage for the automated design search and efficiency theories discussed in Section 2.7, as the efficacy of these systems ultimately hinges on balancing deployment constraints with the quality of the adapter architectures themselves.

The proliferation of PEFT has fundamentally altered the adaptation landscape, shifting the primary bottleneck from training compute to serving infrastructure. As organizations deploy thousands of specialized “expert” models derived from a single base model via techniques like LoRA or adapters,

the challenges of storing, retrieving, and transmitting these modules become critical. A primary concern in scalable serving is the management of concurrent adapters; in high-throughput environments, dynamically loading distinct adapter weights for different user queries can introduce significant latency and memory fragmentation. To mitigate this, recent advancements propose unified paging mechanisms that treat adapter parameters as swappable memory pages, allowing multiple experts to reside in GPU memory simultaneously or be swapped with minimal overhead. However, even with efficient paging, the sheer volume of adapter data transmitted over networks—particularly in federated or edge-cloud continuums—necessitates aggressive compression strategies that preserve the compositional generalization capabilities established in the previous section.

Compression of PEFT modules has emerged as a critical enabler for scalable deployment, with sparsification and quantization serving as the twin pillars of this effort. Traditional quantization approaches often struggle with the high sensitivity of low-rank adaptation parameters, leading to performance degradation. Addressing this, novel frameworks have been developed to compress fine-tuning residuals specifically. For instance, ComPEFT introduces a method that employs sparsification and ternary quantization to reduce the size of PEFT modules without requiring additional retraining [117]. This approach has demonstrated compression ratios ranging from 8x to 50x across various model scales, from T5 to LLaMA-based architectures. Notably, ComPEFT reveals that stronger, larger models exhibit higher compressibility, achieving superior performance on benchmarks like MMLU compared to standard quantized baselines like QLoRA while drastically reducing storage footprints. This finding suggests that the redundancy in large foundation models allows for aggressive compression of their adaptation layers, facilitating the storage of vast libraries of task-specific experts on single devices or within limited memory pools.

Beyond static compression, dynamic and adaptive compression strategies are essential for handling the variability inherent in distributed and federated learning environments. In federated settings, where clients operate on heterogeneous hardware with fluctuating bandwidth, static compression ratios can either throttle convergence or waste communication resources. The Accordion algorithm exemplifies an adaptive approach, identifying “critical learning regimes” where gradient precision is paramount and relaxing compression during less sensitive phases [118]. While originally designed for gradient updates, the underlying principle of adaptive compression is directly applicable to the transmission of PEFT updates in federated continual learning, ensuring that high-fidelity adaptation occurs when necessary without incurring unnecessary communication costs. Furthermore, in scenarios involving distributed transformer pipelines at the edge, systems like QuantPipe utilize adaptive post-training quantization to adjust bit-widths in real-time response to dynamic bandwidth conditions [119]. This ensures that the pipeline performance remains stable even as network conditions deteriorate, a crucial feature for deploying PEFT-enhanced models across unstable edge networks.

The deployment of PEFT in federated and heterogeneous ecosystems also demands robust protocols that can handle diverse device capabilities and data distributions. The challenge of heterogeneity is not merely computational but extends to the communication fabric itself. Systems designed for real-world cross-device federated learning, such as FS-Real, highlight the gap between theoretical homogeneous assumptions and the reality of diverse device scales and software stacks [120]. Effective PEFT deployment in such environments requires orchestration layers that can manage personalization and asynchronous aggregation while accommodating devices with vastly different memory and compute constraints. Complementing this, hardware-aware quantization frameworks like OHQ demonstrate the necessity of tailoring compression strategies to specific on-chip architectures [121]. By constructing an on-chip quantization awareness pipeline, OHQ optimizes bit-width

configurations without relying on external high-performance devices, ensuring that the quantized PEFT modules are immediately deployable on resource-constrained edge hardware with minimal latency overhead.

Moreover, the integration of compression with distributed optimization algorithms offers theoretical guarantees for convergence even under aggressive communication constraints. Research into optimal gradient compression, such as the development of Sparse Dithering and Spherical Compression operators, provides tight lower bounds on the trade-off between bit usage and compression error [122]. These theoretical insights are vital for designing federated PEFT protocols that can provably converge despite the lossy nature of compressed adapter updates. Additionally, advanced techniques like Contractive Error Feedback (ConEF) address the memory bottlenecks in distributed training by leveraging biased, all-reducible gradient compression, achieving significant memory savings with negligible impact on test performance [123]. When applied to the federated updating of PEFT parameters, such methods ensure that the accumulation of compression errors does not lead to model divergence or catastrophic forgetting of previously learned tasks.

Finally, the intersection of generative capabilities and compression opens new avenues for efficient PEFT deployment. The concept of unifying generation and compression, as seen in ultra-low bitrate image coding via multi-stage transformers, suggests a future where PEFT modules might not only be compressed but also partially generated or reconstructed on-device from compact latent representations [124]. This paradigm shift could reduce the communication payload from kilobytes of weights to mere tokens, revolutionizing how adaptable models are distributed in bandwidth-starved environments. As the field moves toward trillion-parameter systems, the synergy between architectural innovations in PEFT, advanced compression algorithms like ComPEFT and SqueezeLLM [125], and robust federated protocols will define the feasibility of lifelong, personalized intelligence at the edge. The ability to serve hundreds of concurrent, compressed adapters with unified paging and adaptive precision establishes the necessary infrastructure for the next generation of adaptation, where the focus shifts to optimizing the design of these modules through automated search and rigorous efficiency theory, as explored in the following section.

2.7 Automated Design Search and Efficiency Theory

2.7 Automated Design Search and Efficiency Theory

While the preceding section established the infrastructural mechanisms for serving and compressing Parameter-Efficient Fine-Tuning (PEFT) modules at scale, the efficacy of these systems ultimately hinges on the quality of the adapter architectures themselves. The rapid proliferation of PEFT methods has shifted the paradigm of adapting Foundation Models from manual hyperparameter tuning to automated architectural search. Although techniques like Low-Rank Adaptation (LoRA) and prompt tuning have demonstrated efficacy, the manual selection of configuration spaces—including rank dimensions, module placement, and adapter architectures—often yields suboptimal performance-efficiency trade-offs. The complexity of these design choices necessitates automated configuration search frameworks capable of navigating the vast combinatorial space of PEFT hyperparameters without incurring prohibitive computational costs. A pioneering approach in this domain is AutoPEFT, which formalizes the search for optimal PEFT configurations as a multi-objective optimization problem. By designing an expressive search space encompassing various PEFT modules and leveraging multi-objective Bayesian optimization, AutoPEFT identifies Pareto-optimal configurations that balance performance against parameter counts and training efficiency [AutoPEFT Automatic Configuration Search for Parameter-Efficient Fine-Tuning]. This work em-

pirically demonstrates that automatically discovered configurations can significantly outperform manually designed baselines on benchmarks like GLUE and SuperGLUE, often matching or exceeding full fine-tuning performance with a fraction of the trainable parameters [AutoPEFT Automatic Configuration Search for Parameter-Efficient Fine-Tuning].

Beyond static architecture search, recent advances have integrated cost-awareness directly into the optimization loop to address the scaling challenges inherent in large models. Traditional hyperparameter tuning often ignores the varying computational costs of different evaluations, leading to inefficient resource allocation. The Cost-Aware Pareto Region Bayesian Search (CARBS) algorithm addresses this by performing local searches around the performance-cost Pareto frontier, effectively learning scaling relationships that allow for robust tuning even as model sizes increase [Tune As You Scale Hyperparameter Optimization For Compute Efficient Training]. Such methods are critical for PEFT, where the interaction between adapter rank, layer selection, and model scale creates a high-dimensional, expensive-to-evaluate objective landscape. Furthermore, the integration of low-fidelity data and transfer learning into Bayesian optimization frameworks offers a pathway to accelerate these searches. Techniques that harness low-fidelity approximations through posterior regularization can drastically reduce the number of expensive function evaluations required to find optimal PEFT settings [Harnessing Low-Fidelity Data to Accelerate Bayesian Optimization via Posterior Regularization]. Similarly, learning search space geometries from historical tuning tasks allows for the automatic design of compact and promising search regions, endowing standard Bayesian optimizers with transfer learning capabilities that boost efficiency across diverse downstream tasks [Learning search spaces for Bayesian optimization Another view of hyperparameter transfer learning][126].

Parallel to the development of automated search algorithms is the critical need for a rigorous theoretical understanding of efficiency and forgetting in PEFT regimes. Contrary to the initial belief that updating fewer parameters inherently mitigates catastrophic forgetting—a concern central to the continual learning themes of this survey—recent theoretical and empirical analyses reveal a more nuanced reality. Studies on scaling laws for forgetting indicate that PEFT strategies, including LoRA, still suffer from significant knowledge degradation. Specifically, research has identified a strong inverse linear relationship between fine-tuning performance and the amount of forgetting, with forgetting increasing according to a shifted power law relative to the number of fine-tuned parameters and update steps [127]. These findings suggest that forgetting cannot be simply avoided by reducing the number of trainable parameters or employing early stopping, highlighting a safety-critical gap in current efficient tuning practices [127]. This theoretical insight underscores the necessity of incorporating explicit regularization mechanisms to preserve pre-trained knowledge during adaptation, bridging the gap between the compression strategies discussed previously and the stability requirements of lifelong learning.

Bayesian regularization techniques have emerged as a principled solution to the stability-plasticity dilemma in PEFT. By treating the pre-trained weights as a prior and the fine-tuned parameters as a posterior, Bayesian methods can constrain the parameter shift to prevent the erosion of foundational knowledge. Research demonstrates that applying Laplace approximations, particularly Kronecker-factored approaches, to PEFT modules can effectively regularize the learning process [Bayesian Parameter-Efficient Fine-Tuning for Overcoming Catastrophic Forgetting]. These techniques enable the calculation of differentiable parameter shifts that maintain the integrity of the pre-trained model’s capabilities while allowing for task-specific adaptation [Bayesian Parameter-Efficient Fine-Tuning for Overcoming Catastrophic Forgetting]. The superiority of Kronecker-factored approximations over diagonal ones in preserving pre-training knowledge further illustrates the importance

of capturing second-order information in the regularization process [Bayesian Parameter-Efficient Fine-Tuning for Overcoming Catastrophic Forgetting].

Moreover, the theoretical limits of efficiency extend beyond forgetting to include memory consumption and inference latency, reinforcing the system-level concerns raised in Section 2.6. Innovations such as reversible model architectures and side-tuning mechanisms aim to decouple trainable modules from the frozen backbone, thereby reducing the activation memory footprint without sacrificing gradient flow fidelity [128][Make Pre-trained Model Reversible From Parameter to Memory Efficient Fine-Tuning]. For instance, making pre-trained models reversible allows for the recomputation of intermediate activations, significantly reducing memory usage during the training of adapters [Make Pre-trained Model Reversible From Parameter to Memory Efficient Fine-Tuning]. Additionally, the development of sparse masks generated in a task-agnostic manner offers a route to storage efficiency that avoids the inference latency penalties often associated with adding new parameters [129]. The convergence of automated design search, scaling laws for forgetting, and Bayesian regularization forms a cohesive theoretical framework for the next generation of PEFT. Future research must continue to bridge the gap between automated configuration discovery and theoretical guarantees, ensuring that efficient adaptation strategies are not only performant but also robust, safe, and scalable across the ever-expanding landscape of Foundation Models. The integration of causal reasoning into search spaces, as seen in robotics configuration, may further refine these automated systems by identifying causally relevant parameters, thus reducing the effective dimensionality of the search problem [130]. Ultimately, the synergy between automated search and efficiency theory will define the trajectory of lifelong intelligence in foundation models, enabling them to adapt continuously without compromising their core competencies.

3 Memory Mechanisms, Replay Strategies, and Generative Synthesis

3.1 Optimal Experience Replay and Strategic Memory Selection

3.1 Optimal Experience Replay and Strategic Memory Selection

Experience replay stands as a cornerstone mechanism in continual learning, designed to mitigate catastrophic forgetting by interleaving current training data with stored representations of past tasks. While the fundamental concept of storing and replaying past experiences is well-established, the efficacy of this approach is heavily contingent upon the strategies employed for buffer population and sample retrieval. Naive approaches, such as uniform random sampling or simple reservoir sampling, often fail to account for the heterogeneous importance of data points, leading to suboptimal convergence and persistent forgetting, particularly in scenarios with limited memory capacity. Consequently, recent research has pivoted towards advanced buffer management techniques that prioritize strategic selection based on interference metrics, cognitive-inspired allocation, and information-theoretic principles. This focus on optimizing *existing* data storage serves as a critical precursor to the generative paradigms discussed in the following section, establishing the necessity of high-quality representative samples before addressing how such samples might be synthesized when storage is impossible.

A primary direction in optimizing experience replay involves identifying and retrieving samples that are most susceptible to forgetting or most critical for maintaining decision boundaries. This “maximally interfered retrieval” paradigm posits that not all past experiences contribute equally to stability; rather, the model should focus on instances where the current parameter updates would cause the most significant degradation in performance. For instance, the method proposed in [131] explicitly formulates a sampling criterion to retrieve samples whose predictions are most

negatively impacted by foreseen parameter updates. By targeting these high-interference examples, the learning process can actively counteract the drift in the loss landscape, resulting in consistent performance gains and significantly reduced forgetting compared to random selection. Similarly, [132] expands this inquiry by evaluating various selective retrieval primitives, highlighting that the strategy for retrieving samples from a buffer is just as critical as the strategy for constructing it. Their work emphasizes the need to prevent duplicate replays and explores whether new samples with low loss values can be learned without intervention, suggesting a dynamic threshold for retrieval necessity.

Beyond interference-based retrieval, cognitive-inspired strategies have emerged to mimic human memory consolidation processes, where the brain adaptively allocates resources based on the difficulty and retention rate of specific information. Human memory does not treat all events equally; instead, it prioritizes emotionally arousing or frequently reviewed information. Translating this to artificial agents, [CORE Mitigating Catastrophic Forgetting in Continual Learning through Cognitive Replay] introduces a framework comprising Adaptive Quantity Allocation and Quality-Focused Data Selection. This approach dynamically modulates the replay buffer allocation for each task based on its specific forgetting rate, ensuring that tasks prone to rapid decay receive more rehearsal. Concurrently, it guarantees the inclusion of representative data that best encapsulates task characteristics, thereby surpassing baseline methods that treat every task and data sample with equal weight. Further bridging the gap between biological and artificial memory, [133] proposes a framework that archives only salient data segments via sparse memory encoding, utilizing associative memory paradigms for content-focused retrieval. This method moves away from storing entire data samples, aligning more closely with the human tendency to retain only the most significant portions of an experience.

Information theory also provides a rigorous foundation for strategic memory selection, particularly through the lens of entropy maximization. The goal is to ensure that the limited slots in a replay buffer capture the maximum diversity and informational content of the data distribution. [AdaER An Adaptive Experience Replay Approach for Continual Lifelong Learning] addresses this by incorporating an entropy-balanced reservoir sampling strategy. This technique enhances the memory buffer by maximizing information entropy, ensuring that the stored samples are not redundant but rather cover the diverse manifold of the class distribution. In parallel, [134] argues that random selection can introduce noisy outliers that harm generalization. Their proposed method, Memory Outlier Elimination (MOE), selectively chooses samples from label-homogeneous subpopulations, effectively filtering out noise to create a more representative and robust memory buffer.

The complexity of strategic selection increases in reinforcement learning and class-imbalanced settings, where the distribution of rewards or class frequencies can skew the learning process. In reinforcement learning, [135] proposes partitioning the buffer into event tables to capture important subsequences of optimal behavior, combining this with prioritized sampling to improve stability. For class-imbalanced data, [136] extends standard class-balancing approaches by considering intra-class diversity, ensuring that the stored instances for a given class are not identical but reflect the varying complexities within that class. Furthermore, [137] introduces a policy that dynamically allocates storage space based on class contribution and difficulty, allowing challenging classes to occupy a larger portion of the buffer.

Finally, the efficiency of replay is not solely dependent on what is selected, but also on how the memory is managed over time. [Balanced Destruction-Reconstruction Dynamics for Memory-replay Class Incremental Learning] theoretically analyzes the inherent destruction-reconstruction dynamics in memory replay, proposing a module that balances the contribution of current and memory

samples to alleviate knowledge destruction. Additionally, [138] highlights the importance of prediction consistency, utilizing future experiences mimicked on current data to preserve acquired knowledge when few previous samples are available. These advanced management techniques collectively demonstrate that the path to robust continual learning lies not merely in the existence of a memory buffer, but in the intelligent, adaptive, and theoretically grounded strategies used to populate and query it. However, these methods inherently rely on the availability of storage space for raw data or features. When privacy constraints, storage limits, or the sheer scale of foundation models preclude explicit memory retention, the field must transition from selecting optimal real samples to synthesizing optimal pseudo-samples, a challenge addressed by the evolution of generative replay mechanisms.

3.2 Advanced Generative Replay with Diffusion Models and Latent Alignment

3.2 Advanced Generative Replay with Diffusion Models and Latent Alignment

Building upon the strategic selection of real data samples discussed in the previous section, the field encounters a fundamental bottleneck: the physical and privacy constraints that limit the capacity to store raw experiences. When memory buffers reach their limits or data retention is prohibited, the paradigm must shift from curating existing data to synthesizing new representations that faithfully emulate past distributions. This necessity has driven a significant evolution in generative replay, moving from early Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs)—which often suffered from mode collapse, training instability, and low-fidelity outputs—to the adoption of diffusion probabilistic models. Diffusion models have emerged as a robust solution, capable of synthesizing high-quality pseudo-data that indistinguishably mimics the statistical properties of original training data. This transition is critical, as the efficacy of generative replay hinges on the synthetic data’s ability to prevent the classifier from distinguishing between real and replayed samples, thereby mitigating catastrophic forgetting. Recent advancements have specifically targeted the integration of class-prototype conditioning and latent space alignment to refine this process, ensuring that generated data not only appears realistic but also preserves the discriminative features necessary for maintaining decision boundaries across sequential tasks.

A primary challenge in applying generative replay to diffusion models is the potential degradation of denoising capabilities when the generative model itself is continually updated. Standard approaches that simply interleave synthetic data from a previous model with new task data can lead to a catastrophic loss in sample quality over time, undermining the replay mechanism. To address this, researchers have proposed “generative distillation,” a technique that distills the entire reverse process of a diffusion model rather than just the final output. By preserving the step-by-step denoising trajectory, this approach ensures that the generative model retains its capacity to synthesize diverse and high-fidelity images even after learning multiple sequential tasks, significantly outperforming standard replay methods with only a moderate increase in computational cost [139]. Furthermore, the hierarchical nature of data representation within diffusion processes has been leveraged to understand forgetting dynamics. Studies reveal a phase transition in the backward diffusion process where high-level semantic features (such as class identity) are reconstructed at specific timesteps, while low-level details evolve smoothly. Understanding this temporal hierarchy allows for more targeted replay strategies that prioritize the preservation of semantic consistency over pixel-perfect reconstruction, which is often sufficient for maintaining classification performance [140].

To enhance the specificity of generated samples for class-incremental scenarios, advanced frame-

works have incorporated class-prototype conditioning directly into the diffusion mechanism. Traditional diffusion models may suffer from semantic drift, where the generated images for old classes gradually lose their defining characteristics as the model adapts to new distributions. The Gradient Projection Class-Prototype Conditional Diffusion Model (GPPDM) addresses this by introducing learnable class prototypes that capture the core characteristics of images within a given class. These prototypes are integrated into the denoising process via cross-attention layers, ensuring that the generated images remain faithful to the original class semantics. Additionally, GPPDM employs a gradient projection technique tailored for these attention layers to maximize the preservation of old task representations, effectively anchoring the model’s knowledge and reducing the risk of forgetting [141]. Similarly, in the domain of semantic segmentation, the DiffusePast framework utilizes a dual-generator paradigm guided by text prompts or edge maps to produce semantically accurate images with reliable masks. This approach ensures that the structural layout and semantic content of old classes are preserved, allowing for precise mask generation that aligns with the distribution of downstream datasets [142].

Beyond conditioning mechanisms, alignment in the latent space has proven to be a crucial factor in stabilizing generative replay. In VAE-based systems, a significant cause of forgetting is the drift of generated features in the latent space, causing them to diverge from the original data distribution. The “Looking through the past” framework proposes a suite of modifications to mitigate this, including latent space distillation between current and previous models to reduce feature drift, and latent matching to improve the alignment of reconstructed features with original data. A key innovation in this work is the cycling of generations through the previously trained model, which forces the generated features to converge closer to the original data manifold, thereby enhancing knowledge retention [143]. This concept of feature-driven replay is further extended in methods like Genifer, where the generative model is explicitly trained to produce images that induce the same hidden features as real samples when passed through the classifier. This bridges the gap between image-level and feature-level replay, ensuring that the synthetic data is optimized for the specific discriminative needs of the continual learning agent [144].

The application of these advanced diffusion-based strategies extends beyond static image classification into complex decision-making and reinforcement learning environments, demonstrating the versatility of synthetic experience generation. In offline reinforcement learning, the “Continual Offline Reinforcement Learning via Diffusion-based Dual Generative Replay” framework decouples the policy into a diffusion-based generative behavior model and a multi-head action evaluation model. By training a task-conditioned diffusion model to mimic the state distributions of past tasks and pairing generated states with responses from the behavior generator, this method achieves high-fidelity replay of the sample space. This dual approach allows the policy to inherit distributional expressivity, encompassing a progressive range of diverse behaviors while mitigating forgetting through behavior cloning regularization [145]. Similarly, for trajectory generation in decision-making tasks, non-autoregressive diffusion models have been employed to avoid the compounding errors typical of autoregressive approaches, achieving state-of-the-art performance in continual learning benchmarks by generating task samples conditioned on trajectory timesteps [146]. These developments collectively demonstrate that the synergy between diffusion models, prototype conditioning, and latent alignment provides a powerful toolkit for synthesizing pseudo-data that effectively preserves the statistical and semantic integrity of past experiences. However, while these methods successfully synthesize data to overcome storage limits, they still typically assume access to the original data distribution for initial model training or require the storage of the generative model itself. In scenarios where even the original data distribution is inaccessible due to strict privacy regulations or legal constraints, the field must advance further toward completely data-free paradigms, a tran-

sition explored in the following section through Data-Free Knowledge Distillation and Synthetic Pseudo-Replay.

3.3 Data-Free Knowledge Distillation and Synthetic Pseudo-Replay

3.3 Data-Free Knowledge Distillation and Synthetic Pseudo-Replay

While the preceding section detailed how advanced diffusion models and latent alignment can synthesize high-fidelity replays to mitigate forgetting, a critical constraint in many real-world deployments is the complete unavailability of original training data due to privacy regulations, copyright restrictions, or data silos. In such scenarios, the paradigm of Data-Free Knowledge Distillation (DFKD) emerges as a vital solution, shifting the focus from replaying stored or generated past data to synthesizing pseudo-data that mimics the statistical properties of an unknown original distribution. Unlike traditional distillation methods reliant on static datasets, DFKD enables the transfer of intelligence from powerful teacher models to compact student architectures without ever exposing sensitive information. This approach is particularly indispensable for foundation models operating in privacy-critical domains such as healthcare and finance, where the direct sharing of patient records or financial transactions is legally constrained.

At the core of DFKD lies the generation of synthetic samples designed to effectively probe the teacher model’s decision boundaries. Early approaches often utilized generative networks trained to match batch normalization statistics or invert network weights to reconstruct input-like features. However, a significant challenge in this domain is “knowledge degradation,” where the distribution shift between synthetic and real data causes the student model’s performance to deteriorate over the course of training. To address this, recent advancements have introduced generative pseudo-replay mechanisms that conceptually bridge the gap between the diffusion-based strategies discussed previously and data-free constraints. For instance, researchers have proposed modeling the distribution of previously observed synthetic samples using a Variational Autoencoder (VAE) customized to learn optimal synthetic data representations [147]. By rehearsing the student model with samples produced by this VAE, the method prevents knowledge degradation without the need to store large buffers of synthetic images, thus eliminating memory overhead and mitigating privacy risks associated with data storage.

The integration of Differential Privacy (DP) into DFKD frameworks further strengthens the privacy guarantees of the distillation process, ensuring that the synthetic generation itself does not become a vector for information leakage. In many practical settings, even synthetic data generated by a teacher model may inadvertently leak information about the sensitive training set if the generator overfits. To counter this, frameworks have been developed where the teacher provides noisy information to the student, ensuring that the distilled knowledge adheres to strict privacy budgets. A notable example is the adaptation of the Private Aggregation of Teacher Ensembles (PATE) framework, where an ensemble of teachers trained on disjoint sensitive datasets votes on unlabeled public or synthetic data, with noise added to the aggregation to guarantee differential privacy [148]. This approach allows a student model to learn from the consensus of teachers without ever accessing individual data points or parameters, providing robust protection even against adversaries who can inspect the student’s internal workings.

Furthermore, the scope of DFKD has expanded beyond static image classification to encompass complex data modalities and specific architectural constraints inherent to modern foundation models. In the realm of Natural Language Processing, generative knowledge transfer techniques have been proposed where student networks are trained using text and output probabilities generated

by teacher models, ensuring that no user data is directly utilized [149]. This is particularly relevant for Large Language Models (LLMs), where recent work has demonstrated novel differentially private knowledge distillation algorithms that exploit synthetic data generated by a DP-LLM [150]. These methods transfer knowledge through both hard labels derived from the synthetic text and soft labels from the teacher’s output distribution, successfully compressing autoregressive models while preserving the privacy of the underlying training corpus. Similarly, in domains characterized by data scarcity and imbalance, such as medical applications, frameworks combining GANs with external memory replay have been utilized to generate synthetic clinical data that captures rare class distributions while maintaining low patient disclosure risk [151]. For structured environments prevalent in financial and demographic studies, composable generative models have been introduced to synthesize high-fidelity tabular data that preserves statistical utility while adhering to privacy constraints [152], leveraging the flexibility of combining various conditional generative components to handle mixed data types.

The security of the distillation process itself remains a focal point of current research, especially as attack vectors become more sophisticated. The “black-box” and “white-box” scenarios present different levels of risk regarding model confidentiality and data leakage. Innovative paradigms such as Absolute Zero-Shot Learning (AZSL) have been proposed, where a teacher model acts as a sentinel to guide the training of a student without any real data exposure, effectively achieving state-of-the-art performance in zero-shot learning tasks under strict security protocols [153]. Additionally, advanced attack vectors like label-only model inversion attacks highlight the vulnerabilities that persist even when only hard labels are exposed, necessitating more robust defense mechanisms within the distillation pipeline [154]. To mitigate such risks, frameworks like Sentinel-Guided Zero-Shot Learning introduce collaborative paradigms where the teacher guides the student through a generator, ensuring efficient collaboration without exchanging models or sensitive data [155]. Moreover, the efficiency of these data-free methods has been significantly improved through techniques like dataset condensation, which serves as a superior alternative to traditional data generators for private data generation. Theoretical analyses suggest that dataset condensation can provide privacy “for free” by ensuring that the existence of any single sample in the original dataset has a negligible impact on the synthesized data distribution [156], offering a scalable path forward for deploying foundation models in resource-constrained environments.

In conclusion, Data-Free Knowledge Distillation and Synthetic Pseudo-Replay represent a convergence of generative modeling, differential privacy, and efficient adaptation strategies that complement the high-fidelity replay mechanisms discussed earlier. By leveraging synthetic data generation through VAEs, GANs, and specialized diffusion approaches, and integrating rigorous privacy mechanisms like PATE, these methods enable the continuous evolution of AI systems without compromising data sovereignty. While this section focuses on classification, language, and structured data, the principles of synthesizing experiences to overcome data limitations naturally extend to sequential decision-making. The following section will explore how these generative strategies are adapted for reinforcement learning and offline settings, where agents must similarly construct synthetic experiences and compress environmental maps to operate effectively under strict data and memory constraints.

3.4 Generative Strategies for Reinforcement Learning and Offline Settings

3.4 Generative Strategies for Reinforcement Learning and Offline Settings

The principles of synthesizing experiences to overcome data limitations, as explored in the preceding section on Data-Free Knowledge Distillation, naturally extend to the domain of sequential decision-making. In Reinforcement Learning (RL), particularly within offline and continual learning settings, the integration of generative strategies represents a paradigm shift in how agents manage memory, mitigate catastrophic forgetting, and optimize sample efficiency. Traditional experience replay mechanisms, which rely on the uniform or prioritized storage of raw state-action transitions, often face significant limitations regarding memory capacity, distributional shifts, and the inability to synthesize novel experiences beyond the collected dataset. To address these challenges, recent research has pivoted toward sophisticated generative frameworks that construct synthetic experiences, compress environmental maps, and selectively curate offline data to facilitate robust sequential decision-making.

A primary avenue of innovation involves dual generative behavior modeling, where the policy architecture is decoupled into distinct components responsible for behavior generation and action evaluation. This approach is exemplified by frameworks designed for continual offline reinforcement learning, which utilize diffusion-based models to retain previous knowledge through the concurrent replay of generated pseudo-data. By training a task-conditioned diffusion model to mimic the state distributions of past tasks, agents can pair generated states with responses from a behavior generator, effectively creating high-fidelity replayed samples that encompass a progressive range of diverse behaviors [145]. This method allows the policy to inherit distributional expressivity, ensuring that the agent does not merely memorize specific trajectories but learns the underlying manifold of feasible behaviors. Such dual generative architectures are critical for preventing the degradation of performance on previously learned tasks when new skills are acquired sequentially, a common pitfall in lifelong RL scenarios.

Complementing these generative approaches are map-based experience compression techniques inspired by cognitive science. These methods move away from storing individual transitions in favor of organizing stored experiences into a concise, environment-model-like network of state-nodes and transition-edges. Drawing upon the principles of self-organizing networks, such as the Grow-When-Required (GWR) model, these systems merge similar samples to drastically reduce memory footprint while increasing the pairwise distance among stored samples to enhance their relevancy [157]. This “mental map” abstraction allows agents to maintain a global understanding of the state space, mitigating the redundancy inherent in naive replay buffers. Furthermore, this compression strategy facilitates more efficient retrieval and replay, as the agent can navigate the topological structure of the environment rather than sifting through a disorganized list of historical events.

In the realm of offline reinforcement learning, where agents must learn exclusively from pre-collected datasets without further environment interaction, the challenge of distributional shift between the replay buffer and the learned policy becomes paramount. Standard experience replay often fails in these Continual Offline Reinforcement Learning (CORL) settings due to the mismatch between stored experiences and the trajectories proposed by the evolving policy. To resolve this, advanced offline experience selection strategies have been developed, employing model-based schemes to filter data that most closely resembles the learned model’s distribution [158]. By retrofitting experience replay with dual behavior cloning architectures, these methods avoid the disturbances typically caused by behavior-cloning losses on the Q-learning process, thereby stabilizing learning in resource-limited scenarios.

Beyond simple selection, the synthesis of synthetic experiences has emerged as a powerful tool for bootstrapping learning and enhancing sample efficiency. Techniques such as Synthetic Experience Replay (SynthER) leverage diffusion-based approaches to flexibly upsample an agent’s collected experience, enabling the training of larger networks and higher update-to-data ratios even when original data is scarce [159]. Similarly, interpolation-based methods like Neighborhood Mixup Experience Replay generate synthetic transitions by interpolating between vicinal state-action features, preserving the locally linear approximation of the transition manifold and encouraging greater policy generalizability [160]. These generative interventions allow agents to “dream” or simulate outcomes in regions of the state space that were sparsely visited during data collection, effectively densifying the learning signal.

Moreover, the concept of “lucid dreaming” has been introduced to refresh past experiences within the buffer by leveraging the agent’s current policy to re-execute actions from stored states, replacing outdated memories with superior trajectories discovered through recent learning [161]. This dynamic updating of the replay buffer ensures that the agent always learns from the best available version of its past, rather than being anchored to suboptimal historical behaviors. In multi-agent and distributed settings, generative strategies also extend to federated distillation, where proxy experience memories are exchanged instead of raw data, preserving privacy while allowing for collective learning [162].

Collectively, these generative strategies for reinforcement learning and offline settings demonstrate that the future of memory mechanisms lies not in static storage but in dynamic, model-based synthesis and compression. By transforming raw data into generative behaviors, cognitive maps, and curated synthetic samples, these approaches enable agents to achieve forward transfer, minimize forgetting, and operate effectively under strict data and memory constraints. The convergence of diffusion models, self-organizing maps, and selective replay protocols marks a significant advancement toward truly lifelong and data-efficient autonomous systems. Building upon these generative strategies for sequential decision-making, the subsequent section explores their critical application in Federated Learning and multi-center contexts, where the focus shifts to addressing data heterogeneity and collaborative lifelong learning under stringent privacy regulations.

3.5 Privacy-Preserving Synthesis in Federated and Multi-Center Contexts

3.5 Privacy-Preserving Synthesis in Federated and Multi-Center Contexts

Building upon the generative strategies for reinforcement learning and offline settings discussed previously, the application of generative replay extends critically into Federated Learning (FL) and multi-center contexts. Here, the paradigm shifts from optimizing sample efficiency and memory compression to addressing the dual challenges of catastrophic forgetting and data heterogeneity under strict privacy constraints. In domains such as healthcare consortia and financial networks, regulations like GDPR and HIPAA prohibit the sharing of raw data, rendering standard parameter averaging or direct data exchange infeasible. Generative synthesis emerges as a vital bridge in these scenarios, enabling collaborative lifelong learning by creating high-fidelity pseudo-data that mimics the statistical properties of local distributions without exposing sensitive patient or user records.

A primary driver for adopting generative synthesis in FL is the mitigation of non-independent and identically distributed (non-IID) data, a condition that frequently causes model divergence and poor generalization in standard federated averaging. By leveraging Generative Adversarial Networks (GANs) locally, clients can capture complex data distributions and generate synthetic samples that enrich the global training pool. For instance, the [163] framework demonstrates how

synthetic data can be generated in a federated manner to address data incompleteness and class imbalance, achieving significant accuracy improvements over traditional baselines without violating privacy requirements. Similarly, [164] proposes a mechanism where clients pretrain local GANs to produce differentially private synthetic data, which is then aggregated by a parameter server to construct a global shared dataset. This approach effectively homogenizes data distributions across clients, fostering consistency among local models and enhancing global aggregation performance.

Beyond simple data augmentation, advanced frameworks have evolved to decouple the generation process from raw data entirely, utilizing server-side synthesis to further harden privacy guarantees. The [165] introduces a novel strategy where a generative model is trained on the server using data-free methods at the conclusion of each task. This allows the server to synthesize samples from past distributions, which are then distributed to clients to mitigate catastrophic forgetting during class-incremental learning. Crucially, this method eliminates the need for clients to store old data or models, offering flexibility for dynamic participation while preventing data leakage risks associated with client-side generation. Complementing this, [166; 167] reinforces the efficacy of server-side generative replay, demonstrating that mimicking past distributions through synthetic samples is a viable alternative to episodic memory storage in privacy-sensitive federated environments.

The intersection of generative synthesis and differential privacy (DP) has also yielded robust frameworks designed to withstand sophisticated inference attacks. While standard FL protects raw data, gradient updates can still leak sensitive information, a vulnerability exploited by gradient inversion attacks using generative priors as highlighted in [168]. To counter this, researchers have integrated DP directly into the generative process. The [169] paper proposes FedDPMS, an algorithm where clients utilize variational auto-encoders to augment local datasets with data synthesized from differentially private means of latent representations. This ensures that the synthetic data itself carries formal privacy guarantees before being used for training. Furthermore, [170] presents a protocol where data generators with strong differential privacy guarantees are shared instead of explicit gradient information. This “Secure Generative Data Exchange” allows for the synthesis of arbitrarily large datasets that retain distinctive features of private samples while providing resilience against poisoning and inference attacks.

In multi-center medical collaborations, where data heterogeneity is extreme and privacy requirements are stringent, generative replay has proven essential for achieving generalizable diagnostic models. [171] addresses the shortcomings of centralized aggregation in medical settings by exploiting experience replay and generative adversarial concepts. FedER integrates features from local nodes to create models that generalize across multiple datasets, such as those for tuberculosis and melanoma classification, while maintaining strict privacy. Similarly, [172] employs a dual-model architecture where an auxiliary “generative replay model” aggregates knowledge from heterogeneous institutions. This auxiliary model regulates the training of the primary task model with an unbiased target distribution, significantly improving prediction accuracy on highly heterogeneous medical data partitions.

The evolution of these techniques also includes the adoption of diffusion models, which offer superior sample quality compared to traditional GANs and align with the high-fidelity synthesis trends observed in reinforcement learning. [173] proposes training Denoising Diffusion Probabilistic Models (DDPM) across multiple data sources, demonstrating that high-quality samples can be generated even under statistical heterogeneity, thereby preserving data diversity and reducing communication overhead. Additionally, [174] introduces a hard feature matching method that synthesizes data by learning essential class-relevant features while discarding redundant ones, effectively erasing information about real features to enhance privacy while tackling non-IID challenges.

Despite these advancements, the deployment of generative synthesis in FL is not without risks. The potential for unintended memorization of rare sequences in generative models remains a concern, as investigated in [175], which notes that while FL clustering reduces memorization, strong user-level differential privacy is required to minimize it further. Moreover, the trade-off between privacy, fairness, and utility is intricate; [176] emphasizes that balancing these factors requires careful consideration of specific contexts. Nevertheless, the convergence of generative replay, differential privacy, and federated architectures provides a promising pathway toward sustainable, lifelong AI systems. These systems not only preserve privacy but also lay the groundwork for the advanced hybrid architectures and consistency regularization techniques discussed in the following section, offering a robust foundation for continuous adaptation in decentralized, sensitive data environments.

3.6 Negative Replay, Consistency Regularization, and Hybrid Architectures

3.6 Negative Replay, Consistency Regularization, and Hybrid Architectures

Building upon the robust privacy-preserving frameworks established in federated and multi-center contexts, the evolution of Continual Learning (CL) now extends beyond simple data rehearsal toward sophisticated strategies that leverage generative capabilities for refining decision boundaries, enforcing structural consistency, and constructing hybrid memory systems. While previous sections highlighted how generative synthesis mitigates data heterogeneity and privacy risks in decentralized settings, this subsection explores how these generative mechanisms are optimized for scenarios where storage is strictly constrained, data distributions are highly non-stationary, or perfect reconstruction is statistically infeasible. A critical innovation in this domain is the paradigm shift from positive exemplar replay to **negative replay**, where generated samples serve as antagonists rather than direct substitutes for past data. Traditionally, generative replay aims to synthesize high-fidelity data mimicking past distributions to prevent forgetting. However, achieving perfect reconstruction in high-dimensional spaces is often computationally prohibitive. An alternative and highly effective approach utilizes generated data as negative examples to sharpen the model’s decision boundaries. This strategy posits that while synthetic data may lack the fidelity to directly improve classification accuracy for old classes, it effectively functions as an “antagonist” to help the model distinguish new classes from old ones, thereby mitigating catastrophic interference. This concept is vividly demonstrated in [177], which shows that in complex class-incremental scenarios with high-dimensional data, generated patterns are most effective when used to define the boundaries between current and previous tasks, especially when learning experiences are small. By treating synthetic outputs as negative signals, the model learns to reject outdated feature configurations, ensuring that new knowledge does not overwrite old representations indiscriminately.

Parallel to negative replay, **consistency regularization** has emerged as a powerful mechanism to stabilize learning without relying solely on raw data storage, complementing the differential privacy techniques discussed previously by reducing the dependency on sensitive historical records. This approach enforces prediction consistency across different views or perturbations of the input, effectively acting as a self-supervised pretext task that preserves the semantic structure of previous tasks. The core intuition is that soft targets or probability distributions from a previous model state contain richer information about the data manifold than hard labels alone. By regularizing the current model to maintain consistency with these soft targets, the system can retain knowledge even when the explicit replay buffer is small or unavailable. Research indicates that [178] demonstrates how casting consistency regularization as a self-supervised objective significantly enhances the performance of ER frameworks. Specifically, stricter consistency constraints have been found to preserve previous task information more effectively than standard replay alone, while simul-

taneously improving model calibration and robustness against natural corruptions. This method transforms the replay buffer from a mere storage unit into a dynamic regularization engine, where the focus shifts from memorizing specific instances to maintaining the invariant relationships within the data distribution.

Furthermore, the integration of these concepts into **hybrid architectures** represents the state-of-the-art in balancing plasticity and stability, synthesizing the generative feedback loops mentioned in earlier sections with traditional experience replay. Hybrid approaches often combine generative feedback with explicit memory to create systems that are both memory-efficient and highly adaptable. One such architectural innovation involves equipping models with generative feedback or backward connections, allowing the network to “dream” or reconstruct past experiences internally without storing raw data. This “Replay-through-Feedback” approach, as detailed in [179], integrates the generative model directly into the main architecture, substantially reducing computational costs and training time while maintaining superior performance across various continual learning scenarios. This is particularly crucial in task-agnostic settings where task identity must be inferred, a scenario where pure regularization-based methods often fail. Additionally, hybrid strategies are being extended to reinforcement learning contexts, where dual generative models are employed to simulate both behavior and state distributions. For instance, [145] proposes decoupling the policy into a diffusion-based behavior model and a multi-head action evaluator, enabling the agent to inherit distributional expressivity and mitigate forgetting through high-fidelity pseudo-sample replay.

The synergy between negative sampling and consistency checks is also evident in advanced sampling strategies that prioritize diverse and meaningful transitions, echoing the feature-matching and diversity-preservation goals seen in federated diffusion models. While standard prioritized replay focuses on temporal-difference errors, newer methods incorporate global context to ensure that the replayed or generated samples provide maximal informational gain. Although [180] primarily addresses RL sampling, its principles of considering relative importance and diversity are directly applicable to designing hybrid CL systems that generate negative examples based on global interference patterns. Moreover, the application of consistency constraints extends to self-predictive representations, where agents are trained to predict their own latent states multiple steps into the future. As shown in [181], combining future prediction objectives with data augmentation forces the agent’s representations to remain consistent across multiple views, a technique that can be adapted for continual learning to ensure that latent spaces do not drift catastrophically during task switches.

Finally, the robustness of these hybrid and consistency-driven architectures is further enhanced by techniques that actively manage the quality of generated data, addressing the fidelity concerns raised in previous discussions on diffusion and GAN-based synthesis. Methods like [182] introduce quality-driven contrastive losses to ensure that generated positive pairs preserve essential semantics while introducing necessary variability. When adapted for continual learning, such mechanisms ensure that the consistency regularization is applied to high-quality synthetic views, preventing the propagation of errors that could arise from low-fidelity generative models. Similarly, the use of augmented negative sampling, as explored in [183], provides a theoretical basis for constructing unconstrained negative samples that refine decision boundaries without introducing excessive noise. By disentangling hard and easy factors and regulating the augmentation direction, these systems can generate precise negative examples that effectively counteract the bias towards new tasks. Collectively, these innovations in negative replay, consistency regularization, and hybrid architectural design mark a significant shift from passive data storage to active, dynamic knowledge preservation.

These advancements not only resolve the limitations of pure replay or regularization but also lay the theoretical and practical groundwork for the emerging field of model editing in foundation models, where precise, localized updates must be performed without disrupting the vast knowledge encoded in pre-trained weights.

4 Precision Model Editing and Targeted Knowledge Manipulation

4.1 Locate-and-Edit Paradigms and Mechanistic Insights

4.1 Locate-and-Edit Paradigms and Mechanistic Insights

As the foundational approach to modifying specific factual knowledge within Large Language Models (LLMs) without full retraining, the “locate-then-edit” framework serves as a critical precursor to the more scalable architectures discussed in subsequent sections. This paradigm operates on the hypothesis that factual associations are not diffusely distributed across the entire network but are instead localized within specific computational modules, particularly the feed-forward networks (FFN) or Multi-Layer Perceptron (MLP) layers of transformer architectures. By identifying these precise locations, researchers can apply targeted weight updates to inject new knowledge or correct errors while minimizing interference with unrelated capabilities. The efficacy of this approach relies heavily on two distinct phases: the mechanistic localization of knowledge storage and the subsequent application of precise mathematical updates to the identified parameters.

The first critical step in this paradigm involves causal tracing, a technique designed to pinpoint the specific model components responsible for recalling factual information. Seminal work in this area demonstrated that factual associations in autoregressive transformers correspond to localized computations within middle-layer feed-forward modules [184]. Through causal intervention experiments, researchers identified that specific neuron activations at the subject token position are decisive in mediating factual predictions. This discovery provided the mechanistic justification for directly manipulating these weights to update specific associations. Further investigations into the internal information flow revealed a three-step mechanism for attribute extraction: an enrichment process driven by early MLP sublayers, the propagation of relation information, and a final query step where attention heads extract the attribute from the enriched subject representation [185]. These mechanistic insights suggest that the storage and retrieval of facts follow a predictable structural pattern, enabling the design of surgical editing algorithms.

Building upon these localization findings, Rank-One Model Editing (ROME) was introduced as a pioneering algorithm that modifies feed-forward weights using a rank-one update constraint. ROME treats the FFN layer as a key-value memory system and computes a closed-form solution to update the weights such that the new fact is memorized while preserving the output for unrelated inputs [184]. This method relies on the assumption that the hidden states acts as keys and the output projections as values, allowing for a direct overwrite of specific knowledge entries. The success of ROME on zero-shot relation extraction tasks validated the viability of the locate-then-edit approach, demonstrating that direct manipulation of computational mechanisms could achieve high specificity and generalization.

However, the assumption that localization perfectly informs editing has been subject to rigorous scrutiny, revealing limitations that motivate the shift toward parameter-preserving strategies. Recent studies have uncovered surprising discrepancies between where knowledge is localized via causal tracing and where edits are most effective. For instance, research indicates that editing weights in layers different from those identified by representation denoising can sometimes yield superior

results, challenging the direct translation of mechanistic understanding into editing strategies [186]. This finding suggests that while causal tracing identifies components critical for recall, the optimal locus for intervention may depend on other factors, such as the plasticity of the layer or the interaction with residual connections. Furthermore, the validity of the locality hypothesis itself has been questioned, leading to the development of benchmarks like KLoB, which systematically evaluate whether existing locating methods can truly pinpoint the exact parameters embedding desired knowledge [187]. These challenges highlight the fragility of direct weight manipulation, setting the stage for architectures that decouple storage from the base model.

To address limitations in precision and the complexity of information flow within the direct editing paradigm, more advanced algorithms have been developed. Precise Model Editing in a Transformer (PMET) represents a significant evolution in this domain by recognizing that transformer layer hidden states comprise information from Multi-Head Self-Attention (MHSA), FFN, and residual connections. PMET argues that previous methods neglect the fact that hidden states contain information not specifically required for the FFN, leading to suboptimal editing performance [188]. By analyzing the distinct roles of MHSA and FFN, PMET optimizes the hidden states of both components but restricts weight updates to the FFN based on the optimized FFN-specific states. This approach acknowledges that MHSA encodes general knowledge extraction patterns that should remain invariant, thereby achieving state-of-the-art performance on counterfactual datasets by preventing the contamination of general reasoning capabilities with specific factual updates [188].

The scope of locate-and-edit paradigms has also expanded beyond standard transformer architectures, demonstrating the versatility of mechanistic insights. Investigations into state-space models like Mamba have revealed that factual recall in these architectures shares similarities with transformers, with specific components in middle layers showing strong causal effects, thus enabling the adaptation of rank-one editing methods to non-transformer models [189]. Additionally, the application of these principles to multimodal systems, such as text-to-image diffusion models, has shown that while knowledge localization differs—often distributed across UNet components rather than isolated MLPs—the core philosophy of identifying causal states for efficient, data-free editing remains applicable [190].

Despite these advancements, the field continues to grapple with the tension between mechanistic interpretability and practical editing efficacy, particularly regarding long-term stability. While methods like ROME and PMET provide powerful tools for targeted modification, the underlying assumption that knowledge is strictly localized and isolated remains an oversimplification of the interconnected nature of neural representations. Direct weight updates, even when precisely located, risk accumulating interference over sequential edits, a challenge that direct parameter modification struggles to overcome indefinitely. This limitation underscores the necessity of exploring alternative architectures that externalize knowledge storage to ensure lifelong adaptability, a transition that leads naturally to the memory-based and parameter-preserving approaches detailed in the following section.

4.2 Memory-Based and Parameter-Preserving Editing Architectures

4.2 Memory-Based and Parameter-Preserving Editing Architectures

While the “locate-and-edit” paradigm detailed in the previous section offers precise mechanistic control over factual associations, its reliance on direct weight manipulation introduces inherent risks of interference and instability when applied to sequential updates. The fragility of direct parameter modification becomes particularly pronounced in lifelong learning scenarios, where the

accumulation of successive edits can lead to catastrophic forgetting or the degradation of general capabilities. To address these limitations, a distinct paradigm has emerged: memory-based and parameter-preserving editing architectures. These approaches fundamentally decouple the storage of new information from the static parameters of the base model. By utilizing explicit external memory modules, discrete adapters, or auxiliary networks to localize changes, these methods ensure that original model weights remain frozen. This architectural separation preserves the integrity of pre-trained knowledge while enabling scalable, reversible, and high-fidelity edits, serving as a critical bridge between the surgical precision of direct editing and the high-throughput demands of batch processing discussed later.

A seminal approach in this domain involves utilizing discrete, local codebooks to manage edits without modifying underlying weights. The GRACE framework exemplifies this strategy by implementing “spot-fixes” on streaming errors through the creation of a discrete memory within the model’s latent space. Rather than updating neural weights, GRACE writes new mappings into a pre-trained model’s latent space, effectively creating a lookup table of corrections that activates only for specific input contexts. This method allows for thousands of sequential edits using only streaming errors, demonstrating state-of-the-art performance in retaining edits while generalizing to unseen inputs across T5, BERT, and GPT models. The success of GRACE highlights a critical insight: by treating edits as discrete additions to a latent codebook rather than continuous shifts in weight space, models can avoid the interference patterns that typically lead to performance degradation in consecutive editing scenarios [191].

Complementing discrete latent memory, other architectures employ explicit external memory stores coupled with retrieval mechanisms to dynamically modulate model behavior. The SERAC (Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model) framework represents a high-capacity alternative that stores edits in an explicit memory and learns to reason over them to adjust the base model’s predictions. Unlike methods that struggle to define the scope of an edit, SERAC effectively handles the intended scope of modifications, consistently outperforming existing approaches in complex tasks such as question answering, fact-checking, and dialogue generation. By decoupling edit storage from model parameters, SERAC achieves high performance even after numerous edits, addressing the expressiveness limitations of direct weight manipulation [192]. Similarly, the PEMA (Plug-in External Memory Adaptation) method introduces a parameter-efficient fine-tuning strategy that integrates external memory to store context representations mapped with target tokens. This approach enables adaptation without accessing all model weights, utilizing weight matrices in the final layer to enhance efficiency while maintaining sentence meaning and generating appropriate styles [193].

Another significant branch of parameter-preserving architectures involves the use of auxiliary networks or “patchers” that operate alongside the main model. The Transformer-Patcher method addresses the challenge of sequential model editing by adding and training a mere few neurons in the last Feed-Forward Network layer. This minimalistic addition allows the model to successively correct thousands of errors while maintaining reliability, generality, and locality. By isolating the edits to these added neurons, Transformer-Patcher prevents the corruption of the original model’s knowledge, achieving state-of-the-art performance in sequential editing scenarios where traditional fine-tuning and HyperNetwork-based methods fail [194]. In a similar vein, the MEND (Model Editor Networks using Gradient Decomposition) approach utilizes a collection of small auxiliary editing networks. These networks learn to transform gradients obtained from standard fine-tuning into low-rank decompositions, enabling fast, local edits to pre-trained models with over 10 billion parameters. MEND demonstrates that auxiliary networks can effectively edit massive models on

a single GPU, providing a scalable solution for post-hoc corrections that leaves the base model otherwise intact [195].

The efficacy of parameter-preserving methods has been further validated through comprehensive evaluations comparing them against parameter-modifying techniques. Recent studies reveal an intriguing phenomenon: while most parameter-modifying methods consistently degrade performance across various tasks after a few sequential edits, parameter-preserving methods effectively maintain the model’s fundamental capabilities. However, these studies also note that while parameter-preserving approaches excel at stability, they may occasionally struggle with recalling edited knowledge when presented in formats significantly different from the training instances. This nuanced understanding underscores the importance of continued research into improving the generalization of memory-based edits without compromising their inherent stability advantages [44].

Furthermore, the integration of memory mechanisms extends beyond text to address specific domain challenges, such as hallucination correction. The PURR model leverages the power of large language models to generate synthetic corruptions, training compact editors to denoise these errors. While not strictly an external memory in the traditional sense, it operates on a principle of preserving the base model’s integrity by training specialized, efficient editors that act as a corrective layer, achieving faster execution times and improved attribution compared to prompt-based methods [196]. Additionally, theoretical advancements in differentiable random access memory using lattices suggest that augmenting standard architectures with single memory layers can scale parameter counts to billions of entries with negligible computational overhead, offering a promising direction for future memory-augmented editors that require $O(1)$ performance regardless of memory size [197].

In conclusion, memory-based and parameter-preserving editing architectures represent a robust solution to the stability-plasticity dilemma in Foundation Models, effectively overcoming the interference limitations inherent in the direct weight manipulation strategies discussed previously. By externalizing knowledge updates into discrete codebooks, explicit memory stores, or auxiliary networks, these methods enable the realization of truly lifelong learning systems capable of absorbing vast amounts of new information without erasing the past. However, while these approaches excel at maintaining stability during sequential, isolated updates, they face distinct challenges when the objective shifts to massive, simultaneous knowledge integration. Addressing this need for high-throughput, batch-oriented editing requires a further evolution in architectural design, leading naturally to the meta-learning strategies and hypernetwork-based solutions detailed in the following section.

4.3 Hypernetworks and Scalable Batch Editing

4.3 Hypernetworks and Scalable Batch Editing

While the memory-based and parameter-preserving architectures discussed in the previous section excel at maintaining stability through thousands of sequential, isolated updates, a distinct challenge arises when the objective shifts from incremental corrections to massive, simultaneous knowledge integration. Real-world scenarios often demand the rectification of thousands of outdated or incorrect facts concurrently, a task where traditional locate-and-edit methods—optimized for single facts or small triplets—face severe scalability bottlenecks. When applied to large batches, these sequential approaches suffer from computational costs that grow linearly or super-linearly, while the interference between simultaneous updates frequently leads to catastrophic forgetting or optimization instability. To bridge the gap between the stability of parameter-preserving methods

and the need for high-throughput updates, recent research has increasingly turned to meta-learning strategies employing hypernetworks. These approaches enable efficient, scalable batch editing by decoupling the generation of parameter shifts from the base model’s backbone.

Hypernetworks, defined as neural networks that generate weights or parameter shifts for a main target network, offer a powerful architectural solution for this domain. In the context of model editing, a hypernetwork acts as a meta-learner: it takes a description of the desired knowledge update (e.g., a subject-predicate-object triplet) and outputs a specific weight modification vector for the foundation model. This mechanism allows for the amortization of editing costs; once trained, the hypernetwork can generate edits for new facts via a single forward pass, bypassing the need for iterative gradient descent on the massive backbone for each individual update. However, standard hypernetwork architectures have historically struggled with scalability during synchronous editing operations involving large volumes of facts. The primary limitation lies in the aggregation of parameter shifts; simply summing the outputs of a hypernetwork for thousands of distinct facts often results in suboptimal updates due to non-linear interactions and memory constraints.

A breakthrough in overcoming these scalability limits is the formulation of parameter shift aggregation as a least-squares problem, a strategy exemplified by the MAssive Language Model Editing Network (MALMEN) [198]. This approach fundamentally rethinks how multiple edits are combined. Instead of relying on heuristic summation or sequential application, MALMEN treats the collective update as an optimization problem where the goal is to find a parameter shift that minimizes the error across all targeted facts simultaneously. By utilizing the normal equation to solve this least-squares aggregation, the method computes an optimal update direction that explicitly accounts for the interdependencies among thousands of facts. This mathematical formulation enables the system to scale operations from single facts to massive batches without the exponential increase in computational overhead typically associated with batch editing.

Furthermore, to accommodate the editing of multiple facts simultaneously within limited memory budgets, advanced frameworks have introduced architectural separations between the hypernetwork computation and the main language model updates. As demonstrated in recent works, separating these computations enables the system to handle arbitrary batch sizes on both the hypernetwork and the target model [198]. This decoupling is crucial for managing the memory footprint of trillion-parameter models, where storing intermediate gradients for thousands of simultaneous edits would be prohibitive. By processing the hypernetwork generation in batches and aggregating the resulting shifts via efficient linear algebraic operations before applying them to the main model, these methods achieve a level of throughput that is orders of magnitude higher than previous baselines. Empirical evaluations have shown that such hypernetwork-based approaches can edit hundreds of times more facts than strong baseline methods while maintaining or even surpassing the performance of editors specifically designed for single-fact updates on architectures like GPT-J and T5 [198].

The integration of hypernetworks also opens avenues for more sophisticated meta-learning dynamics within the editing process. For instance, the concept of learning optimal update directions and scaling factors, rather than just raw weight shifts, aligns with broader meta-learning principles where the “learning to learn” paradigm is applied to the editing trajectory itself. Techniques that involve learning task-specific parameter shifts with respect to initialization changes, as seen in large-scale meta-learning with continual trajectory shifting, provide theoretical underpinnings for why hypernetworks can effectively generalize across diverse editing tasks [199]. By estimating the required shift of task-specific parameters relative to initialization changes, these methods increase the frequency of effective meta-updates, thereby improving the convergence quality of the learned

editing functions.

Moreover, the application of hypernetworks in batch editing benefits from advances in handling non-mutually exclusive tasks, a common scenario in continual learning where new knowledge must be integrated without erasing old capabilities. Standard meta-learning often fails when tasks are not mutually exclusive because the model tends to memorize task identifiers rather than learning adaptable representations. However, regularization techniques that enforce mutual exclusivity or utilize information-theoretic objectives can be adapted to the hypernetwork training phase to ensure that the generated edits are specific to the intended facts and do not inadvertently alter unrelated knowledge domains [200; 201]. This ensures that the scalable batch editing process maintains the integrity of the model’s general reasoning abilities while injecting vast amounts of new information.

In addition to efficiency, hypernetwork-based batch editing offers enhanced flexibility for dynamic environments. The ability to generate parameter shifts on the fly allows for real-time updates in response to streaming data, a capability that is essential for applications requiring up-to-date knowledge in non-stationary environments. When combined with mechanisms for detecting task switches or distribution shifts, hypernetworks can dynamically adjust the magnitude and direction of edits to preserve in-distribution knowledge while rapidly acquiring out-of-distribution facts [202]. This adaptability makes hypernetwork-driven editing a robust solution for lifelong intelligence systems that must continuously evolve without the latency of full retraining.

In summary, the deployment of hypernetworks for scalable batch editing marks a significant paradigm shift from localized, sequential corrections to massive, simultaneous knowledge manipulation. By framing parameter aggregation as a solvable least-squares problem and decoupling computational stages to optimize memory usage, these methods overcome the traditional bottlenecks of model editing. While the parameter-preserving methods of the previous section prioritize long-term stability through external memory, and the following section will explore the complexities of temporal dynamics and event-level updates, hypernetwork-based strategies uniquely address the critical need for high-volume, efficient adaptation. The synergy between meta-learning theories and hypernetwork architectures not only enhances the scale and speed of updates but also ensures the stability and specificity required for reliable foundation model adaptation. As foundation models continue to grow in size and complexity, hypernetwork-based strategies will likely become the standard mechanism for maintaining their accuracy and relevance in an ever-changing information landscape.

4.4 Sequential Editing, Temporal Dynamics, and Event-Level Updates

4.4 Sequential Editing, Temporal Dynamics, and Event-Level Updates

While Section 4.3 demonstrated how hypernetworks can efficiently handle massive, simultaneous batches of updates, real-world deployment often demands a different capability: the absorption of a continuous, non-stationary stream of information over time. This necessitates a paradigm shift from static, one-off corrections or bulk updates to continuous, lifelong model editing. The core challenge in this domain is maintaining sequential stability, where a model must integrate a flow of new events without succumbing to performance degradation, toxicity buildup, or the catastrophic forgetting of previously edited knowledge. This subsection explores the evolution from simple triplet manipulation to complex event-level updates and temporal knowledge retention, addressing the critical need for systems that can evolve alongside a changing world while preserving the logical consistency discussed in the following section.

A primary obstacle in lifelong editing is the phenomenon of performance collapse under sequential operations. Traditional parameter-modifying methods, which directly alter model weights to inject new facts, often exhibit a rapid decline in reliability as the number of edits increases. Research indicates that these methods can lead to “toxicity buildup” and “toxicity flash,” where the accumulation of edits causes the model to generate harmful outputs or lose coherence. This degradation is frequently attributed to “pattern mismatch,” where the structural characteristics of new knowledge conflict with the latent representations established by prior edits. To mitigate this, advanced strategies such as Wise-Layer Knowledge Editor (WilKE) have been proposed, which dynamically select editing layers based on the degree of pattern matching between the new knowledge and the existing model state. By aligning the edit location with the model’s intrinsic structural patterns, WilKE significantly improves performance in lifelong editing scenarios, demonstrating that layer selection is a critical variable for maintaining stability over time [203].

Furthermore, the nature of real-world knowledge updates often transcends the modification of isolated subject-relation-object triplets. In reality, knowledge evolves through the occurrence of complex events that imply a network of related factual changes. Conventional editing methods, which treat facts as independent entities, struggle to capture these interdependencies, leading to incomplete updates and logical inconsistencies that exacerbate the reasoning failures detailed in Section 4.5. To address this limitation, the concept of Event-Level Knowledge Editing has been introduced. This approach posits that editing a single event should automatically propagate updates to all entailed factual triplets and even influence the model’s understanding of future trends associated with that event. Benchmarks such as ELKEN have been constructed to evaluate this capability, revealing that existing triplet-based methods face significant challenges in handling the efficiency and completeness required by event-driven updates. The shift toward event-level editing represents a move from mechanical fact replacement to a more holistic integration of contextual knowledge, ensuring that models remain coherent when faced with cascading information changes [204]. Complementing this, other frameworks like EVEDIT emphasize the importance of defining “deductive editing boundaries” to prevent ambiguity, pairing facts with event descriptions to ensure that the scope of an edit is logically sound and consistent with unchanged background knowledge [205].

Temporal dynamics introduce another layer of complexity, particularly when distinguishing between correcting intrinsic errors and updating outdated information due to worldly shifts. A critical insight in recent literature is that updating outdated knowledge should not necessitate the erasure of historical context. Models deployed in time-sensitive domains must retain the ability to recall historical states while simultaneously integrating current realities. However, standard editing techniques often induce catastrophic forgetting of this historical knowledge. To resolve this, the task of Temporal Knowledge Editing (TKE) has been formalized, accompanied by benchmarks like AToKe to assess a model’s ability to manage time-dependent facts. Innovative frameworks such as Multi-Editing with Time Objective (METO) have been developed to concurrently edit both historical and new knowledge, optimizing the model’s predictions based on the specific temporal context of each fact. This approach ensures that a model can answer questions about the past accurately while reflecting the most recent developments, thereby preserving the richness of temporal evolution rather than flattening it into a single current state [25].

Achieving true lifelong editing also requires architectural innovations that decouple the storage of new knowledge from the base model parameters to prevent interference, bridging the gap between the hypernetwork approaches of Section 4.3 and the memory-based strategies discussed earlier. Methods like GRACE (Lifelong Model Editing with Discrete Key-Value Adaptors) exemplify this

by implementing spot-fixes through a discrete, local codebook of edits written into the model’s latent space. By avoiding direct weight modification, GRACE enables thousands of sequential edits with minimal impact on unrelated inputs, effectively solving the scalability issues inherent in weight-based approaches. Similarly, Larimar introduces a brain-inspired architecture utilizing distributed episodic memory to facilitate dynamic, one-shot updates without retraining, offering significant speed advantages and flexibility in sequential editing setups. These parameter-preserving strategies highlight a growing consensus that sustainable lifelong intelligence relies on external or modular memory mechanisms rather than the continuous plasticity of the core network weights [191; 206].

Finally, the robustness of sequential editing is contingent upon the model’s ability to generalize edited knowledge without compromising its fundamental capabilities. Comprehensive evaluations reveal that while parameter-preserving methods excel at maintaining general abilities, they sometimes struggle with recalling edited knowledge in varied formats. Conversely, parameter-modifying methods may achieve high success rates initially but degrade rapidly across tasks after multiple edits. This duality underscores the necessity for unified frameworks that can balance specificity with generalization, ensuring that the pursuit of up-to-date knowledge does not come at the cost of the model’s broader reasoning and linguistic proficiencies [44]. As the field advances, the integration of event-level semantics, temporal awareness, and memory-augmented architectures will be paramount in developing Foundation Models capable of enduring, adaptive, and safe lifelong learning. However, even with these temporal and event-based advancements, a fundamental barrier remains: ensuring that these localized updates propagate consistently throughout the model’s reasoning capabilities without inducing logical contradictions, a challenge that leads directly to the issues of generalization and the reversal curse explored in the next section.

4.5 Generalization, Consistency, and the Reversal Curse

4.5 Generalization, Consistency, and the Reversal Curse

While Section 4.4 established the necessity of handling sequential updates, temporal dynamics, and event-level cascades in lifelong editing, a fundamental barrier remains: ensuring that these localized updates propagate consistently throughout the model’s reasoning capabilities without inducing logical contradictions. Achieving true editability requires more than merely inserting a fact; it demands that the model generalize this knowledge across diverse contexts, particularly in multi-hop scenarios and bidirectional relationships. A critical manifestation of this limitation is the “reversal curse,” a phenomenon where auto-regressive decoder models, even after being explicitly trained or edited on a statement “A is B,” fail to infer the reverse relationship “B is A.” This failure highlights a structural deficiency in the logical deduction capabilities of current architectures, raising serious concerns for applications requiring symmetric knowledge retrieval, such as knowledge graph construction [207]. The curse suggests that while models may successfully store edited facts in a unidirectional manner, they often lack the structural understanding to access this knowledge when the query structure is inverted.

The severity of the reversal curse has been extensively documented, revealing that even state-of-the-art models like GPT-4 struggle significantly with reverse queries compared to forward ones. For instance, if a model is edited to know that “Uriah Hawthorne is the composer of Abyssal Melodies,” it frequently fails to answer “Who composed Abyssal Melodies?” correctly, despite the explicit training on the forward statement [208]. This issue is not merely a data scarcity problem but is deeply rooted in the next-token prediction objective inherent to causal language models,

which restricts comprehension to preceding context. Consequently, standard editing techniques that do not account for bidirectional attention mechanisms often result in edits that are fragile and non-generalizable. Recent research indicates that mitigating this curse requires more than simple fine-tuning; it necessitates novel training objectives, such as bidirectional causal modeling optimization, which can improve accuracy on reversed tasks from near zero to substantial levels [209]. Furthermore, semantic-aware permutation training has been proposed to address the root cause by allowing models to predict antecedent words, thereby fostering a more robust understanding of relational symmetry [210].

Beyond bidirectional recall, the consistency of model editing is severely tested in multi-hop reasoning scenarios, where the model must chain multiple pieces of information to reach a conclusion. Editing a single node in a reasoning chain often fails to update the derived conclusions, leading to entailed knowledge propagation failures. Investigations into multi-hop factual shortcuts reveal that LLMs often rely on direct co-occurrences between initial and terminal entities rather than performing genuine multi-step reasoning. When knowledge editing is applied, these shortcuts can lead to approximately 20% of failures, as the model may revert to pre-editing statistical associations rather than utilizing the newly edited logical path [211]. This reliance on shortcuts means that editing a fact does not guarantee that the model will correctly update all downstream inferences that depend on that fact. The problem is exacerbated by the fact that many models exploit dataset artifacts to produce correct answers without genuinely connecting information across multiple supporting facts, a behavior known as disconnected reasoning [212].

To rigorously evaluate these limitations, new benchmarks and metrics have been developed to assess the reversibility and consistency of edited models. The Bidirectional Assessment for Knowledge Editing (BAKE) benchmark, for example, specifically targets the ability of edited models to recall knowledge in the reverse direction of the edit, exposing serious deficiencies in current methods [213]. Similarly, the Multi-hop Reasoning Evaluation of LLMs by Knowledge Edition (MRKE) benchmark addresses the issue of data contamination and the lack of reasoning chain evaluation in standard benchmarks. MRKE utilizes edited versions of existing datasets to create unprecedented knowledge scenarios, revealing that even advanced models like GPT-4 achieve correct reasoning chains in only a small fraction of cases, highlighting the fragility of their compositional reasoning abilities [214]. These evaluations underscore that successful model editing requires not just the insertion of a fact but the restructuring of the model’s internal reasoning pathways to ensure logical consistency.

Furthermore, the challenge of maintaining consistency extends to the preservation of general abilities and the avoidance of knowledge conflicts. When models are edited to correct specific facts or remove biases, there is a risk of degrading general reasoning capabilities or introducing new hallucinations. The interplay between editing and multi-hop reasoning is complex; for instance, injecting memories into specific attention heads can rectify multi-hop failures, yet this approach must be carefully calibrated to avoid disrupting existing knowledge structures [215]. The prevalence of attention glitches, where the Transformer architecture intermittently fails to capture robust reasoning over long dependencies, further complicates the reliability of edited models in complex inference tasks [216].

Addressing these challenges requires a shift from unidirectional, fact-level editing to frameworks that explicitly model bidirectional relationships and logical entailment. Approaches such as Bidirectionally Inversible Relationship moDeling (BIRD) attempt to incorporate reverse relationships directly into the editing objectives, demonstrating improvements across various model sizes [213]. Additionally, strategies like Reverse Training, which involve training models on both forward and reversed strings, have shown promise in resolving the reversal curse by doubling the effective train-

ing signal for relational symmetry [217]. Ultimately, achieving true generalization and consistency in model editing demands a holistic view of the model’s knowledge graph, ensuring that edits are not isolated updates but are integrated into a coherent, logically sound framework capable of robust multi-hop and bidirectional reasoning. These insights into the internal logical coherence of edited models provide a crucial foundation for the subsequent discussion on deploying these systems in constrained environments, where access to internal parameters may be limited and domain-specific consistency is paramount.

4.6 Black-Box, Multilingual, and Domain-Specific Editing

4.6 Black-Box, Multilingual, and Domain-Specific Editing

Building upon the internal logical coherence challenges discussed in Section 4.5—specifically the reversal curse and multi-hop consistency failures—the practical deployment of Foundation Models (FMs) introduces a new set of constraints: the inability to access internal parameters and the necessity for cross-lingual and domain-specific reliability. While previous sections established the theoretical barriers to consistent reasoning, real-world applications often operate in environments where traditional gradient-based “locate-and-edit” methods like ROME are inapplicable due to proprietary restrictions or architectural opacity. This section surveys the evolving landscape of model editing techniques designed for inaccessible black-box systems, multilingual synchronization challenges, and high-stakes domain-specific environments such as healthcare and software engineering. These scenarios demand editing strategies that operate without direct weight manipulation, ensuring safety, consistency, and linguistic alignment across diverse contexts, thereby bridging the gap between theoretical editability and deployable robustness.

Black-Box Editing via In-Context and Prompt-Based Strategies In many commercial and proprietary settings, models are accessed solely through inference APIs, rendering standard parameter-modifying methods obsolete. To address this constraint, researchers have pivoted toward In-Context Learning (ICL) as a viable mechanism for temporary yet effective knowledge injection. Empirical studies demonstrate that providing demonstration contexts containing corrected facts allows large language models to override parametric knowledge without any gradient updates. For instance, research indicates that In-Context Knowledge Editing (IKE) can achieve competitive success rates comparable to gradient-based methods on models like GPT-J, while significantly reducing side effects such as over-editing unrelated facts or catastrophic forgetting [218]. This approach is particularly valuable for black-box scenarios where the internal architecture remains opaque, offering a non-invasive alternative to the structural interventions discussed previously.

Beyond simple demonstration, advanced prompting strategies have been developed to enhance the robustness of black-box editing against the fragility observed in parameter-space updates. The EREN framework, for example, leverages instruction-finetuned models to prioritize contextual notes over parametric memory, enabling the integration of multiple edits and ensuring the model correctly responds to syntactically similar but semantically unrelated inputs [219]. Furthermore, for tasks requiring controlled text generation in specific domains, score-based progressive editors like ScoPE have been introduced. These methods iteratively edit intermediate tokens during the auto-regressive process to align outputs with target domain attributes, effectively guiding black-box models toward desired behaviors without parameter access [220]. Similarly, approaches like CrossTune utilize label-enhanced cross-attention networks to adapt black-box models for few-shot classification tasks without the computationally expensive search for task-specific prompts [221]. These methods collectively offer a pathway to mitigate the stability issues highlighted in the fol-

lowing section by decoupling the edit from the base weights entirely.

Multilingual Synchronization and Cross-Lingual Consistency A significant challenge in global model deployment, which exacerbates the consistency issues noted in Section 4.5, is ensuring that knowledge edits propagate uniformly across languages. Traditional editing techniques often operate in a monolingual vacuum, leading to situations where a fact is corrected in English but remains outdated or hallucinated in Spanish or Chinese, thereby creating logical asymmetries across linguistic boundaries. To mitigate this, recent work has formalized the task of cross-lingual model editing, where an edit in a source language must naturally transfer to target languages. Studies have shown that naive application of monolingual editing baselines fails to propagate changes effectively across linguistic boundaries. Consequently, frameworks employing language anisotropic editing have been proposed, which amplify specific parameter subsets for each language to ensure synchronized updates [222].

Further advancing this frontier, researchers have explored the use of multilingual patch neurons to store cross-lingual knowledge explicitly. By training these specialized neurons, it is possible to enhance existing editing methodologies to handle multilingual synchronization with minimal architectural modification [223]. Additionally, the development of knowledge-anchored biomedical multilingual models, such as KBioXLM, illustrates how incorporating entity, fact, and passage-level alignments during pre-training can significantly improve cross-lingual zero-shot and few-shot performance, laying the groundwork for more robust downstream editing in non-English domains [224]. These efforts highlight the necessity of moving beyond English-centric editing paradigms to support truly global AI systems, ensuring that the logical consistency sought in previous sections is maintained regardless of the query language.

Domain-Specific Precision: Healthcare and Code Generation In critical domains like healthcare and software engineering, the margin for error is negligible, necessitating editing techniques that offer high precision and explainability to avoid the safety pitfalls detailed in Section 4.7. In the medical field, where hallucinations can have life-threatening consequences, model editing is being tailored to integrate external knowledge bases directly into the inference process. For example, MedEdit proposes a comprehensive retrieval strategy to extract medical facts from external databases and incorporates them into the query prompt, significantly improving accuracy in medical question answering without retraining the underlying model [225]. Beyond factual correction, recent studies have addressed the editing of explanatory abilities in medical LLMs. The MedLaSA strategy employs causal tracing to identify precise neuronal locations for medical knowledge and introduces layer-wise scalable adapters, allowing for targeted updates to both facts and their explanations while preserving irrelevant knowledge [226].

Similarly, the domain of code generation presents unique editing challenges, as modifications often require understanding complex logical dependencies and syntax, where inconsistency can lead to system failures. Research into instructional code editing has revealed significant gaps between open and closed models in their ability to follow natural language instructions for code modification. To bridge this gap, curated datasets and fine-tuning strategies have been developed to enhance the code editing capabilities of open models, enabling them to perform feature additions, bug fixes, and solution refactoring with greater reliability [227]. Moreover, efficient general editing models like G-SPEED have been proposed to handle diverse editing requirements through a single, sparse architecture, demonstrating that smaller models can surpass massive LLMs in specific editing-oriented tasks when optimized correctly [228].

Collectively, these advancements underscore a paradigm shift from monolithic, parameter-heavy

editing toward flexible, context-aware, and domain-specialized strategies. Whether through black-box prompt engineering, cross-lingual neuron patching, or medical-grade causal interventions, the field is rapidly evolving to meet the demands of dynamic, real-world applications where static pre-training is no longer sufficient. However, as these methods push the boundaries of what can be edited, they also bring into sharper focus the trade-offs between localization and global stability, setting the stage for a critical examination of safety pitfalls, bias removal, and the preservation of general abilities in the subsequent section.

4.7 Safety Pitfalls, Bias Removal, and Preservation of General Abilities

4.7 Safety Pitfalls, Bias Removal, and Preservation of General Abilities

While the preceding section highlighted the adaptability of editing techniques for black-box, multilingual, and domain-specific scenarios, the practical deployment of these methods introduces a critical dichotomy: the tension between localized precision and global stability. As foundation models are increasingly tasked with dynamic updates, a rigorous examination reveals that the pursuit of targeted corrections often compromises the model’s broader cognitive architecture. This section explores the safety pitfalls inherent in current methodologies, focusing on the degradation of general abilities, the emergence of logical inconsistencies, the complexities of social debiasing, and the vulnerabilities to adversarial exploitation. These challenges underscore that effective editing requires not just successful knowledge injection, but the preservation of the intricate web of reasoning and safety guardrails established during pre-training.

Degradation of General Abilities and Cognitive Stability One of the most alarming findings in the domain of model editing is the potential for significant degradation of general abilities following even minor parameter modifications. Research indicates that while editing methods effectively improve factuality for specific targets, they frequently fail to maintain the model’s proficiency in unrelated tasks such as reasoning, question answering, and language understanding. For instance, [229] systematically analyzed popular editing methods across multiple large language models and task categories, revealing that current techniques struggle to simultaneously enhance factuality and preserve general capabilities. In extreme cases, a single edit applied to a model like LLaMA-1 resulted in a catastrophic performance collapse, reducing accuracy to nearly zero across all evaluated tasks. This phenomenon underscores the fragility of high-dimensional parameter spaces, where localized updates can propagate destructive ripple effects through the network, destabilizing the foundational skills that make these models versatile.

Logical Consistency, Knowledge Distortion, and the Reversal Curse Beyond the erosion of general utility, model editing introduces profound challenges regarding knowledge consistency and logical coherence. The “butterfly effect” of editing implies that modifying a single fact can trigger cascading inconsistencies throughout the model’s knowledge base. [230] highlights two critical pitfalls: knowledge conflict and knowledge distortion. When editors attempt to update groups of facts that logically clash, they may exacerbate inherent inconsistencies within the model, leading to contradictory outputs. Furthermore, the act of altering parameters to inject new factual knowledge can irreversibly warp the innate knowledge structure, causing the model to generate distorted responses for queries only tangentially related to the edit. This issue is further compounded by the “reversal curse,” where models fail to recall edited knowledge in bidirectional contexts. As demonstrated in [213], while models may successfully recall an edited fact in the direction of the prompt

(e.g., “Capital of France is London”), they often fail to infer the reverse relationship (e.g., “London is the capital of France”), revealing a lack of deep semantic integration and logical symmetry in the edited representations.

Challenges in Social Debiasing and Stylistic Integrity The application of model editing for social debiasing presents another layer of complexity, offering both potential solutions and significant risks. While editing provides a targeted mechanism to mitigate stereotype biases without the data hunger of fine-tuning, its effectiveness is often limited by poor generalization. [231] found that while existing methods can mitigate biases in specific instances, the debiasing effect frequently fails to generalize to semantically equivalent sentences, limiting its real-world utility. Moreover, the robustness of these edits varies significantly by method; parameter-preserving approaches like SERAC show resilience in sequential editing, whereas internal parameter-modifying methods tend to degrade rapidly as the number of edits increases. Compounding these technical limitations is the risk of unintended stylistic alterations and misinformation. [232] observed that conversational models attempting to debias news content often introduce unnecessary changes that alter the author’s original style or, worse, generate new forms of misinformation, indicating that current models lack the nuanced understanding required for responsible editorial intervention.

Adversarial Robustness and Malicious Exploitation Safety concerns are further amplified when model editing is viewed through the lens of adversarial robustness and malicious exploitation. The very mechanisms designed to correct models can be repurposed to inject durable backdoors or unsafe behaviors. [233] investigates this paradox, revealing that injecting accurate information can destabilize a model’s foundational ethical framework, leading to unpredictable and potentially unsafe behaviors. The study proposes using model editing as a tool for topical red-teaming, demonstrating how targeted edits can bypass safety guardrails and induce hazardous outputs within specific domains. Similarly, [234] identifies a “ripple effect” in the hidden space that is difficult to detect but significantly impairs model efficacy, suggesting that current evaluation metrics may overlook subtle yet dangerous deviations in model behavior.

Towards Robust, Reversible, and Safe Editing Frameworks To address these multifaceted challenges, the community is moving towards more robust frameworks that prioritize stability and reversibility, bridging the gap between the specialized strategies discussed previously and the unified evaluations detailed in the following section. Approaches such as [192] (SERAC) demonstrate that decoupling edits from the base model weights via explicit memory modules can achieve high performance while mitigating the degradation of general abilities, offering a safer alternative to direct parameter manipulation. Additionally, [235] proposes StableKE, which shifts the paradigm from knowledge localization to knowledge augmentation, employing semantic paraphrasing and contextual enrichment to prevent the forgetting of related information and maintain stability. Furthermore, the concept of machine unlearning is being re-evaluated as a complementary strategy for removing undesirable influences, as discussed in [236], which emphasizes the need for rigorous assessment frameworks to ensure that unlearning processes do not compromise the integrity of essential knowledge. Ultimately, ensuring the safety and reliability of model editing requires a holistic approach that accounts for the interconnected nature of knowledge, the limits of localizability, and the critical need to preserve the general reasoning faculties that make Foundation Models valuable.

4.8 Unified Frameworks, Benchmarks, and Future Directions in Editing

4.8 Unified Frameworks, Benchmarks, and Future Directions in Editing

Building upon the critical safety challenges and stability pitfalls identified in the preceding section, the trajectory of model editing research is now pivoting from isolated, fact-based updates toward comprehensive, unified frameworks. These emerging paradigms aim to handle complex reasoning, stylistic transformations, and self-correcting behaviors while rigorously preserving the global stability that direct parameter modifications often jeopardize. While early approaches predominantly focused on modifying specific factual triples within large language models (LLMs), the field increasingly recognizes that true adaptability requires mechanisms capable of maintaining logical consistency and semantic coherence across diverse domains. A pivotal step in this evolution is the development of unified evaluation protocols that broaden the definition of editing beyond simple knowledge injection to include the refinement of reasoning chains and the modulation of writing styles. Recent studies have highlighted the limitations of evaluating edits solely on factual recall, noting that parameter-modifying methods often degrade fundamental capabilities like logical reasoning and reading comprehension after only a few sequential updates [44]. This underscores the necessity for benchmarks that assess not just the success of an edit, but its stability and lack of interference with the model’s broader cognitive functions, directly addressing the degradation issues discussed previously.

To address these challenges, emerging frameworks are redefining the scope of editing tasks to ensure holistic integrity. For instance, the concept of “Quantifiable Sequence Editing” (QuaSE) proposes a framework where editing involves altering specific numerical outcomes, such as review ratings or click-through rates, while maintaining the core content of the input [237]. This moves the field toward precision control over abstract properties rather than just token replacement, offering a more nuanced alternative to the blunt instrument of direct weight modification. Similarly, the introduction of event-based knowledge editing challenges the traditional triplet-based approach by incorporating contextual event descriptions, thereby addressing the issue of uncertain editing boundaries and ensuring that related knowledge remains logically consistent [205]. These approaches suggest that future unified frameworks must treat editing as a holistic process that respects the intricate dependencies between facts, reasoning steps, and stylistic nuances, thereby mitigating the logical inconsistencies and knowledge distortion highlighted in Section 4.7.

The creation of robust benchmarks is paramount for tracking progress in these expanded domains and verifying the safety claims of new methods. The “EditEval” benchmark represents a significant leap forward, offering an instruction-based suite designed to evaluate modular editing skills such as improving text cohesion, paraphrasing, and updating outdated information [238]. Unlike previous metrics that focused merely on generation quality, EditEval specifically targets the capability to iteratively refine text, revealing that many current models struggle with neutralizing biases or updating facts without hallucination. Furthermore, the “DepEdit” protocol introduces a rigorous assessment of dependency awareness, testing whether edited facts correctly propagate to their logical implications without disrupting irrelevant knowledge [239]. These benchmarks collectively illustrate a growing consensus: effective editing must be evaluated on its ability to maintain global consistency and logical entailment, not just local accuracy, providing the necessary metrics to validate the reversible and safe frameworks proposed earlier.

Looking toward future directions, the integration of editing with self-correcting systems emerges as

a pivotal research avenue to enhance reliability. The vision of “self-correcting” AI involves models that can autonomously detect errors in their reasoning or outputs and apply precise edits to rectify them without external intervention. Research into self-correction has shown that while LLMs often struggle to identify logical mistakes independently, they possess a strong capacity to correct errors when provided with accurate feedback or specific mistake locations [240]. This suggests a hybrid future where editing mechanisms are coupled with verifiers or “critique” modules. For example, frameworks that fine-tune models to generate natural language critiques can assist in identifying flaws, which are then resolved through targeted editing or refinement loops [241]. Additionally, methods like “Learning to Check” demonstrate that training models on specific checking-correction datasets can enhance their intrinsic self-checking capabilities, reducing the reliance on external ground truth [242]. Such autonomy is crucial for developing systems that can continuously maintain their own safety and correctness over time.

Another promising direction is the development of black-box editing techniques that do not require access to model parameters, making them applicable to proprietary systems and addressing the accessibility constraints of internal parameter manipulation. The “postEdit” framework addresses this by utilizing downstream post-processing to resolve privacy concerns and maintain stylistic consistency, achieving significant improvements in style retention for black-box models [243]. Similarly, “DEEPEDIT” treats knowledge editing as a constrained decoding problem, using depth-first search to adaptively substitute reasoning steps, which allows for flexible application to any black-box LLM [244]. These approaches indicate that future unified frameworks will likely operate at the inference or decoding level, leveraging constrained generation and retrieval-augmented mechanisms to achieve editable behavior without invasive weight modifications, thus offering a potential pathway to avoid the catastrophic forgetting associated with direct editing.

Finally, the pathway toward truly lifelong intelligent systems requires editing mechanisms that are scalable, reversible, and ethically aligned. The “Learning to Edit” (LTE) framework exemplifies this by combining an alignment phase for reliable in-scope edits with a retrieval-based inference phase for mass editing, demonstrating superior robustness in sequential editing scenarios [245]. As the field advances, the synthesis of these diverse strands—reasoning-aware benchmarks, style-controllable frameworks, and self-correcting architectures—will be essential. Future research must focus on creating systems that not only update knowledge but also understand the causal impact of those updates on the model’s entire knowledge graph, ensuring that the pursuit of dynamic adaptation does not come at the cost of reliability or truthfulness. The ultimate goal is a paradigm where models are not static repositories of information but dynamic entities capable of continuous, safe, and precise self-evolution, effectively bridging the gap between the safety concerns outlined in the previous section and the broader vision of adaptable foundation models.

5 Advanced Learning Scenarios: Federated, Online, and Open-World Settings

5.1 Federated Continual Learning under Time-Evolving Data

5.1 Federated Continual Learning under Time-Evolving Data

The convergence of Federated Learning (FL) and Continual Learning (CL) represents a critical frontier in distributed artificial intelligence, particularly when addressing the complexities of time-evolving data. In real-world deployments, the assumption of stationary data distributions across decentralized clients is rarely valid. Instead, systems must contend with concept drift, where the underlying statistical properties of data streams shift dynamically over time, and non-stationary

heterogeneity, where these shifts occur asynchronously across different devices. This subsection examines frameworks designed to navigate these challenges, ensuring that models can adapt to new concepts while preserving historical knowledge and strict privacy constraints. These fundamental issues of data dynamics provide the necessary context for understanding the asynchronous operational protocols discussed in the subsequent section, as the nature of the data stream often dictates the timing and availability of client updates.

The fundamental challenge in this domain is the dual nature of distributional shifts: spatial heterogeneity across clients and temporal evolution within individual client streams. Traditional FL algorithms, such as Federated Averaging (FedAvg), often struggle in these environments because they assume independent and identically distributed (IID) data or static non-IID conditions. When data distributions evolve, standard aggregation mechanisms can lead to significant performance degradation. Research has demonstrated that prior solutions relying on a single global model are ill-suited for staggered drifts that arise at different times and locations across the network. Consequently, effective adaptation requires treating the problem as a time-varying clustering task, where clients experiencing similar concept drifts are grouped dynamically to facilitate more relevant knowledge sharing [246].

To address the specific issue of concept drift in asynchronous federated settings, novel frameworks have been proposed that integrate local drift detection with adaptive regularization. For instance, the FedConD approach introduces an adaptive mechanism that leverages the historical performance of local models to detect drift without requiring labeled data streams. Upon detection, the framework adjusts the regularization parameters of the local objective functions to balance plasticity for new concepts and stability for old ones, thereby minimizing the negative impact of inconsistency between existing and upcoming data [247]. This is crucial because traditional chunk-based or ensemble learning methods, while effective in centralized streams, often fail to scale or maintain privacy in heterogeneous federated ecosystems.

A deeper theoretical understanding reveals that the phenomena of client drift (caused by data heterogeneity) and catastrophic forgetting (caused by sequential task shifts) are intrinsically linked. Recent unified analysis frameworks suggest that the root causes of performance deterioration in both scenarios are connected, with empirical studies showing a high correlation between the magnitude of client drift and the severity of forgetting. Interestingly, this interplay can sometimes yield a “Generalization Bump,” where a moderate combination of both shifts improves model robustness compared to scenarios where only one type of shift occurs. This insight necessitates a holistic approach to algorithm design that simultaneously mitigates client drift and forgetting rather than treating them as isolated problems [248]. Notably, while this theoretical connection is established here in the context of evolving data distributions, its practical mitigation becomes even more complex under the asynchronous update schedules detailed in the following subsection, where timing mismatches can exacerbate these divergences.

Privacy preservation adds another layer of complexity to learning from time-evolving streams. In infinite data streams, the cumulative privacy leakage can become unbounded if standard differential privacy (DP) mechanisms are applied naively at every time step. To counter this, advanced techniques such as the “Differentially Private data streaming over Infinite disclosure” (DPI) have been developed. These methods effectively bound the total privacy leakage for each user even in infinite streams by managing the sensitivity of outputs in evolving distributions, ensuring that privacy guarantees do not degrade as the data stream continues indefinitely [249]. Furthermore, in scenarios involving time-series data where individuals cannot participate at every time step, privacy amplification can be achieved through subsampling in the time domain. By recognizing

that a single participant’s influence is reduced when they do not contribute to every timestamp, mechanisms can be designed to add less noise while maintaining rigorous privacy standards, thus preserving the utility of trend analysis in dynamic environments [250].

The interplay between privacy and utility in heterogeneous, evolving settings has also led to the development of specialized algorithms like DP-SCAFFOLD. This approach incorporates differential privacy constraints into variance-reduced optimization techniques to handle highly heterogeneous user data. It is particularly effective in “honest-but-curious” server settings, where the goal is to protect privacy not just from external adversaries but also from the central aggregator. Theoretical analysis confirms that such methods offer superior convergence rates and privacy-utility trade-offs compared to standard DP-FedAvg, especially as the level of data heterogeneity and the number of local updates increase [251].

Moreover, the nature of concept drift in streaming data implies temporal dependence, challenging the common assumption of independently distributed instances in stream learning literature. Ignoring this temporal structure can lead to destructive adaptation strategies that are computationally expensive and inefficient. Instead, continuous adaptation methods, such as gradient-descent-based approaches that do not rely on explicit drift detection, have shown major advantages in accuracy and efficiency for concept-drifting streams [252; 253]. In predictive scenarios where underlying environmental factors drive the drift, frameworks like DDG-DA go a step further by forecasting future data distributions. By training predictors to estimate the evolution of data distribution and generating synthetic training samples accordingly, these models can proactively adapt to predictable concept drifts, significantly outperforming reactive methods in tasks such as stock price forecasting and energy load prediction [254].

Finally, the fairness implications of time-evolving data cannot be overlooked. Group-specific concept drift, where certain demographic groups experience distributional shifts while others do not, can lead to increasing unfairness even if overall accuracy remains stable. Addressing this requires multi-model approaches and continuous clustering that account for distributed, group-specific drifts to ensure that federated systems remain equitable over time [255]. Collectively, these advancements illustrate a paradigm shift from static federated optimization to dynamic, lifelong intelligence systems capable of evolving alongside the data they process. However, realizing this vision in practice requires overcoming the systemic bottlenecks of synchronization; as clients generate these evolving data streams on independent schedules, the transition to asynchronous protocols becomes not just an option, but a necessity, a topic we explore in depth in the next section.

5.2 Asynchronous Protocols and Client Availability Dynamics

5.2 Asynchronous Protocols and Client Availability Dynamics

Building upon the challenges of time-evolving data distributions discussed in the previous section, the practical realization of Federated Continual Learning (FCL) in real-world deployments necessitates a fundamental rethinking of aggregation protocols and client participation dynamics. While Section 5.1 established that data streams shift asynchronously across devices, the operational reality of massive Internet-of-Things (IoT) networks and mobile edge ecosystems introduces a parallel constraint: clients operate on independent schedules dictated by heterogeneous hardware capabilities, fluctuating battery levels, and intermittent network connectivity. The standard assumption of synchronized rounds, where the server waits for all or a fixed subset of clients before updating the global model, becomes a severe bottleneck in such dynamic settings, exacerbating the latency issues inherent in evolving data streams. Consequently, Asynchronous Federated Continual Learning

(AFCL) has emerged as a critical paradigm, allowing the server to aggregate updates immediately upon receipt. This shift eliminates the waiting time imposed by stragglers but introduces complex interdependencies between gradient staleness, client drift, and privacy preservation that require sophisticated mitigation strategies.

A primary driver for this transition is the “straggler” problem, where devices with limited computational power or poor connectivity delay the entire training process in synchronous frameworks, effectively stalling the model’s ability to adapt to the concept drifts described earlier. To address this, researchers have developed fully asynchronous training paradigms that completely eliminate waiting times. For instance, the FedFa framework proposes a fully asynchronous training paradigm that utilizes buffered results on the server to guarantee model convergence while removing synchronization barriers, demonstrating significant speedups compared to semi-asynchronous strategies [256]. Similarly, production-scale systems like Papaya have demonstrated that asynchronous federated learning can converge significantly faster than synchronous counterparts when training across nearly one hundred million devices, effectively handling the variability in system characteristics and reducing communication overhead [257]. These approaches highlight that asynchronous protocols are not merely an optimization but a necessity for scaling to massive, heterogeneous client populations where data evolution cannot wait for the slowest node.

However, the decoupling of client schedules introduces the challenge of stale updates, a phenomenon particularly detrimental in continual learning where the relevance of historical knowledge decays rapidly. In an asynchronous regime, a client may train on a version of the global model that is several iterations old, leading to updates that are misaligned with the current state of the global optimizer and the most recent data distribution. This phenomenon can degrade convergence stability and accuracy, effectively amplifying the forgetting risks identified in time-evolving scenarios. To mitigate the impact of stale gradients without discarding valuable data from slower clients, novel aggregation policies have been proposed. The FedHist framework, for example, enhances stability by performing a weighted fusion of local gradients with historical global gradients cached on the server, effectively using hindsight to assign multi-dimensional aggregation weights that account for both non-IID data and gradient staleness [258]. Furthermore, theoretical frameworks have been established to analyze asynchronous optimization with delays, leading to methods like FedFix, which extends standard aggregation schemes with stochastic weights to preserve the stability of synchronous aggregation while retaining the efficiency of asynchronous updates [259].

Beyond staleness, the issue of client drift—where local models diverge from the global objective due to data heterogeneity and independent update frequencies—is significantly exacerbated in asynchronous continual learning. When clients learn sequences of tasks with different orderings and timeframes, the risk of catastrophic forgetting and interference increases, creating a compound effect with the temporal drifts discussed in Section 5.1. The FedSpace approach addresses this by introducing a novel setting where continual learning occurs at each client with different task orderings in asynchronous time slots, utilizing prototype-based learning and modified aggregation policies to handle these uncorrelated data streams [260]. Additionally, the interplay between client drift and catastrophic forgetting has been shown to be deeply connected; a unified analysis framework reveals that performance drops from client drift are highly correlated with those from forgetting, suggesting that mitigation strategies must address both simultaneously [248]. Adaptive optimization techniques have also been employed to specifically target client drift, showing that adaptive federated optimization can significantly reduce its adverse impact in continual learning systems where task orders vary independently across clients [261].

Resource constraints and communication dynamics further complicate client availability, creating

a volatile environment where update reception is irregular. In wireless environments, channel conditions are time-varying and often unknown, leading to non-uniform communication failures that can bias the global model if not properly managed. The Federated Postponed Broadcast (FedPBC) algorithm addresses this by postponing global model broadcasts to enable implicit gossiping among clients, ensuring convergence even when link activation probabilities are non-uniform and time-varying [262]. Moreover, to optimize resource usage in such unpredictable environments, strategies like AsyncDrop utilize dropout regularization to create submodels tailored to specific device capabilities, allowing heterogeneous clients to contribute effectively without excessive communication or training overhead [263].

Security and privacy remain paramount in these dynamic asynchronous settings, where the lack of synchronization complicates threat detection and budget accounting. The inherent delay in update propagation can expose the system to poisoning attacks, yet traditional Byzantine-robust methods designed for synchronous rounds often fail here. AFLGuard bridges this gap by proposing a Byzantine-robust asynchronous method that remains effective against various adaptive poisoning attacks despite substantial delays in model updates [264]. Furthermore, ensuring differential privacy in asynchronous flows is challenging due to the difficulty of tracking privacy budgets across irregular updates, a concern that compounds the infinite stream privacy issues noted previously. Recent advances have introduced algorithms that add Gaussian noise in a manner that significantly reduces the aggregated noise required for privacy while maintaining rigorous theoretical guarantees for strongly convex objectives [265].

In conclusion, asynchronous protocols and client availability dynamics represent a cornerstone of scalable federated continual learning, serving as the operational bridge between the theoretical data dynamics of Section 5.1 and the structural scalability solutions of Section 5.3. By decoupling client updates from global synchronization barriers, these methods enable learning in realistic, time-varying environments. However, while asynchronous protocols effectively mitigate latency bottlenecks caused by stragglers in flat, star-topology networks, they do not fully resolve the communication bottlenecks created by the sheer scale of Foundation Model deployments. As the number of devices grows into the billions, even asynchronous updates to a single central parameter server can become inefficient. Consequently, the field is shifting toward Hierarchical Federated Learning and decentralized architectures, which introduce intermediate aggregation layers to better manage the flow of asynchronous updates across complex cyber-physical systems, a structural evolution we explore in the following section.

5.3 Hierarchical Architectures and Multi-Device Ecosystems

5.3 Hierarchical Architectures and Multi-Device Ecosystems

While asynchronous protocols (Section 5.2) effectively mitigate the latency bottlenecks caused by stragglers in flat, star-topology networks, the sheer scale and complexity of Foundation Model deployments in cyber-physical systems (CPS) and the Internet of Things (IoT) often necessitate a more structural evolution. Relying solely on a single central parameter server creates communication bottlenecks that impede scalability and exacerbate the “straggler” effect, even with asynchronous updates. Consequently, the field is shifting toward Hierarchical Federated Learning (HFL) and decentralized architectures, which introduce intermediate aggregation layers—typically edge servers or gateways—between end devices and the cloud. This structural reorganization enables a more efficient balance between global knowledge sharing and local personalization, allowing models to adapt to regional data patterns while maintaining global coherence. The rise of complex cyber-

physical systems, such as smart farming and smart energy management, demands such hierarchical approaches to optimize processing across the cloud-edge continuum, balancing generalized model robustness with locality-dependent precision [266].

In massive mobile edge computing networks, the volume of user equipments (UEs) renders conventional two-layer FL frameworks inefficient due to administrative complexity and communication overhead. To mitigate this, Hierarchical Personalized Federated Learning (HPFL) has been proposed, wherein UEs are clustered and forward updates to edge servers synchronously, while edge servers communicate with the cloud semi-asynchronously. This multi-layer approach allows for the joint optimization of bandwidth allocation and server scheduling, effectively trading off training loss minimization against round latency minimization [267]. By decomposing the aggregation process, these architectures can significantly reduce the communication rounds required between edge nodes and the central server, with some optimized solutions reporting reductions of 75-85% while simultaneously improving classification accuracy by 4-6% in heterogeneous IoT systems [268].

A critical advantage of hierarchical structures is their synergistic ability to handle system heterogeneity, complementing the asynchronous strategies discussed previously. Where asynchronous methods manage timing variances, hierarchical frameworks address capability variances by clustering devices based on computing capacity. This allows for synchronous updates within homogeneous groups while managing asynchronous aggregation across groups. For instance, the FedHiSyn framework clusters devices by capacity and employs a ring topology for intra-cluster communication, effectively mitigating straggler effects and addressing data heterogeneity without sacrificing accuracy [269]. Furthermore, the integration of edge association and resource allocation algorithms, such as those found in the HFEL framework, enables the system to minimize global costs by dynamically assigning devices to the most suitable edge servers, thereby achieving better training performance and energy efficiency compared to conventional cloud-based FL [270].

Beyond mere efficiency, hierarchical architectures facilitate a nuanced trade-off between global generalization and local personalization, which is essential for applications where data distributions are geographically or contextually skewed. In scenarios like smart agriculture, where soil conditions vary significantly by region, a purely global model may fail to capture local anomalies. Hierarchical structures allow for the creation of regional models that consider local dependencies while still benefiting from global anomaly robustness [266]. This is further enhanced by client-edge-cloud collaborative paradigms, such as Agglomerative Federated Learning (FedAgg), which recursively organizes computing nodes to allow models to grow in size and strength as they move from end devices to the cloud. By utilizing bridge sample-based online distillation, FedAgg enables knowledge transfer between tiers with different model capacities, overcoming the limitation where the global model size is constrained by the weakest end device [271].

The theoretical underpinnings of these multi-stage hybrid architectures have also been rigorously explored. Research into Multi-Stage Hybrid Federated Learning (MH-FL) demonstrates that orchestrating device-to-device (D2D) interactions within clusters, combined with multi-stage parameter relaying across a tree-shaped hierarchy, can guarantee convergence to a global optimum under specific spectral radius conditions of the network topology [272]. Moreover, the introduction of intermediate edge servers in client-edge-cloud hierarchies has been shown to simultaneously reduce model training time and the energy consumption of end devices, providing a viable pathway for sustainable AI deployment in resource-constrained environments [273].

However, the transition to hierarchical and decentralized ecosystems introduces new complexities regarding coordination and security, extending the challenges identified in asynchronous settings.

While hierarchical models improve efficiency, they require sophisticated protocols to manage the flow of information between layers without inducing client drift or compromising privacy. Approaches like HierarchyFL utilize self-distillation and ensemble libraries to enable heterogeneous models at different levels to learn from one another intelligently, maximizing knowledge sharing across diverse AIoT devices [274]. Additionally, the move towards decentralized structures within these hierarchies, where devices communicate only with neighbors rather than a central coordinator, offers resilience against single points of failure. Yet, this requires advanced initialization strategies that account for network topology and eigenvector centralities to ensure efficient training in uncoordinated environments [275].

Ultimately, the evolution toward hierarchical and multi-device ecosystems represents a pivotal advancement in enabling Foundation Models to operate effectively in the real world. By leveraging the structural advantages of cloud-edge-device collaboration, these architectures not only resolve the scalability and latency issues inherent in massive IoT deployments but also provide the flexibility needed to address the dual challenges of data and system heterogeneity. As cyber-physical systems become increasingly pervasive, the ability to balance global knowledge aggregation with localized adaptation through hierarchical design will be indispensable for achieving robust, efficient, and privacy-preserving lifelong intelligence, setting the stage for the advanced client selection and clustering techniques discussed in the following section to further refine heterogeneity mitigation.

5.4 Advanced Client Selection and Clustering for Heterogeneity Mitigation

5.4 Advanced Client Selection and Clustering for Heterogeneity Mitigation

Building upon the structural foundations of hierarchical and multi-device ecosystems established in Section 5.3, which organize devices into efficient cloud-edge-tiered topologies to manage latency and scalability, the focus now shifts to the granular optimization of participant dynamics within these structures. While hierarchical architectures provide the necessary framework to balance global aggregation with local personalization, the efficacy of Federated Continual Learning (FCL) ultimately hinges on *which* clients participate and *how* they are grouped during each communication round. Data heterogeneity—specifically non-independent and identically distributed (non-IID) data across clients—remains a primary obstacle to convergence and model performance. Traditional client selection strategies, which often rely on uniform random sampling, fail to account for the significant variance in data distributions, computational capabilities, and communication constraints inherent in large-scale networks. This indiscriminate sampling can lead to the aggregation of unrepresentative model updates, thereby slowing down convergence and exacerbating the stability-plasticity dilemma in lifelong learning scenarios. To address these challenges, recent research has pivoted towards intelligent client selection algorithms and uncertainty-based clustering techniques that dynamically group clients based on their data distribution similarity or system capabilities, serving as a critical bridge between high-level architectural design and the low-level resource constraints discussed in the subsequent section.

A fundamental approach to mitigating heterogeneity involves clustering clients to ensure that the selected subset in each communication round is representative of the global data distribution. Standard random selection often results in high variance of aggregated updates, particularly when the participating subset is small or biased. To counteract this, clustering-based selection schemes have been proposed to accelerate convergence by explicitly reducing this variance. By grouping clients with similar data characteristics, the server can select a subset that better captures the diversity of the entire network, leading to more stable training dynamics [276]. Furthermore, theoretical

analyses have demonstrated that such clustered sampling strategies not only improve client representativeness but also reduce the variance of stochastic aggregation weights, resulting in consistently better training convergence compared to standard sampling approaches in non-IID and unbalanced scenarios [277].

Beyond simple statistical clustering, advanced methods leverage the structural and behavioral heterogeneity of clients to optimize selection, directly complementing the resource-aware scheduling introduced in hierarchical frameworks. In wireless edge networks, where bandwidth and latency are critical constraints, client selection algorithms must balance data representativeness with communication efficiency. Some approaches introduce a two-phased selection and scheduling mechanism that exploits device heterogeneity. Initially, clients are scheduled based on their round latency and bandwidth availability, allowing the server to perform model averaging and cluster clients based on predefined similarity thresholds. Once a cluster approaches a stationary point, a greedy scheduling algorithm prioritizes clients with lower latency within that specific group, significantly accelerating the convergence rate of specialized models [278]. This strategy ensures fairness while capturing diverse data distributions, effectively mitigating the time-consuming nature of capturing global data statistics in resource-constrained environments [279].

The integration of uncertainty estimation into client selection and clustering provides another powerful dimension for handling heterogeneity, particularly vital for continual learning where data distributions shift over time. Quantifying the uncertainty of client updates allows the system to identify and prioritize informative participants, ensuring the model adapts to new concepts without forgetting old ones. For instance, in vertical heterogeneous networks such as those enabled by high-altitude platform stations, strategic client selection based on predicted network traffic behavior can foster a more uniform data distribution across the network, effectively tackling non-IIDness [280]. Moreover, frameworks that estimate the diversity of predictive knowledge contributed by clients have shown promising results. By introducing a weighted averaging and client selection framework that emphasizes updates from high-diversity clients while diminishing the influence of low-diversity ones, systems can achieve significantly faster convergence rates [281].

Graph-based and topology-aware clustering methods further refine client selection by modeling the relationships between clients as a dynamic graph, extending the decentralized principles discussed in Section 5.3 to the client interaction level. In decentralized settings, where a central server may not be available or efficient, algorithms can dynamically adjust the communication graph to connect mutually beneficial tasks and isolate those that might negatively impact each other. By exchanging gradients to calculate task correlations automatically, these methods facilitate personalized decentralized multi-task learning that adapts to the underlying data heterogeneity [282]. Additionally, cellular automata-based approaches have been developed to capture spatio-temporal changes in fast-evolving environments, considering both computational resources and inter-client interactions to enable intelligent selection for online federated processes [283].

Addressing the issue of intermittent client availability is also crucial for robust heterogeneity mitigation, ensuring that the theoretical benefits of clustering are realized in practice despite device volatility. Biased client selection strategies, such as those employing greedy Shapley value approximations, can identify and select the most contributing clients in each round, ensuring fast and stable convergence even under strict timing constraints and high degrees of heterogeneity [284]. Conversely, unbiased algorithms that dynamically learn availability-dependent selection strategies can minimize the impact of client-sampling variance, preventing the global model from being biased towards clients with higher availability [285].

Finally, the concept of clustering extends to the latent space of user preferences and behaviors, particularly in recommendation systems and mobility contexts, enabling a finer granularity of personalization. Hierarchical contextual bandit frameworks that integrate clustering with expert advice can efficiently utilize hierarchical user information to generate representative cluster centroids, substantially reducing regret in personalized learning scenarios [286]. Similarly, in large-scale IoT networks, graph neural networks are employed to embed social relationships and device characteristics, facilitating effective clustering analysis that surpasses traditional topology-based methods [287]. These advanced selection and clustering mechanisms collectively form a robust toolkit for managing the complexities of heterogeneity in federated continual learning. By optimizing *who* learns and *how* they are grouped, these strategies prepare the ecosystem for the next critical challenge: adapting the models themselves to the severe physical and computational constraints of the edge devices, as explored in the following section on resource-constrained adaptation and model heterogeneity.

5.5 Resource-Constrained Adaptation and Model Heterogeneity at the Edge

5.5 Resource-Constrained Adaptation and Model Heterogeneity at the Edge

While advanced client selection and clustering strategies (discussed in Section 5.4) effectively mitigate data heterogeneity and improve convergence dynamics, the practical deployment of Federated Continual Learning (FCL) in real-world Internet of Things (IoT) ecosystems faces a parallel, equally critical bottleneck: the severe resource limitations of edge devices and the inherent heterogeneity of the network infrastructure. Unlike centralized data centers, edge environments comprise a diverse array of hardware ranging from microcontrollers with kilobytes of memory to powerful embedded GPUs, all operating under fluctuating energy budgets and volatile wireless channel conditions. Addressing these physical constraints requires a paradigm shift from static, one-size-fits-all models to adaptive frameworks that dynamically tailor computation, communication, and model architecture to local capabilities. This transition is essential not only for efficiency but also for enabling the continuous adaptation of foundation models in lifelong learning scenarios where device resources may vary over time.

A primary strategy for enabling such adaptation is variable pruning, which allows models to compress themselves according to the specific capabilities of the hosting device. Recent advancements have moved beyond simple post-training compression to integrated pipelines that generate diverse ensembles of pruned models, subsequently selecting optimal representatives for distributed deployment. For instance, multi-phase pruning pipelines have been proposed to create lightweight learners that achieve significant model size reductions—up to 90%—while maintaining or even improving predictive accuracy through ensemble techniques, thereby making deep learning feasible on resource-constrained IIoT devices [288]. Furthermore, complexity-driven pruning algorithms now exploit the inherent layer-level complexities of Convolutional Neural Networks (CNNs) to select filters for removal based on their contribution to overall network complexity, offering versatile compression modes that are aware of parameters, floating-point operations, or memory constraints [289].

Complementing pruning, split learning adaptations have emerged as a critical mechanism for off-loading intensive computational tasks from weak edge nodes to more capable edge servers while preserving data privacy. In this paradigm, the neural network is partitioned at a specific “cut layer,” with the initial layers executing on the device and the remainder on the server. However, static partitioning fails to account for the dynamic nature of edge resources, particularly in continual learning settings where workload characteristics evolve. Consequently, adaptive split learning

frameworks have been developed to dynamically control the model splitting point and client-side aggregation strategies, effectively balancing communication-computing latency against training convergence rates [290]. These systems often employ decentralized schemes where devices proactively determine their optimal cut layers and resource requirements based on local multidimensional configurations, minimizing training latency without requiring global knowledge of other devices' states [291]. To further enhance efficiency in wireless environments, cluster-based parallel split learning schemes have been introduced, which partition devices into clusters to enable parallel training of device-side models before sequential aggregation across clusters, significantly reducing the straggler effect caused by device heterogeneity [292]. Additionally, innovative U-shaped split federated learning architectures leverage edge servers to assist IoT devices by splitting the network into three parts, thereby reducing local computation overhead while integrating noise mechanisms to resist reconstruction attacks and ensure robust privacy preservation [293].

The challenge of model heterogeneity extends beyond computation to the architectural diversity required to serve clients with vastly different capabilities. Traditional federated learning assumes a homogeneous global model, which excludes weaker devices unable to support the full parameter count. To overcome this, heterogeneous federated learning frameworks utilize local parameter sharing, dividing local models into shareable encoders and task-specific heads. This approach, combined with channel-wise pruning, allows multiple tasks to coexist on a single device while accommodating data and system heterogeneity, significantly reducing computational resource consumption compared to state-of-the-art homogeneous frameworks [294]. Similarly, personalized federated learning strategies employ partial model pruning and personalization, splitting the model into a global pruned part for shared representation learning and a personalized part for fine-tuning. By jointly optimizing the pruning ratio and bandwidth allocation using Karush-Kuhn-Tucker conditions, these frameworks can reduce computation and communication latency by approximately 50% while improving accuracy on non-IID data distributions [295]. Another promising direction involves stochastic parameter updates, where devices maintain the full model architecture but randomly freeze a subset of neurons during training. This method, known as FedSPU, avoids the bias introduced by federated dropout and enhances robustness against divergent parameters from other clients, outperforming dropout-based strategies in both accuracy and training time [296].

Furthermore, the integration of early exits and hierarchical inference provides a flexible mechanism for adapting to varying difficulty levels of input data and device capabilities, a feature particularly valuable when handling the non-stationary data streams characteristic of continual learning. Hierarchical inference systems allow edge devices to perform initial inference using small, local models and only offload samples to larger server-side models when the local confidence is low or the task is out-of-distribution. Online meta-learning frameworks have been developed to predict the correctness of local inferences in real-time, enabling dynamic offloading decisions that maximize the benefits of both local low-latency processing and server-side high-accuracy modeling [297]. This concept is further extended in unified approaches that merge early exits with split computing, determining the optimal splitting layer and offloading decision online to adapt to environmental distortions without ground truth labels, achieving substantial cost reductions with marginal performance degradation [298]. Finally, to address the energy constraints inherent in mobile and IoT environments, joint computation-and-communication resource management frameworks have been designed for heterogeneous CPU-GPU architectures. These systems optimize bandwidth allocation, workload partitioning, and speed scaling to minimize total energy consumption, leveraging equilibrium states in energy rates to devise efficient scheduling algorithms that scavenging spectrum holes and balance loads across heterogeneous processing units [299]. Collectively, these innovations in variable pruning, adaptive split learning, and heterogeneous model aggregation form the backbone

of sustainable, lifelong intelligence at the network edge. By resolving the physical and economic constraints of distributed IoT ecosystems, these methods lay the necessary groundwork for the next evolution of FCL: operating in task-agnostic and open-world scenarios where task boundaries are undefined and novel classes emerge dynamically, as explored in the following section.

5.6 Task-Agnostic and Open-World Federated Scenarios

5.6 Task-Agnostic and Open-World Federated Scenarios

Building upon the resource-constrained and heterogeneous frameworks established in Section 5.5, which enable foundation models to operate across diverse edge devices, the next evolutionary step for Federated Continual Learning (FCL) addresses the semantic uncertainty of real-world data streams. While Section 5.5 resolved the physical and economic constraints of distributed IoT ecosystems, practical deployment rarely adheres to the structured constraints of predefined task boundaries or synchronized task arrivals assumed in early FCL research. Instead, systems must operate in task-agnostic and open-world settings where task identities are unknown, boundaries are blurry or non-existent, and novel classes emerge dynamically across heterogeneous clients. In these scenarios, the central challenge shifts from merely preventing catastrophic forgetting within known parameters to actively inferring task structures on-the-fly and detecting novel categories without violating privacy constraints or requiring global coordination. This transition sets the stage for the critical security and robustness challenges discussed in Section 5.7, as the very mechanisms enabling such plasticity can become vulnerabilities if not properly secured.

In task-agnostic federated settings, the assumption that all clients learn the same sequence of tasks in a synchronized manner is fundamentally flawed. Clients operate independently, receiving data streams that may vary in order, timing, and content distribution. This asynchronicity necessitates robust mechanisms for on-the-fly task inference that do not rely on explicit task identifiers. Traditional approaches that depend on task-ID supervision fail in these environments, prompting the development of methods that leverage statistical deviations in loss landscapes or representation shifts to detect task transitions. For instance, the concept of asynchronous federated continual learning has been formalized to address scenarios where clients encounter tasks in uncorrelated orders and timeframes. Frameworks like FedSpace tackle this by employing prototype-based learning and modified aggregation policies that accommodate independent client schedules, effectively managing the divergence caused by asynchronous task arrivals [260]. Similarly, the identification of “client drift” as a critical weakness in vanilla federated averaging highlights the necessity for adaptive optimization strategies that can handle independent task sequences without degrading global model performance [261].

Beyond asynchronous task arrival, the open-world nature of federated streams requires systems to distinguish between known and novel classes dynamically. In an open-world federated scenario, a client may encounter data belonging to entirely new categories that the global model has never seen, necessitating a mechanism for novel class detection that functions without centralized labeled data. This problem is exacerbated by the non-IID nature of federated data, which can mask the distinction between domain shift and the emergence of new classes. To address this, researchers have explored the integration of out-of-distribution (OOD) detection capabilities directly into the continual learning pipeline. Methods that combine OOD detection with CL techniques, such as ODEx, demonstrate that reliable maintenance of performance on earlier tasks is possible even when facing smooth population shifts and undefined task boundaries [300]. Furthermore, the theoretical underpinning of class-incremental learning suggests that strong OOD detection is not merely an

auxiliary tool but a necessary and sufficient condition for achieving robust performance in open-ended settings, as it enables the model to decompose the problem into within-task prediction and task-ID prediction [301].

The complexity of open-world federated learning is further amplified when considering the stochastic and blurry nature of real-world task boundaries. Unlike idealized benchmarks with disjoint tasks, real-world data streams often exhibit gradual transitions where samples from old and new tasks intermingle. Addressing this requires sophisticated memory management and sampling strategies that can adapt to stochastic boundary conditions. Approaches like MVP (Mask and Visual Prompt tuning) have been proposed to handle stochastic blurry task boundaries by utilizing instance-wise logit masking and contrastive losses to consolidate previous knowledge while adapting to new classes [302]. In a federated context, this implies that local clients must possess the autonomy to manage these blurry transitions locally before aggregating updates, ensuring that the global model does not collapse due to conflicting gradients arising from ill-defined task frontiers.

To facilitate knowledge transfer in these undefined settings without sharing raw data, recent frameworks have leveraged selective inter-client transfer and generative synthesis. The FedWeIT framework, for example, decomposes network weights into global and sparse task-specific parameters, allowing clients to selectively receive knowledge from others based on task similarity, thereby minimizing interference between incompatible tasks in an open-world setting [303]. Moreover, in scenarios where storing past data is impossible due to privacy or memory constraints—constraints often dictated by the resource-limited environments described in Section 5.5—generative models trained via data-free methods on the server can synthesize samples from past distributions. This allows clients to rehearse old concepts and maintain decision boundaries for known classes while detecting new ones, effectively mitigating forgetting without accessing private historical data [166; 167]. The integration of such generative replay with task-agnostic inference mechanisms creates a robust defense against the dual threats of forgetting and catastrophic interference from novel classes.

Furthermore, the detection of novel classes in federated settings often benefits from hierarchical or prototypical representations that capture the semantic structure of the data across clients. Prototypical networks have shown promise in task-free data-incremental learning by adapting continuously to new sensor data streams, suggesting that similar centroid-based approaches could be adapted for federated open-world detection to identify clusters of novel instances [304]. Additionally, the use of prompt-based learning in federated environments, such as Fed-CPrompt, introduces contrastive losses that help handle heterogeneous data distributions and asynchronous task arrivals, providing a mechanism to align representations across clients even when task boundaries are unclear [305].

Ultimately, achieving true task-agnosticism and open-world capability in federated learning requires a holistic rethinking of the aggregation and inference protocols. It demands systems that can implicitly infer task structures through representation analysis, detect novelty through confidence calibration and OOD metrics, and transfer knowledge selectively to avoid negative transfer. However, as the field moves towards these more realistic and dynamic deployments, the increased flexibility and autonomy granted to clients introduce new attack surfaces. The synthesis of asynchronous protocols, generative replay, and robust OOD detection creates a complex landscape where the mechanisms enabling plasticity can be exploited by malicious actors. This necessity to secure lifelong intelligent systems against adversarial manipulation in decentralized, open-ended data streams directly motivates the discussion in the following section on robustness, security, and fault tolerance.

5.7 Robustness, Security, and Fault Tolerance in Distributed Streams

5.7 Robustness, Security, and Fault Tolerance in Distributed Streams

While the task-agnostic and open-world scenarios discussed in Section 5.6 highlight the algorithmic complexities of inferring dynamic task structures and detecting novel classes, they also expose a critical vulnerability: the very mechanisms enabling plasticity and adaptability can be exploited by malicious actors. The deployment of Continual Learning (CL) within Federated Learning (FL) environments thus introduces a confluence of security vulnerabilities, fault tolerance requirements, and robustness challenges distinct from static or centralized settings. In distributed streams where data distributions evolve and clients join or leave dynamically, the system must simultaneously mitigate Byzantine attacks, handle unreliable communication links, and prevent the propagation of adversarial perturbations across sequential tasks. The fundamental tension lies in maintaining the model plasticity required for new tasks while ensuring stability against malicious actors who may exploit continuous update mechanisms to inject persistent backdoors or degrade global performance.

A primary concern in these distributed continual streams is susceptibility to Byzantine attacks, where malicious clients upload arbitrary or carefully crafted gradients to disrupt global model convergence. Traditional robust aggregation rules often struggle in non-Identically and Independently Distributed (non-IID) settings, which are inherent to federated scenarios. Research has revealed that the curse of dimensionality and gradient heterogeneity significantly degrade the performance of standard robust aggregators when data is non-IID, necessitating advanced adaptation strategies [306]. To address these limitations, novel approaches such as gradient splitting have been proposed to adapt existing robust aggregation rules to heterogeneous data distributions, ensuring convergence even when local datasets vary significantly across clients [306]. Furthermore, the dynamic nature of continual learning requires defenses capable of handling time-varying behaviors; for instance, spatial-temporal analysis of local model updates has been employed to detect and exclude incorrect updates by leveraging their geometric properties and adapting learning rates based on momentum-based speculation [307].

Beyond static Byzantine threats, distributed streams face the challenge of dynamic adversaries who may switch identities or alter their attack strategies over time. Most existing techniques assume a static set of Byzantine machines, failing to capture real-world scenarios involving transient malfunctions or targeted temporal attacks. To counter this, dynamic Byzantine-robust learning methods have been developed that can withstand rounds of Byzantine identity alterations while matching the asymptotic convergence rates of static settings [308]. These methods often incorporate multi-level Monte Carlo gradient estimation and fail-safe filters to limit bias from dynamic strategies, eliminating the need to know the exact percentage of malicious workers in advance [308]. Additionally, the threat of colluders—groups of malicious clients conspiring to upload vicious updates—requires spectral analysis frameworks that formulate detection as a community detection problem in weighted graphs, effectively discerning attackers from benign participants through strong consistency and temporal coherence analysis [309].

Fault tolerance in distributed streams extends beyond adversarial resistance to include resilience against network failures and unreliable clients, a prerequisite for the asynchronous operations described in open-world settings. In large-scale hierarchical systems, packet-dropping link failures and communication bottlenecks can severely impede information propagation. Hierarchical robust push-sum algorithms have been proposed to achieve average consensus despite frequent link failures, interleaving consensus updates with dual averaging to ensure provable convergence guarantees [310]. Moreover, asynchronous protocols are critical for handling client availability dynamics, where de-

vices operate on independent schedules. Byzantine-robust asynchronous FL methods, such as those utilizing trust bootstrapping or adaptive client selection, have been designed to update models immediately upon receiving updates without waiting for stragglers, thereby maintaining robustness against poisoning attacks in asynchronous settings [264].

The propagation of adversarial robustness across continual tasks presents another layer of complexity, particularly when models must adapt to new classes without forgetting old ones. In federated adversarial learning, the multi-local step updating before aggregation exposes unique vulnerabilities, making convergence analysis significantly harder due to min-max optimization complexities and inter-client heterogeneity [311]. Direct application of adversarial training in non-IID federated settings can severely compromise clean accuracy, necessitating techniques like logit calibration and feature contrast to align local representations with unbiased global features [312]. Furthermore, the risk of internal evasion attacks, where adversarial clients craft inputs at test time to deceive other clients, highlights the need for combining adversarial training with personalized federated learning frameworks to enhance relative internal attack robustness [313].

Privacy and robustness must also be achieved simultaneously, a goal often neglected by defenses focusing on only one aspect. Protocols that integrate differential privacy with Byzantine robustness via client momentum have been developed to reduce the variance of honest clients while exposing small malicious perturbations that accumulate over time [314]. Such approaches ensure that privacy noise does not mask Byzantine signals, achieving a superior privacy-utility trade-off. Additionally, ensemble-based approaches have shown promise in providing provable security guarantees, where multiple global models are learned from random subsets of clients, and predictions are made via majority vote to ensure that the output is not affected by a bounded number of malicious clients [315].

Finally, the efficacy of these robustness mechanisms cannot be assumed; it requires rigorous empirical validation, bridging the gap between theoretical guarantees and practical deployment. Recent studies utilizing scalable benchmark suites have re-evaluated representative attacks and defenses, revealing that many state-of-the-art methods fail under specific attack settings or data skewness, thus highlighting the urgent need for more rigorous and diverse evaluation protocols in federated continual learning [316]. These findings underscore that robustness cannot be an afterthought but must be a core metric in any comprehensive assessment. This necessity for standardized, realistic evaluation directly motivates the discussion in the following section, which details the specific benchmarking frameworks and real-world deployment insights required to validate FCL systems under extreme heterogeneity and adversarial conditions.

5.8 Benchmarking Frameworks and Real-World Deployment Insights

5.8 Benchmarking Frameworks and Real-World Deployment Insights

The rigorous evaluation protocols demanded at the conclusion of Section 5.8 underscore a critical reality: the efficacy of robustness mechanisms in Federated Continual Learning (FCL) cannot be assumed but must be validated through comprehensive benchmarking. Consequently, the transition from theoretical formulations to robust real-world deployment is heavily contingent upon the availability of simulation platforms that accurately reflect the complexities of dynamic, distributed environments. While early research often relied on simplified assumptions regarding data distribution and client availability, recent advancements have produced a suite of sophisticated tools designed to bridge the gap between algorithmic innovation and practical applicability. These frameworks are now essential for evaluating FCL systems under conditions of extreme heterogeneity,

resource constraints, and non-stationary data streams, revealing significant discrepancies between idealized models and operational realities.

A primary challenge in benchmarking FCL lies in simulating the inherent heterogeneity of real-world edge devices and data distributions. Traditional benchmarks often partition standard datasets uniformly, failing to capture the “isolated data island” characteristics where data quantity and quality vary drastically across clients. To address this, platforms like [317] have been developed, offering domain-specific benchmarks in medical, financial, and AIoT sectors that explicitly configure for data distribution heterogeneity and privacy-preserving scenarios. Similarly, [318] provides a modular framework with open-source federated datasets derived from real-world sources such as wearable devices and mobile phones, enabling rigorous evaluation of algorithms against the obstacles of practical federated environments. These tools are essential for moving beyond independent and identically distributed (IID) assumptions, as demonstrated by studies showing that models trained on realistic, non-IID splits suffer from significant performance degradation compared to their theoretical counterparts [319].

Furthermore, the dynamic nature of continual learning introduces the dimension of time-evolving heterogeneity, where client data distributions shift, and clients join or leave the network asynchronously—dynamics that standard federated learning simulators often struggle to model efficiently. The [320] work highlights the necessity of frameworks capable of capturing concept drift and intermittent client participation, proposing Continual Federated Learning (CFL) as a flexible approach to approximate local objective functions in such volatile settings. Complementing this, [120] offers a scalable prototyping system that supports heterogeneous device runtimes and asynchronous aggregation, directly addressing the mismatch between homogeneous simulation environments and the diversity of real-world hardware. The insights from [120] underscore that ignoring device heterogeneity and scale can lead to unclear effectiveness of FCL algorithms, necessitating systems that can quantify the impact of varying device capabilities on training dynamics.

Resource constraints and system-level inefficiencies represent another critical gap between theory and deployment, particularly given the strict energy budgets and computational limits of real-world edge devices that static benchmarks frequently overlook. [321] addresses this by allocating dedicated GPU resource budgets to simulate workload heterogeneity, revealing that existing frameworks without such constraints may misleadingly evaluate new algorithms. Additionally, [322] introduces a tier-based selection policy to mitigate straggler problems caused by resource and data heterogeneity, demonstrating that adaptive client selection is vital for maintaining training efficiency in realistic scenarios. The importance of efficient simulation is further emphasized by [323], which enhances parallelism and GPU utilization for large-scale experiments, allowing researchers to profile client resources and estimate execution times more accurately. Without such resource-aware simulations, theoretical convergence rates may not translate to viable deployment strategies, as seen in the analysis of [324], which shows that participants with insufficient computation budgets require specialized strategies like model estimation to avoid performance decline.

Crucially, security and robustness considerations discussed in the previous section must be integrated directly into these benchmarking suites. The distributed nature of FCL makes it susceptible to unique attack vectors, including Byzantine failures and poisoning attacks that accumulate over time. [316] provides a scalable and extensible platform specifically designed to validate novel defense strategies against a wide range of adversarial settings. By re-evaluating representative attacks and defenses across approximately 1,500 trials, [316] highlights previously overlooked limitations in current FCL methods, reinforcing the conclusion that robustness cannot be an afterthought but

must be a core metric in any comprehensive benchmark.

Despite these advancements, significant gaps remain between existing benchmarks and the full spectrum of real-world deployment challenges. Many current frameworks still rely on synthetic data partitions or fail to fully capture the complexity of open-world scenarios where task boundaries are undefined. For instance, while [325] proposes a framework to disentangle performance gaps from unseen clients versus unseen distributions, it notes that dataset synthesis strategies are crucial for realistic simulations, yet many benchmarks still fall short in this regard. Moreover, the interplay between client drift and catastrophic forgetting in dynamic streams requires more nuanced evaluation protocols, as suggested by [248], which reveals that moderate levels of both shifts can sometimes lead to a “Generalization Bump,” a phenomenon that standard benchmarks might miss.

In conclusion, while platforms like [326] and [327] have made strides in providing realistic datasets and system-level evaluations, the field still lacks a unified standard that seamlessly integrates continual learning dynamics, extreme system heterogeneity, and adversarial robustness. Future benchmarking efforts must prioritize the simulation of asynchronous, task-agnostic environments and provide deeper insights into the energy and communication trade-offs inherent in lifelong federated systems. Only by closing these gaps can the community ensure that FCL algorithms are not only theoretically sound but also ready for deployment in the unpredictable and resource-constrained landscapes of the real world.

6 Domain-Specific Applications and Multimodal Extensions

6.1 Edge-Deployed LLMs for Healthcare and Privacy-Critical Domains

6.1 Edge-Deployed LLMs for Healthcare and Privacy-Critical Domains

While the preceding sections have explored the theoretical foundations of adapting foundation models, the practical realization of these systems in high-stakes environments necessitates a shift from centralized cloud processing to decentralized edge intelligence. Nowhere is this transition more critical than in healthcare and other privacy-sensitive domains. Although Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language processing and diagnostic support, their traditional cloud-based deployment faces significant barriers in medical settings, including prohibitive latency, excessive bandwidth consumption, and, most critically, severe risks to data privacy. Migrating these models to edge devices—ranging from consumer GPUs to embedded microcontrollers—is therefore essential to realizing the promise of real-time, secure, and accessible medical assistance. This transition demands a rigorous re-evaluation of model architecture, optimization techniques, and system design to accommodate the stringent resource constraints of edge environments while maintaining clinical accuracy.

The primary driver for edge deployment in healthcare is the imperative to preserve patient data locality. In traditional cloud-centric architectures, sensitive medical records must be transmitted to remote servers, exposing them to potential interception and violating strict regulatory frameworks such as HIPAA or GDPR. To address this, recent initiatives have focused on developing on-premise healthcare chatbots that operate entirely within local infrastructure. For instance, the MedAide system exemplifies this approach by leveraging tiny-LLMs integrated with LangChain to provide preliminary medical diagnostics on edge devices without relying on server infrastructure [328]. By employing model optimizations such as Low-Rank Adaptation (LoRA) and training on diverse medical datasets via Reinforcement Learning from Human Feedback (RLHF), such systems achieve competitive accuracy on benchmarks like the USMLE while ensuring that no patient data leaves

the local device [328]. This localization not only alleviates privacy concerns but also ensures functionality in remote areas with limited connectivity, thereby democratizing access to high-quality medical support.

However, deploying LLMs on resource-constrained edge hardware introduces substantial technical challenges regarding memory footprint and inference latency. The astronomical size of modern foundation models often exceeds the memory capacity of portable devices, necessitating advanced compression and quantization strategies. Activation-aware Weight Quantization (AWQ) has emerged as a hardware-friendly approach that protects salient weights based on activation observations, significantly reducing quantization error without requiring backpropagation [329]. When combined with efficient inference frameworks like TinyChat, AWQ enables the deployment of massive models, such as the 70B Llama-2, on mobile GPUs, achieving speedups of over 3x compared to standard implementations [329]. Furthermore, research into the compressibility of quantized LLMs highlights that data compression techniques can be jointly optimized with quantization to mitigate I/O latency bottlenecks, which become critical when models must be partially loaded from storage on memory-constrained devices [330].

Beyond pure on-device inference, hybrid architectures that distribute computation between the edge and the cloud offer a viable pathway for balancing performance and privacy. The PrivateLoRA framework proposes a novel service paradigm where privacy-sensitive computations are executed on edge devices, while shared computations are offloaded to the cloud, transmitting only activations rather than raw data [331]. This approach exploits the low-rank nature of residual activations to reduce communication overhead by over 95%, enabling throughput rates that significantly exceed device-only solutions while maintaining strict data locality [331]. Similarly, methods that enhance small medical learners through privacy-preserving contextual prompting allow smaller, edge-deployable models to leverage the knowledge of larger LLMs without exposing patient data; this is achieved by extracting keywords to generate medical context locally, bridging the performance gap between small language models (SLMs) and their larger counterparts [332].

The integration of LLMs into the emerging 6G mobile edge computing (MEC) ecosystem further expands the horizon for real-time medical applications. Position papers on pushing LLMs to the 6G edge emphasize that next-generation networks can resolve critical issues of response time and bandwidth costs through techniques like split learning and parameter-efficient fine-tuning [333]. In such ecosystems, edge devices can perform initial triage and diagnostics, offloading only complex reasoning tasks when necessary, thus optimizing the trade-off between computational load and diagnostic precision. Additionally, the development of decentralized systems utilizing consumer-level GPUs, as envisioned in frameworks like FusionAI, suggests that distributed edge resources can be aggregated to train and deploy LLMs with privacy protection, overcoming the limitations of individual hardware constraints [334].

Despite these advancements, operational challenges remain prevalent in the widespread adoption of edge AI in healthcare. The “TinyMLOps” landscape reveals that beyond computational limits, practitioners must navigate complex issues related to monitoring, managing distributed applications, and protecting model intellectual property in physically accessible devices [335]. Security mechanisms such as TransLinkGuard are being developed to safeguard transformer models against stealing attacks in edge deployments by utilizing secure Enclaves for authorization, ensuring that proprietary medical models remain protected even in white-box environments [336]. Moreover, the feasibility of transmitting large medical imaging data over 5G networks supports the backbone required for these edge-cloud collaborations, with studies demonstrating effective throughput rates sufficient for real-time diagnostic support [337].

In conclusion, the deployment of compact, optimized LLMs on edge devices for healthcare represents not merely a technical optimization but a fundamental requirement for ethical and practical AI adoption in medicine. By leveraging techniques such as quantization, parameter-efficient fine-tuning, hybrid split-computing, and decentralized training, researchers are successfully addressing the triad of latency, bandwidth, and privacy. As evidenced by systems like MedAide and frameworks like PrivateLoRA, the future of medical AI lies in intelligent, localized systems that empower clinicians and patients alike without compromising the sanctity of sensitive health data. Looking forward, as these edge-deployed models become more capable and interconnected, the focus naturally shifts toward enhancing their ability to process and align multimodal inputs—such as medical imaging and clinical text—within unified vision-language systems, a challenge explored in the subsequent section.

6.2 Multimodal Alignment and Semantic Grounding in Vision-Language Systems

6.2 Multimodal Alignment and Semantic Grounding in Vision-Language Systems

Building upon the localized, text-centric capabilities of edge-deployed LLMs discussed in the previous section, the next frontier in foundation models involves expanding perceptual horizons beyond language to encompass the visual world. The rapid proliferation of Vision-Language Models (VLMs) represents a fundamental paradigm shift, enabling systems to perceive, reason, and interact through a synthesis of visual and textual modalities. However, the efficacy of these models hinges critically on bridging the inherent “modality gap”—the semantic and structural dissonance between continuous visual signals and discrete linguistic tokens. Effective multimodal alignment is not merely a preprocessing step but a core architectural challenge that dictates a model’s capacity for fine-grained semantic grounding: the precise mapping of language concepts to specific visual entities, regions, or temporal intervals. As the field moves from simple global feature matching toward sophisticated unified frameworks, dynamic interaction mechanisms, and hierarchical alignment strategies, it lays the essential groundwork for the complex physical interactions required in embodied AI, which will be explored in the subsequent section.

A primary strategy for overcoming the modality gap involves the development of unified frameworks that treat vision and language as interchangeable or jointly optimizable sequences. Traditional approaches often relied on separate encoders fused via late interaction, which frequently resulted in information loss and misalignment. In contrast, modern architectures advocate for deep, early fusion and task-agnostic designs. For instance, the OFA framework unifies diverse cross-modal and unimodal tasks within a simple sequence-to-sequence learning paradigm, demonstrating that a single model can effectively handle image generation, visual grounding, and captioning without task-specific layers [338]. Similarly, BEiT-3 advances this convergence by introducing Multiway Transformers that enable both deep fusion and modality-specific encoding, achieving state-of-the-art performance by performing masked modeling on images, texts, and image-text pairs in a unified manner [339]. These unified approaches suggest that breaking down modality-specific scaffolds allows for more fluid information flow, a prerequisite for the complex reasoning tasks where visual and textual contexts are deeply intertwined.

Beyond architectural unification, dynamic visual interaction mechanisms have emerged as a critical component for capturing high-level dependencies between modalities. Static fusion methods often fail to account for the varying importance of visual regions relative to specific query words. To address this, researchers have proposed dynamic fusion schemes that facilitate intra- and inter-modality attention flows. A notable example is the Dynamic Fusion method, which alternatively

passes dynamic information between visual and language modalities, allowing the system to robustly capture high-level interactions and significantly improve performance in Visual Question Answering (VQA) [340]. Furthermore, the Multimodal Unified Attention Network (MUAN) extends this concept by simultaneously capturing intra-modal and inter-modal interactions, stacking unified attention blocks to create deep networks that excel in both VQA and visual grounding tasks [341]. These dynamic mechanisms ensure that the model does not merely correlate global features but actively attends to relevant spatial and semantic details conditioned on the linguistic input, a capability essential for the real-time perceptual feedback loops required in robotics.

Semantic grounding, particularly at a fine-grained level, remains a persistent challenge, especially when models must distinguish between similar objects or understand complex relational structures. Standard pre-training objectives often lack the granularity required for tasks like referring expression comprehension or spatial localization. To mitigate this, recent works introduce hierarchical and multi-level alignment strategies. The HiVG framework proposes a hierarchical multimodal fine-grained modulation approach, utilizing a multi-layer adaptive cross-modal bridge and low-rank adaptation to address inconsistencies between visual features and grounding requirements from shallow to deep layers [342]. Complementing this, MVPTR emphasizes multi-level semantic alignment by introducing concepts as high-level semantics and enforcing interactions via coarse-grained and fine-grained tasks, thereby facilitating synergistic representation learning [343]. Such hierarchical approaches are vital for reducing perceptual errors and ensuring that models can ground language to specific image regions with the high precision necessary for safety-critical applications.

The necessity for rigorous evaluation of semantic grounding capabilities has also gained traction, as large models often exhibit hallucinations or misgrounding in real-world scenarios. Studies have designed pipelines to assess grounding abilities across fine-grained attributes such as color, material, and number, revealing prevalent misgrounding issues in popular Large Vision-Language Models (LVLMs) [344]. In response, data-centric enhancement methods involving multimodal instruction tuning on fine-grained conversations have been proposed to rectify these deficiencies. Moreover, the integration of scene graph knowledge has proven effective in enhancing structured representations, allowing models to distinguish between reversed relational statements (e.g., “astronaut rides horse” vs. “horse rides astronaut”) by leveraging structured negative examples and knowledge-enhanced encoders [345]. This structured understanding is a precursor to the scene affordance reasoning and constraint enforcement discussed in the context of autonomous agents.

Finally, the extension of grounding to video and temporal domains introduces additional complexity regarding spatiotemporal alignment, bridging the gap between static perception and dynamic action. Frameworks like STVGFormer address spatio-temporal video grounding by employing static and dynamic branches that model intra-frame appearance and inter-frame motion respectively, facilitating precise localization of events in time [346]. Additionally, innovative strategies such as transforming video frames into image grids allow single VLMs to perform zero-shot video question answering by retaining temporal information within a spatial structure, bypassing the need for specialized video training [347]. Collectively, these advancements in unified frameworks, dynamic interactions, and hierarchical grounding mechanisms represent a significant leap toward creating vision-language systems that possess true semantic understanding. By mastering the nuances of diverse visual domains with human-like precision, these models provide the perceptual backbone required for the next evolution of AI: embodied agents capable of translating semantic understanding into physical action.

6.3 Embodied AI and Autonomous Agents in Robotics

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Building upon the sophisticated multimodal alignment and semantic grounding mechanisms established in vision-language systems, the integration of Large Language Models (LLMs) and Vision-Language Models (VLMs) into robotic platforms marks a transformative leap in Embodied AI. This evolution transitions agents from rigid, pre-programmed executors into flexible, reasoning-capable autonomous entities capable of comprehending high-level natural language instructions, decomposing complex goals into actionable sub-tasks, and adapting to dynamic physical environments. However, bridging the gap between the abstract semantic understanding discussed in previous sections and the concrete demands of physical interaction introduces unique challenges: resolving misalignment between symbolic plans and continuous control, enforcing strict safety constraints in unstructured settings, and harmonizing high-level cognition with low-level motor execution.

A primary obstacle in deploying LLMs for robotics is plan-execution misalignment, where syntactically correct high-level plans fail due to environmental perturbations or imperfect low-level controllers. To address this disconnect, recent frameworks have introduced closed-loop mechanisms that leverage the multimodal perception capabilities detailed in Section 6.2 for real-time correction. For instance, the DoReMi framework utilizes LLMs to generate constraints alongside plans, employing VLMs to continuously detect violations and trigger immediate recovery strategies, thereby significantly improving task success rates in complex manipulation scenarios [348]. Similarly, the RePlan framework harnesses the physical grounding capabilities of VLMs to facilitate online re-planning, allowing robots to adapt to unforeseen obstacles during long-horizon tasks where static plans would otherwise fail [349]. These approaches underscore the necessity of integrating perceptual feedback directly into the planning loop, moving beyond open-loop generation to systems capable of introspection and self-correction, as seen in frameworks like HiCRISP which actively monitors and adapts execution at both high and low levels [350].

Beyond adaptability, the deployment of autonomous agents in human-centric environments demands rigorous safety constraint enforcement, a critical consideration that parallels the reliability concerns in medical AI. While LLMs excel at generating “dos” for task completion, they often lack an inherent understanding of “don’ts,” posing risks in collaborative settings. To mitigate this, researchers have proposed modular safety architectures, such as queryable safety constraint modules based on Linear Temporal Logic (LTL), which translate natural language restrictions into verifiable temporal constraints to prune unsafe actions before execution [351]. Furthermore, uncertainty quantification has emerged as a critical component for safe operation; introspective planning methods guide LLMs to recognize ambiguity in instructions or environmental states, prompting the agent to seek clarification rather than hallucinating confident but potentially dangerous plans [352]. This aligns with findings that leveraging scene affordance scores can significantly reduce harmful hallucinations and align the model’s confidence with the actual probability of task success [353].

Bridging the gap between high-level semantic reasoning and low-level cognition remains a formidable barrier, requiring a translation of discrete linguistic tokens into continuous control signals. Traditional approaches often rely on predefined motion primitives, limiting the robot’s dexterity and adaptability. Innovative solutions now utilize LLMs to generate dense trajectories or even low-level control code directly. The VoxPoser framework, for example, composes 3D value maps by interacting with VLMs, enabling zero-shot synthesis of closed-loop robot trajectories that are robust to dynamic perturbations without relying on fixed primitives [354]. In a similar vein, the “Code as Policies” paradigm repurposes code-writing LLMs to synthesize reactive policy code

that incorporates spatial-geometric reasoning and accesses real-time perception outputs, effectively closing the loop between language and control [355]. Extending this to dynamic locomotion, recent work has demonstrated that LLMs can function as low-level feedback controllers by autoregressively generating control commands based on few-shot prompts from the physical environment, challenging the assumption that foundation models are limited to high-level planning [356].

The complexity of embodied tasks often necessitates hierarchical architectures that distribute cognitive load across multiple specialized agents, mirroring the multi-agent tendencies seen in advanced clinical decision support. Systems like REAL employ LLMs within the autonomy stack of aerial robots to interpret diagnostic logs and adapt control inputs, enhancing resilience against unmodeled dynamics and parameter errors [357]. For multi-robot collaboration, dialectic approaches enable agents to discuss strategies and collectively reason about task allocation, utilizing LLMs for both communication and path planning to resolve deadlocks and coordinate complex maneuvers [358] [359]. Moreover, the concept of “Action Contextualization” allows robots to modify motion strategies based on specific task contexts, utilizing online feedback to rectify errors autonomously and eliminate planning redundancies [360].

Despite these advancements, challenges persist in ensuring that generated plans are environmentally aware and executable. Methods like NLMap provide open-vocabulary, queryable scene representations that ground LLM planners in the immediate surroundings, allowing them to verify object availability before committing to a plan [361]. Additionally, frameworks such as AutoTAMP improve feasibility by translating natural language into intermediate task representations consumed by traditional Task-and-Motion Planning (TAMP) algorithms, using autoregressive re-prompting to correct syntactic and semantic errors [362]. As the field progresses, the focus is shifting towards lifelong learning capabilities, where agents can acquire new skills through verbal corrections and integrate them into their repertoire for future tasks, marking a significant step toward truly autonomous and adaptable embodied intelligence [363] [364]. The convergence of these methodologies suggests a future where embodied agents possess not only the reasoning prowess of foundation models but also the precise, safe, and adaptive control necessary for real-world deployment, setting the stage for their application in highly specialized domains such as healthcare.

6.4 Specialized Medical Multimodal Models and Clinical Decision Support

6.4 Specialized Medical Multimodal Models and Clinical Decision Support

Bridging the gap between the physical adaptability of embodied agents and the high-stakes logical rigor required in technical domains, the deployment of specialized medical multimodal models represents a critical frontier in clinical decision support. While the robotic systems discussed in Section 6.3 focus on translating semantic plans into physical actions, and the software engineering applications in Section 6.5 leverage code generation for digital automation, medical AI demands a unique synthesis of visual diagnostics and linguistic reasoning where errors carry immediate human consequences. Unlike general-purpose foundation models that often struggle with the nuanced semantics and safety constraints of healthcare data, domain-specific adaptations utilize instruction tuning and targeted pre-training to achieve expert-level performance in fields such as radiology and ophthalmology. These models are designed not merely to classify images but to synthesize complex multimodal inputs into coherent diagnostic narratives, functioning as autonomous agents that augment rather than replace human expertise.

In the realm of radiology, the adaptation of Large Language Models (LLMs) has evolved from generic fine-tuning to sophisticated instruction-tuning frameworks that align model outputs with

strict clinical reporting standards. Notable examples include Radiology-Llama2, which utilizes instruction tuning on extensive datasets of radiology reports to generate clinically useful impressions from findings, achieving state-of-the-art performance on benchmarks like MIMIC-CXR and OpenI [365]. Similarly, Radiology-GPT demonstrates superior versatility in radiological diagnosis and communication by leveraging an instruction-tuning approach on domain-specific knowledge, effectively localizing generative capabilities for hospital-specific needs while adhering to privacy standards [366]. The effectiveness of these approaches is further underscored by systems like shs-nlp, which employ large-scale medical text data for domain-adaptive pre-training of instruction-tuned LLMs, outperforming conventional pretrain-and-finetune methods in zero-shot settings for impression generation [367]. Furthermore, innovative frameworks such as ImpressionGPT utilize in-context learning with dynamic prompts constructed from individualized data, enabling iterative optimization of report summarization without requiring additional training data or fine-tuning [368].

Beyond text-only adaptations, the integration of vision encoders has given rise to powerful multimodal foundation models capable of interpreting chest X-rays and other imaging modalities with unprecedented accuracy. CheXagent exemplifies this trend by combining a clinical LLM with a dedicated vision encoder and a bridging network, trained on the large-scale CheXinstruct dataset to perform automated CXR interpretation across eight clinically relevant tasks [369]. This model not only surpasses general-domain foundation models but also incorporates fairness evaluations to address potential disparities across demographic groups. Complementary approaches, such as Med-KLIP, enhance visual-language pre-training by extracting medical triplets and utilizing knowledge graph embeddings to implicitly build relationships between medical entities, resulting in robust performance in disease classification and grounding tasks [370]. Additionally, methods that refine radiology reports to emphasize key semantic knowledge have shown significant improvements in fine-grained analysis tasks like region-of-interest segmentation [371]. The potential of these models extends to error-checking, where multimodal LLMs have demonstrated the ability to identify discrepancies in radiology reports, sometimes even outperforming human clinicians in detecting overlooked errors [372].

Parallel advancements are evident in ophthalmology, a domain characterized by its reliance on multimodal data including fundus images and patient reports. The Ophtha-LLaMA2 model illustrates the efficacy of fine-tuning LLaMA2 on ophthalmic report data, achieving high accuracy in disease diagnosis despite smaller fine-tuning datasets compared to general models [373]. Taking this further, OphGLM introduces visual capabilities into large language models through instruction-following and dialogue fine-tuning, creating a comprehensive assistant capable of disease assessment, diagnosis, and lesion segmentation based on fundus images [374]. These developments highlight a broader trend toward unified transformer models that can handle multi-task procedures, such as the customized instruction tuning approach used for chest radiograph interpretation, which unifies image- and pixel-level tasks to enhance clinical interpretability [375].

A critical challenge in deploying these specialized models globally is ensuring cross-lingual correctness and adaptability, a necessity that mirrors the localization requirements seen in diverse robotic environments. Medical AI systems must function effectively across diverse linguistic landscapes to serve non-English speaking populations equitably. Research indicates that while English-centric models exhibit strong cross-lingual transfer capabilities in instruction tuning, they often suffer from reduced factuality and fluency errors in other languages, necessitating careful hyperparameter tuning and diverse training data [376]. The development of multilingual models for radiology report summarization has shown promising results, with systems capable of generating high-quality impressions in English, Portuguese, and German, often matching or exceeding human-written sum-

maries in blind tests [377]. However, evaluations such as XlingEval reveal pronounced disparities in model performance across languages like Spanish, Chinese, and Hindi, emphasizing the urgent need for benchmarks like XlingHealth to drive improvements in cross-lingual healthcare capabilities [378]. Strategies such as translation-based methods combined with domain-specific encoders or cross-lingual encoders plus adapters have proven effective for clinical phenotyping in low-resource languages, compensating for data sparseness [379]. Moreover, efforts to localize in-domain adaptation for less-resourced languages, such as Italian, demonstrate that while data quantity is a significant constraint, high-quality native corpora can substantially improve model performance [380].

The trajectory of specialized medical multimodal models points toward increasingly autonomous and generalizable systems, setting a precedent for the agent-based architectures explored in subsequent technical domains. Autonomous agents capable of coordinating specialized tools for text, image, and genomic data analysis are emerging as viable solutions for complex clinical scenarios, such as oncology decision-making [381]. These systems leverage the generalist reasoning of LLMs to deploy domain-specific models dynamically, simplifying regulatory compliance and enhancing patient-tailored care. As the field progresses from these highly specialized, safety-critical medical applications to the broader technical automation of software engineering and cybersecurity, the focus shifts toward how these reasoning capabilities can be codified into precise programming logic and defensive mechanisms, a transition detailed in the following section.

6.5 Code Generation, Software Engineering, and Cybersecurity Applications

6.5 Code Generation, Software Engineering, and Cybersecurity Applications

Transitioning from the domain-specific adaptations in healthcare to the broader landscape of technical automation, the integration of Large Language Models (LLMs) into software development life-cycles has fundamentally altered code generation, software engineering practices, and cybersecurity operations. Unlike the specialized medical models discussed previously, which focus on diagnostic narratives and clinical reasoning, these technical applications leverage the inherent logical structure of programming languages to transform foundation models from static knowledge repositories into dynamic agents. This triad of applications—spanning collaborative pair-programming, precise robot policy generation, and complex security assessment—highlights both the transformative potential and the critical safety challenges inherent in deploying LLMs within high-stakes technical domains, setting the stage for the resource-constrained architectures explored in subsequent sections.

In the realm of software engineering, LLMs have emerged as powerful pair-programming assistants capable of accelerating development cycles and lowering barriers to entry for non-experts. These models excel at generating foundational code structures, assisting with syntax correction, and debugging errors, thereby enhancing the productivity of development teams. Empirical studies involving academic software engineering projects indicate that students utilizing LLMs can effectively integrate AI-generated code into their workflows, particularly during the early stages of project development where establishing structure is paramount [382]. Furthermore, the educational paradigm is shifting towards “learning-by-teaching,” where students act as teaching assistants to debug code generated by LLM-powered agents. Systems like HypoCompass demonstrate that this collaborative approach not only generates high-quality training materials but also significantly improves student performance in debugging tasks, suggesting a future where human-AI collaboration is central to computer science education [383]. However, the reliance on these models introduces new dynam-

ics in conversational programming; research shows a gap between the user’s mental model of the system’s understanding and the AI’s actual state. Approaches such as Programming with Representations (PwR) aim to bridge this gap by having the system explicitly convey its understanding back to the user, thereby instilling a greater sense of agency and verification capability in developers [384].

Beyond traditional software development, a frontier application of LLMs lies in the generation of precise robot policy code, effectively translating natural language instructions into executable control logic for embodied agents. This capability mirrors the autonomous agent trends seen in medical oncology but applies them to physical interaction, allowing non-experts to direct complex robot behaviors without requiring deep knowledge of low-level control theory. The “Code as Policies” paradigm repurposes code-writing LLMs to synthesize Python programs that process perception outputs and parameterize control primitives, enabling robots to exhibit spatial-geometric reasoning and generalize to new instructions [355]. Advances in this field have extended to high-precision, contact-rich manipulation tasks. By reparameterizing action spaces to include compliance with interaction forces, models like GenCHiP have demonstrated the ability to generate policies that succeed in tight tolerance scenarios, significantly outperforming non-compliant action spaces [385]. Furthermore, frameworks such as CodeBotler leverage domain-specific languages to generate programs for service mobile robots, ensuring accurate sequencing of actions crucial for task success [386]. The evolution of these systems includes lifelong learning capabilities, where LLM-based planners can query and teach robots new skills in a data-efficient manner, reusing acquired skills for future open-world tasks [364]. Despite these advances, challenges remain regarding the “forgetfulness” of LLMs during execution, where models may lose track of key prompt information, necessitating sophisticated prompt engineering tactics to ensure robustness [387]. These robotic applications inherently demand efficient computation and adaptability, foreshadowing the necessity for elastic modality management and resource-constrained adaptation discussed in the following section.

Parallel to these constructive applications, the role of LLMs in cybersecurity presents a dual-use dilemma, serving as both potent defensive tools and potential vectors for malicious activity. On the defensive front, LLMs are increasingly employed for automated vulnerability detection and patching. Approaches like Feedback-Driven Solution Synthesis (FDSS) utilize static analysis tools to provide feedback to LLMs, which then iteratively refine code to resolve security vulnerabilities, outperforming baseline methods [388]. Multi-role consensus frameworks, which simulate discussions between developers and testers, have shown significant improvements in precision and recall for vulnerability detection compared to single-perspective approaches [389]. Comprehensive benchmarks such as CyberSecEval and CyberSecEval 2 have been developed to rigorously evaluate the propensity of LLMs to generate insecure code and their compliance when asked to assist in cyberattacks, revealing that even advanced models often struggle with prompt injection and may suggest insecure code [390; 391].

However, the offensive potential of LLMs cannot be overlooked. Research indicates that LLMs can be leveraged as attacking agents in reinforcement learning environments, performing comparably to human testers in network security scenarios without additional training [392]. They possess the capability to reason about threats, automate cyber campaigns, and potentially democratize access to dual-use biotechnology and chemical weapons by providing detailed protocols to non-experts [393; 394]. Evaluations using benchmarks like WMDP (Weapons of Mass Destruction Proxy) highlight the urgent need for unlearning techniques to remove hazardous knowledge from models while preserving their general utility [395]. Moreover, studies on smart contract generation reveal that while LLMs can produce valid code, they frequently introduce critical security bugs,

underscoring the necessity for rigorous validation before deployment in financial systems [396]. The reliability of LLMs in identifying security vulnerabilities remains a concern, with findings showing significant non-robustness where minor changes in variable names can lead to incorrect reasoning, suggesting that LLMs are not yet ready to serve as fully autonomous security assistants [397].

In conclusion, the application of LLMs in code generation, robotics, and cybersecurity represents a paradigm shift characterized by immense opportunity and substantial risk. While these models empower developers and engineers to solve complex problems and automate intricate tasks, their deployment requires careful consideration of safety constraints, verification mechanisms, and ethical guidelines to mitigate the potential for misuse and error. As these systems grow more autonomous and integrated into physical and digital infrastructures, the computational demands of running such sophisticated agents become a primary bottleneck. This reality necessitates a shift from monolithic deployments to more flexible, efficient architectures capable of operating under strict hardware limitations, a challenge directly addressed by the resource-constrained adaptation and elastic modality management strategies detailed in the next section.

6.6 Resource-Constrained Adaptation and Elastic Modality Management

6.6 Resource-Constrained Adaptation and Elastic Modality Management

The computational demands of the autonomous agents and sophisticated security systems described in the previous section create a significant bottleneck for deployment in real-world scenarios, particularly within embedded systems, robotics, and edge computing environments. While Section 6.5 highlighted the transformative potential of LLMs in code generation and robotic control, it also underscored the “forgetfulness” and robustness issues that arise when these heavy models operate under strict hardware limitations. Traditional approaches to multimodal integration often rely on static architectures where all modalities are processed simultaneously through fixed, heavy-weight encoders and fusion layers. This rigidity is incompatible with dynamic environments where sensor availability fluctuates, and resource budgets vary temporally. Consequently, recent research has pivoted toward architectural innovations that enable elastic modality management and parameter-efficient adaptation, allowing models to dynamically adjust their computational graph based on runtime context and available hardware resources. These advancements bridge the gap between the high-performance agents of the previous section and the rigorous reliability requirements explored in the subsequent discussion on distribution shifts.

A primary strategy for addressing these constraints is the development of elastic modality adaptation mechanisms that avoid the prohibitive costs of retraining or maintaining fixed connections between encoders and the core language model. The concept of “Modality Plug-and-Play” introduces a paradigm where unimodal encoders are connected to a flexible set of the large language model’s blocks, with these latent connections remaining fully trainable at runtime. This approach, exemplified by the mPnP-LLM framework, allows the system to adaptively involve only the useful modalities depending on current environmental contexts and task requirements. By decoupling the encoders from fixed input layers, such systems achieve significant efficiency gains; for instance, experiments have demonstrated up to a 3.7x reduction in FLOPs and a 30% reduction in GPU memory usage while retaining accuracy comparable to static schemes [398]. This elasticity is crucial for embodied AI applications where an agent might need to switch between visual, auditory, or tactile inputs without incurring the latency of loading full model weights for every modality, directly addressing the efficiency needs of the robotic policies discussed earlier.

Complementing elastic connectivity, parameter-efficient adaptation techniques have emerged as a

vital tool for robustness against missing or corrupted modalities, a common occurrence in resource-constrained settings. Instead of training dedicated networks for every possible combination of available sensors, which scales exponentially, researchers have proposed modulation of intermediate features to compensate for missing data. This method requires tuning an extremely small number of parameters—often fewer than 0.7% of the total model size—to bridge the performance gap caused by absent modalities. Such lightweight adaptation procedures have proven effective across diverse tasks including multimodal semantic segmentation and sentiment analysis, outperforming independent networks trained for specific modality combinations while maintaining a minimal inference footprint [399]. Furthermore, the integration of adapter experts and attention gating mechanisms, as seen in the PILL architecture, enables the decoupling of complex modalities. By utilizing a Mixture-of-Modality-Adapter-Expert to handle different inputs independently and a Modality-Attention-Gating mechanism to control token contributions, these models achieve efficient fine-tuning and robust fusion without compromising the expressive capability of the original foundation model [400].

Managing vast action spaces in embodied agents under resource constraints also necessitates novel architectural designs that break down complex decision-making into manageable units, echoing the need for structured reasoning in robot programming. The “Unit-Grained Hybrid Training Framework” addresses the challenge of long action sequences in vision-language decision making by dividing task episodes into fine-grained units, each containing distinct navigation and interaction phases. By employing a Unit-Transformer with an intrinsic recurrent state that maintains unit-scale cross-modal memory, agents can actively explore environments with reduced exposure bias and lower computational overhead during training and inference. This modular approach allows for more flexible resource allocation, as the model only needs to process the relevant context for the current unit rather than the entire episode history [401]. Similarly, hierarchical learning architectures that factor problems into slow reinforcement learning policies for strategy and fast adaptive control policies for motion optimization have shown promise. These bio-inspired frameworks, such as those utilizing adaptive force-impedance action spaces, drastically improve sample efficiency and reduce energy consumption, making them suitable for real-time robotic manipulation on limited hardware [402].

The challenge of heterogeneity in multimodal data further complicates resource management, particularly when dealing with high-modality scenarios involving diverse sensor types. To address this, information-theoretic metrics have been proposed to quantify modality and interaction heterogeneity, enabling the automatic prioritization of fusion for modalities that provide unique information. The High-Modality Multimodal Transformer (HighMMT) leverages these metrics to scale efficiently up to ten modalities, demonstrating that performance can continue to improve with each added modality while maintaining a favorable trade-off between efficiency and accuracy. This scaling behavior is critical for systems that must operate in environments with varying sensor suites, ensuring that computational resources are not wasted on redundant information processing [403]. Additionally, representing collections of features as unordered sets rather than fixed-size concatenated vectors, as proposed in Deep Multi-Modal Sets, offers a scalable framework that is invariant to the cardinality and permutation of modalities. This approach mitigates the memory explosion associated with adding new modalities and allows for interpretable feature performance observation during inference [404].

Finally, the extension of these principles to federated and distributed settings highlights the necessity of selective communication and client-side adaptation, broadening the scope from single-device efficiency to network-wide optimization. In multimodal federated learning, devices often

face diverse modality availability and strict communication limits. Frameworks like FedMFS and mmFedMC introduce modality selection criteria that weigh the impact of a modality against its model size and update frequency, significantly reducing communication overhead by over 4x to 20x while maintaining competitive accuracy. These strategies ensure that edge devices can contribute to global model improvement without transmitting redundant or overly large modality-specific updates [405], [406]. Collectively, these architectural innovations signify a shift from static, monolithic multimodal models to dynamic, elastic systems capable of thriving in resource-constrained, open-world environments. However, while these methods optimize for efficiency and adaptability, they introduce new challenges regarding reliability when deployed in non-stationary real-world domains. The transition from controlled, resource-optimized settings to operational environments with shifting data distributions necessitates a rigorous re-evaluation of model robustness, a topic central to the domain-specific benchmarks and generalization challenges discussed in the following section.

6.7 Domain-Specific Evaluation Benchmarks and Generalization Challenges

6.7 Domain-Specific Evaluation Benchmarks and Generalization Challenges

While the architectural innovations discussed in Section 6.6 enable foundation models to operate efficiently within resource-constrained and dynamic environments, deploying these elastic systems in high-stakes real-world scenarios introduces a distinct set of challenges centered on distribution shifts. The transition from controlled laboratory settings to operational domains such as healthcare, finance, and scientific discovery is frequently hindered by the divergence between the statistical properties of training data and those encountered during inference. In these specialized fields, where data distributions are inherently non-stationary and the cost of failure is severe, robustness becomes as critical as efficiency. Consequently, the research focus has expanded from merely optimizing computational footprints to synthesizing domain-specific evaluation benchmarks that rigorously test out-of-distribution (OOD) generalization. These emerging benchmarks move beyond static accuracy metrics to evaluate model resilience under covariate shift, concept shift, and temporal drift, providing a realistic assessment of reliability in the very environments where resource-constrained adaptation is most needed.

In the realm of tabular data, which underpins critical applications in finance, public policy, and industrial operations, the lack of robust evaluation frameworks has historically obscured the true extent of model brittleness. Recent efforts have introduced comprehensive benchmarks designed to expose these vulnerabilities, revealing that efficiency gains must not come at the expense of generalization. For instance, the [407] benchmark provides a large-scale testbed derived from industrial datasets involving weather prediction and power consumption. Extensive evaluations on this platform reveal that many state-of-the-art OOD methods struggle to maintain performance on unseen data, often failing to surpass the robustness of simple Empirical Risk Minimization (ERM) baselines. Similarly, [408] addresses the scarcity of high-quality benchmarks by offering 15 binary classification tasks across diverse sectors including healthcare and education. Their large-scale study highlights a concerning linear trend between in-distribution and OOD accuracy, demonstrating that while domain robustness methods can reduce shift gaps, they often do so at the expense of in-distribution performance. Furthermore, [409] emphasizes the necessity of distinguishing between different types of shifts, noting that conditional label shifts ($Y|X$) are particularly prevalent in tabular settings and require distinct algorithmic interventions compared to standard covariate shifts.

The healthcare domain presents unique challenges due to the heterogeneity of patient populations, medical devices, and clinical protocols across different institutions, complicating the deployment

of the adaptive models described previously. The [410] study benchmarks domain generalization methods on multi-site clinical time series and imaging data, finding that current methods do not consistently outperform ERM on real-world medical imaging, though limited gains are observable in specific induced-shift scenarios for time series. This skepticism is echoed in [411], which argues that access to OOD data does not automatically translate to improved performance, pointing to inherent limitations in data harmonization and sampling biases. To quantify these behaviors specifically for Electronic Health Records (EHR), [412] was introduced to measure erratic model behaviors, such as high-confidence incorrect predictions, under OOD settings, highlighting an urgent need for improved robustness in clinical decision support systems. Additionally, [413] extends these concerns to computational chemistry, constructing benchmarks that reveal significant performance degradation in retrosynthesis models when faced with OOD molecules, thereby motivating the development of model-agnostic techniques for chemical domain generalization.

Temporal dynamics introduce another layer of complexity, as real-world data streams often evolve gradually over time, rendering static adaptation mechanisms insufficient. The [414] benchmark curates datasets reflecting temporal shifts in applications like patient prognosis and news classification. Evaluations on Wild-Time demonstrate an average performance drop of 20% from in-distribution to OOD data, with existing continual learning and domain generalization methods failing to close this gap effectively. This underscores the difficulty of extrapolating from past trends to future distributions, a challenge further explored in [415], which proposes frameworks to handle continuously evolving domain drift in non-stationary environments like autonomous driving and sensor monitoring. These findings suggest that the “elasticity” required for future foundation models must extend beyond hardware resource management to include temporal adaptability.

Beyond specific verticals, there is a growing recognition of the need for unified evaluation protocols that can dissect the nuances of distribution shifts across modalities, bridging the gap between the multimodal architectures of Section 6.6 and their real-world reliability. The [416] initiative provides a curated collection of datasets reflecting natural shifts in wildlife monitoring, satellite imaging, and tumor identification, standardizing evaluations to facilitate method development. Complementing this, [417] specifically targets graph-structured data, distinguishing between covariate and concept shifts to provide a systematic evaluation of graph neural networks. In the NLP domain, [418] introduces the BOSS benchmark suite, revealing that while large language models excel in in-context learning for OOD instances, fine-tuned smaller models often struggle, indicating that model scale and adaptation paradigms interact complexly with distribution shift types. Moreover, [419] cautions that model robustness can be inconsistent across varying degrees of shift, suggesting that evaluations fixed to a single shift magnitude may lead to biased conclusions about model reliability.

The synthesis of these benchmarks reveals a critical gap: while numerous algorithms claim state-of-the-art performance on static splits, they frequently fail to generalize when confronted with the multifaceted, evolving shifts characteristic of real-world deployment. Persistent challenges include the trade-off between in-distribution accuracy and OOD robustness, the difficulty of detecting silent failures in high-stakes domains, and the lack of standardized metrics for quantifying “competence regions” where models can be trusted. As highlighted in [420], developing reliable incompetence scores to reject OOD inputs remains an open problem, as current proxies often fail to offer a favorable accuracy-rejection trade-off across all domains. Consequently, future research must prioritize the development of dynamic, context-aware evaluation frameworks that not only measure average performance but also characterize the boundaries of model reliability under diverse and severe distribution shifts, ensuring that the efficient models of tomorrow are also trustworthy.

7 Evaluation Protocols, Security Robustness, and Ethical Considerations

7.1 Benchmarking Forgetting, Plasticity, and Editing Specificity

7.1 Benchmarking Forgetting, Plasticity, and Editing Specificity

The rigorous evaluation of Continual Learning (CL) and Model Editing (ME) systems necessitates a paradigm shift from static accuracy metrics to dynamic assessments that capture the intricate interplay between stability, plasticity, and precision. This transition is critical because traditional benchmarks often fail to account for the nuanced degradation mechanisms inherent in lifelong learning scenarios, leading to an overestimation of model robustness. A primary challenge lies in quantifying catastrophic forgetting (CF) in a manner that reflects real-world deployment constraints rather than idealized laboratory conditions. Standard evaluation protocols frequently permit access to previous data during model selection or retraining control, which masks the true severity of forgetting. Recent investigations have demonstrated that when model selection is restricted to the initial dataset alone, many methods previously deemed effective at mitigating forgetting actually fail to prevent performance collapse across class-incremental visual problems [421]. This highlights a critical gap in current benchmarking practices: the need for application-oriented evaluation procedures that strictly enforce data isolation to reveal the genuine limitations of incremental learning algorithms.

Furthermore, the relationship between forgetting and forward transfer—the ability of a model to leverage past knowledge to learn new tasks more efficiently—has been subject to significant misinterpretation due to flawed metric definitions. Historically, a reduction in catastrophic forgetting has been treated as the primary objective, yet studies suggest this metric does not always correlate with improved forward transfer. It is argued that forward transfer should be measured by the ease of learning new tasks given the representations produced by continual learning, rather than being conflated with retention metrics. Under this refined definition, empirical evidence indicates that representations exhibiting less forgetting are indeed more diverse and discriminative, thereby facilitating better learning efficiency on subsequent tasks [422]. Consequently, comprehensive benchmarks must decouple these two phenomena to accurately assess whether a CL strategy merely preserves old knowledge or actively constructs a reusable feature space.

In the domain of model editing, particularly for Large Language Models (LLMs), the specificity of edits has emerged as a paramount concern, serving as the conceptual bridge between benign knowledge updates and potential security vulnerabilities. Specificity refers to the ability of an editing method to update a specific fact without altering unrelated knowledge or introducing unwanted side effects. Existing benchmarks, such as CounterFact, have been instrumental in this area but are increasingly recognized as insufficient for detecting subtle yet dangerous degradations in model behavior. Recent work has extended the CounterFact benchmark to include a dynamic component, dubbed CounterFact+, and introduced principled KL divergence-based metrics to better measure specificity. Evaluations using this improved framework reveal that many state-of-the-art editing techniques suffer from low specificity, introducing large unwanted side effects that static benchmarks fail to detect [423]. This finding underscores the necessity for next-generation benchmarks that can identify and prevent the propagation of errors resulting from localized parameter updates, a capability that becomes indispensable when distinguishing between legitimate corrections and the stealthy backdoor injections discussed in the following section.

The complexity of benchmarking is further exacerbated by the influence of task similarity and overparameterization on forgetting dynamics. Analytical models have shown that the joint effect of

these factors creates non-monotonic forgetting patterns; for instance, in highly overparameterized models, intermediate task similarity can cause the most severe forgetting, whereas near the interpolation threshold, forgetting decreases monotonically with similarity [424]. Therefore, effective benchmarks must systematically vary task similarity and model scale to map these complex landscapes, rather than relying on single-point estimates of performance. Additionally, in multi-modal large language models (MLLMs), catastrophic forgetting manifests uniquely during fine-tuning, where improvements on unseen tasks often lead to significant drops in original capabilities. Novel post-training adjustment methods like Model Tailor have been developed to address this, utilizing sparse masks to identify “model patches” and compensation mechanisms to decorate them, achieving high effectiveness on both original and new tasks [425]. Benchmarks for MLLMs must therefore evaluate the preservation of pre-trained multi-modal alignment alongside task-specific adaptation.

Beyond simple accuracy drops, the reliability of evaluation metrics themselves is under scrutiny, particularly in the presence of class imbalance and domain shifts. Common metrics such as the Area Under the Receiver Operating Characteristic (AUROC) can provide overly optimistic views of classifier performance when data is imbalanced, whereas Precision-Recall Curves (PRC) offer more representative insights into minority class performance [426]. However, even the superiority of PRC over AUROC in imbalanced settings has been challenged by recent mathematical analyses suggesting that AUPRC may inadvertently favor subpopulations with frequent positive labels, potentially heightening algorithmic disparities [427]. This debate necessitates the adoption of calibrated metrics that are invariant to class priors, ensuring that performance variations across subpopulations are interpreted correctly [428]. In medical and safety-critical domains, where context shifts between test benchmarks and production data are common, robust evaluation requires dynamic benchmarking that captures the underlying data generation process rather than static snapshots [429].

Finally, the detection of unwanted side effects in continual and editing scenarios requires metrics that go beyond global accuracy to examine representational geometry and readout alignment. Research indicates that a significant portion of accuracy loss in continual learning is attributable to readout misalignment rather than the irreversible loss of information, suggesting that metrics focusing on representational stability could provide earlier warnings of degradation [14]. Similarly, in semantic segmentation under adverse weather conditions, forgetting is primarily driven by changes in low-level features, implying that benchmarks should include representational analysis of early network layers to diagnose the root causes of performance drops [430]. By integrating these diverse perspectives—ranging from dynamic specificity checks and calibrated imbalance metrics to representational analyses—the community can develop a more holistic and rigorous framework for evaluating the lifelong intelligence of foundation models. Such a framework is essential not only for optimizing model performance but also for establishing the forensic baseline required to detect when these same mechanisms are exploited for malicious purposes.

7.2 The Convergence of Model Editing and Backdoor Injection

7.2 The Convergence of Model Editing and Backdoor Injection

Building upon the rigorous evaluation frameworks discussed in the previous subsection, which highlighted the critical need for specificity metrics to distinguish benign updates from unwanted side effects, this section examines how the very mechanisms of model editing are being co-opted to execute sophisticated backdoor injections. The rapid advancement of model editing techniques has fundamentally altered the landscape of Large Language Model (LLM) maintenance, offering efficient mechanisms to correct hallucinations, update outdated facts, and align model behaviors without

the prohibitive costs of full retraining. However, this technological leap introduces a profound security paradox: the attributes that make model editing attractive for benign maintenance—namely, its efficiency, locality, and ability to induce precise behavioral changes with minimal data—also render it an ideal vector for malicious backdoor injection. This subsection analyzes the dual-use nature of model editing, examining how frameworks designed for lightweight parameter updates are increasingly repurposed to inject durable backdoors, thereby blurring the critical line between legitimate correction and covert manipulation.

Traditionally, backdoor attacks against deep learning systems relied on data poisoning during the training phase or required substantial fine-tuning datasets to alter model weights significantly. These methods often incurred high computational costs and risked degrading the model’s performance on benign inputs, making them detectable and impractical for large-scale foundation models. The emergence of model editing has dismantled these barriers. By formulating backdoor injection as a lightweight knowledge editing problem, adversaries can now directly alter LLM parameters to incorporate malicious triggers with unprecedented efficiency. A seminal demonstration of this convergence is the **BadEdit** framework, which exploits editing algorithms to inject backdoors using as few as 15 samples. Unlike traditional poisoning methods that demand massive datasets, **BadEdit** adjusts only a subset of parameters, resulting in a dramatic reduction in time consumption while ensuring the model’s overarching performance remains uncompromised. Crucially, backdoors injected via such editing techniques exhibit remarkable robustness, persisting even after subsequent fine-tuning or instruction-tuning procedures that would typically cleanse standard poisoned models [431].

The convergence of these fields is further complicated by the challenge of maintaining consistency on clean data while embedding malicious logic, a direct counterpoint to the specificity metrics emphasized in Section 7.1. In standard backdoor scenarios, large parameter shifts often trigger anomaly detection systems. However, recent research indicates that effective backdoor injection via editing involves Adversarial Weight Perturbations (AWPs)—variations in parameters that are sufficiently small to evade detection yet sufficient to alter behavior under specific triggers. Theoretical analyses suggest that maintaining global and instance-wise consistency on clean data is achievable through specialized loss functions, such as anchoring losses, which freeze model behaviors on benign inputs while allowing the injection of trigger-specific logic. This capability allows attackers to achieve high attack success rates with negligible impact on the model’s utility, effectively hiding the backdoor within the normal operational parameters of the model [432].

Furthermore, the duality extends beyond simple trigger insertion to the manipulation of model ownership and verification mechanisms. Model watermarking, often implemented by embedding specific backdoor behaviors to verify copyright, faces new vulnerabilities when editing techniques are applied. Research has shown that the parameter space surrounding a watermarked model contains numerous variants where the watermark behavior is removed yet the model utility is preserved. Attackers can leverage editing-inspired optimization strategies, such as mini-max formulations, to navigate this space and strip watermarks, or conversely, to inject robust, hard-to-remove backdoors that mimic ownership signatures [433]. This interplay highlights how the tools for protecting intellectual property and those for compromising model integrity are becoming increasingly indistinguishable.

The threat landscape is further expanded by the application of editing concepts to federated learning and multimodal systems, serving as a conceptual bridge to the distributed vulnerabilities discussed in the following section. In federated settings, the “Neurotoxin” attack demonstrates how adversaries can target parameters that change less in magnitude during training, ensuring that back-

doors remain durable even as the global model evolves through continual learning rounds. This approach effectively doubles the durability of state-of-the-art backdoors by exploiting the same stability mechanisms that continual learning algorithms rely on to prevent catastrophic forgetting [434]. Similarly, in multimodal contrastive learning, attacks like **BadCLIP** utilize dual-embedding guidance to align visual triggers with textual target semantics. By optimizing visual patterns to approximate target features in the embedding space, these attacks create backdoors that are resistant to standard detection and fine-tuning defenses, leveraging the same alignment principles used in benign multimodal adaptation [435].

Perhaps most concerning is the stealthiness afforded by editing-based attacks in black-box and in-context learning scenarios. Attackers can now distribute multiple trigger keys across different prompt components or conversational turns, creating composite backdoors that activate only when a specific sequence of conditions is met. This complexity makes detection exceedingly difficult, as the backdoor remains dormant during standard single-turn evaluations. Moreover, training-free lexical backdoor attacks have emerged, where adversaries manipulate the tokenizer’s embedding dictionary directly rather than retraining the model, achieving universal attack capabilities across different tasks with sparse, explainable rules that mimic legitimate vocabulary updates [436][437][438].

The convergence of model editing and backdoor injection necessitates a reevaluation of current security protocols. The assumption that model editing is inherently a corrective, benign process is no longer tenable. As demonstrated by the efficacy of frameworks like **BadEdit** and the persistence of editing-induced vulnerabilities in federated and multimodal contexts, the boundary between updating knowledge and compromising safety has effectively dissolved. Future defense mechanisms must therefore account for the possibility that any localized parameter update, regardless of its claimed intent, could serve as a vector for durable, stealthy malicious behavior. The community must develop forensic capabilities capable of distinguishing between legitimate knowledge integration and adversarial manipulation, recognizing that the mathematical operations underpinning both are increasingly identical [230][233]. This realization sets the stage for the subsequent analysis of how these vulnerabilities are amplified within the dynamic, iterative streams of continual and federated learning, where the temporal dimension adds further complexity to the persistence and propagation of such attacks.

7.3 Vulnerabilities in Continual and Federated Learning Streams

7.3 Vulnerabilities in Continual and Federated Learning Streams

While the previous subsection highlighted how model editing provides a potent mechanism for injecting durable backdoors through direct parameter manipulation, this section examines how the dynamic, iterative nature of Continual Learning (CL) and the decentralized architecture of Federated Learning (FL) further amplify these threats. The convergence of CL and FL introduces a distinct threat landscape where non-stationary data distributions, sequential task updates, and heterogeneous client participation create fertile ground for sophisticated adversarial manipulations that extend beyond single-shot edits. A primary concern in these streams is the phenomenon of accumulative poisoning, where malicious updates injected over time compound to degrade model integrity or embed persistent backdoors that evade standard aggregation defenses. The incremental nature of CL allows adversaries to exploit the temporal dimension of learning, injecting poisoning samples at critical junctures to affect not only the current task but also unknown previous or subsequent tasks. Research has empirically demonstrated that incremental learners are highly vulnerable to poisoning-based backdoor attacks, where an adversary can influence the model’s

behavior across the entire task sequence by injecting an extremely small fraction of backdoor samples, sometimes as low as 0.1% of the data stream [439]. This cross-task generalization of backdoor knowledge poses a severe risk, as the malicious functionality persists and propagates through the learning lifecycle, compromising the system’s long-term reliability.

In the realm of federated learning, the durability of backdoors presents a distinct challenge compared to centralized settings, mirroring the persistence issues seen in editing-based attacks but arising from aggregation dynamics. Standard backdoor attacks in FL often suffer from fragility; the malicious functionality tends to vanish once the attacker stops uploading poisoned updates, as subsequent rounds of aggregation by benign clients dilute the adversarial signal. To counteract this, advanced attack strategies have been developed to ensure persistence by targeting the structural stability of the model, a concept analogous to the parameter-specific targeting in model editing. A notable example is the “Neurotoxin” attack, which targets model parameters that undergo minimal magnitude changes during standard training updates. By focusing on these stable parameters, attackers can implant backdoors that exhibit significantly enhanced durability, effectively doubling the lifespan of the malicious functionality compared to state-of-the-art baseline attacks [434]. Similarly, the “PerDoor” technique leverages adversarial perturbations to target specific parameters that deviate least in successive FL rounds, achieving an average persistence improvement of over ten times compared to traditional methods [440]. These neurotoxin-style attacks underscore the difficulty of purging malicious influences in continuous learning streams, where the model is constantly evolving yet retains latent vulnerabilities in its structural core.

Furthermore, the distributed nature of FL enables complex attack vectors such as distributed and cross-task poisoning, where multiple adversaries or a single strategic attacker can coordinate to compromise the global model. The “Chameleon” attack illustrates how contrastive learning can be weaponized to amplify the relationship between benign and poisoned images, resulting in backdoors that remain effective for significantly longer durations [441]. In scenarios involving multiple malicious agents, the interaction between different poisoning strategies can lead to unexpected outcomes; however, cooperative multi-agent attacks can also synchronize to maximize impact, although some studies note that non-cooperative settings may lead to backfiring effects where collective success rates drop [442]. Nevertheless, the potential for coordinated attacks remains a critical vulnerability, particularly when attackers can anticipate the behavior of other clients to ensure their malicious updates survive the aggregation process [443].

The vulnerability extends beyond simple classification tasks to more complex architectures including reinforcement learning and graph neural networks within federated streams, demonstrating the ubiquity of these risks across diverse foundation model applications. In reinforcement learning, sparse targeted backdoor attacks like “BadRL” can degrade agent performance with minimal poisoning efforts by strategically selecting state observations with high attack values, demonstrating that even infrequent interventions can have catastrophic consequences in decision-making streams [444]. Similarly, in federated graph neural networks, the compounded complexity of graph structures and distributed learning creates additional attack surfaces. The “Bkd-FedGNN” benchmark highlights how factors such as trigger size, position, and poisoning rate interact in node-level and graph-level tasks, revealing that backdoor attacks can be highly effective across diverse domains and data distributions [445].

Moreover, the heterogeneity of clients in FL, often characterized by non-IID data distributions, can be exploited to mask malicious updates, adding another layer of stealth comparable to the subtle parameter shifts discussed in model editing. Attacks like “BapFL” demonstrate that even personalized federated learning methods, which were previously thought to offer robustness through

parameter decoupling, remain vulnerable when attackers diversify their classifiers to mimic benign clients [446]. The use of adaptive aggregation strategies by defenders is often met with equally adaptive attack mechanisms, such as those that modify local model weights directly via sign flips to inject triggers while jointly optimizing the trigger pattern for stealth [447]. Additionally, the emergence of model compression techniques like the Lottery Ticket Hypothesis in edge devices does not inherently protect against these threats; backdoor attacks have been shown to influence the structure of extracted tickets, maintaining their efficacy even in compressed models [448].

The cumulative effect of these vulnerabilities suggests that security in continual and federated learning streams cannot be treated as a static property but must be viewed as a dynamic arms race. The ability of adversaries to inject durable, cross-task, and distributed backdoors challenges the fundamental assumptions of trust in collaborative learning systems. As learning streams become more open and continuous, the risk of accumulative poisoning increases, necessitating defense mechanisms that can detect not only instantaneous anomalies but also long-term deviations in parameter evolution and task performance. The interplay between task boundaries, client availability, and data heterogeneity creates a complex attack surface where traditional defenses based on outlier detection or simple aggregation rules are increasingly insufficient. Consequently, future research must focus on developing robust verification protocols capable of auditing the temporal consistency of model updates and identifying subtle, persistent manipulations that threaten the integrity of lifelong intelligent systems. This evolution toward more stealthy and environmentally integrated threats sets the stage for the following discussion on physical realizability, where triggers transcend digital abstractions to manifest in the real world.

7.4 Stealthy Triggers and Physical Realizability in Foundation Models

7.4 Stealthy Triggers and Physical Realizability in Foundation Models

Building upon the discussion of accumulative poisoning and parameter-specific persistence in continual and federated streams, this section examines how adversaries have evolved from injecting overt digital anomalies to crafting stealthy, physically realizable triggers. While previous attacks often relied on visible patches or statistical outliers that could be filtered by aggregation defenses, modern strategies leverage natural phenomena, semantic attributes, and frequency-domain manipulations to embed malicious functionality that evades both human inspection and automated detection. This transition is particularly critical for foundation models deployed in safety-critical domains such as autonomous driving and medical diagnostics, where poisoned knowledge embedded via subtle triggers can persist through fine-tuning and adaptation phases, effectively bridging the gap between the temporal vulnerabilities of learning streams and the static robustness required for deployment.

A primary avenue for achieving this stealthiness involves the exploitation of natural objects and environmental transformations rather than artificial noise patterns. Traditional backdoor attacks utilizing visible pixels are increasingly obsolete; instead, recent advancements demonstrate the efficacy of using everyday items as triggers. For instance, researchers have successfully implemented physical backdoor attacks on object detectors by utilizing specific clothing, such as a T-shirt, to induce a “cloaking” effect where persons become invisible to the detection system [449]. Similarly, the concept of “reflection backdoor” leverages the physical properties of light reflection to plant triggers that appear as intrinsic image features, making them highly resistant to standard filtering

techniques that typically target unnatural artifacts [450]. These approaches highlight a critical vulnerability: when triggers align seamlessly with the natural data distribution, distinguishing between legitimate features and malicious injections becomes exponentially difficult, complicating the defense of models operating in dynamic physical environments.

In the textual domain, which is central to Large Language Models (LLMs), stealth is achieved through linguistic subtleties that preserve fluency and grammatical correctness, mirroring the parameter-level subtlety discussed in neurotoxin-style attacks. Attacks such as “hidden backdoors” utilize homograph replacement or generate trigger sentences that mimic natural text generation patterns, thereby fooling both human administrators and automated scanners [451]. Furthermore, the emergence of clean-label attacks has significantly lowered the barrier for successful infiltration. Techniques like “Narcissus” demonstrate that effective backdoors can be implanted with minimal information and extremely low poisoning rates by synthesizing triggers that share persistent features with the target class, ensuring that defense mechanisms focusing on label inconsistencies fail [452]. This is further compounded by methods that exploit attribute-based triggers, where content-relevant features serve as the activation signal, rendering the attack indistinguishable from normal semantic variations [453].

The challenge of physical realizability extends beyond static images and text to dynamic, multi-modal environments essential for embodied AI. In video recognition and autonomous systems, attacks must withstand temporal distortions, motion blur, and varying lighting conditions. Studies have shown that universal adversarial triggers can be extended temporally to attack video models effectively, even under strict clean-label constraints [454]. Moreover, the integration of audio and visual modalities introduces complex attack surfaces; “dual-key” backdoors require simultaneous triggers in both modalities to activate, significantly enhancing stealth and reducing the probability of accidental activation or detection [455]. In the context of autonomous vehicles, robust physical attacks have been developed against traffic sign recognition and lane detection systems. These attacks employ variable-size triggers and malicious adversarial training to ensure effectiveness across different distances and illumination levels, posing direct threats to public safety [456][457].

Advanced methodologies have also explored the frequency domain and generative capabilities to construct imperceptible triggers that bypass spatial-domain defenses. Low-frequency black-box attacks manipulate the spectral components of images, creating perturbations that are invisible in the spatial domain yet robust against compression and filtering [458]. Additionally, generative models are increasingly used to automate the creation of physical backdoor datasets, suggesting suitable triggers and refining them for plausibility, thereby lowering the expertise required to launch sophisticated physical attacks [459]. The versatility of these attacks is further exemplified by methods that utilize spatial attention mechanisms to embed triggers without causing noticeable feature loss, ensuring high attack success rates while maintaining model utility [460].

The convergence of these stealthy and physically realizable techniques underscores a fundamental gap in current defense protocols, setting the stage for the forensic analysis discussed in the following section. Traditional defenses often assume static, visible triggers or rely on the presence of label noise, assumptions that are systematically invalidated by modern attack vectors like “BELT,” which enhances backdoor exclusivity to evade state-of-the-art countermeasures [461]. Furthermore, the ability to inject backdoors into pre-trained encoders used in self-supervised learning via dynamic, invisible triggers exacerbates the risk, as these compromised encoders can propagate vulnerabilities to numerous downstream tasks [462]. As foundation models continue to scale and integrate into the physical world, the development of adaptive, context-aware defense mechanisms capable of detecting semantic anomalies and physical inconsistencies becomes paramount. The sophistication

of these invisible and physically grounded threats necessitates a shift from simple outlier detection to advanced forensic techniques capable of reverse-engineering these hidden triggers, a topic explored in the subsequent section.

7.5 Detection Mechanisms and Forensic Analysis of Poisoned Models

7.5 Detection Mechanisms and Forensic Analysis of Poisoned Models

Building upon the sophisticated stealthy and physically realizable triggers discussed in the previous section, the defense landscape has necessitated a parallel evolution in detection mechanisms and forensic analysis. As adversaries transition from conspicuous digital patches to naturalistic, semantic, and frequency-domain manipulations, traditional signature-based defenses often fail. Consequently, defenders must adopt a multi-faceted approach that leverages adversarial perturbations, interpretability insights, and statistical anomaly detection to uncover these hidden threats. This subsection synthesizes the state-of-the-art in identifying poisoned models, categorizing strategies into those utilizing universal adversarial perturbations, interpretability-driven feature analysis, and advanced forensic methods capable of distinguishing malicious injections from natural model behaviors, thereby setting the stage for the remediation techniques detailed in the subsequent section.

A prominent line of defense exploits the intrinsic relationship between backdoor triggers and universal adversarial perturbations (UAPs), capitalizing on the hypothesis that backdoored models contain specific “shortcuts” that render them uniquely susceptible to certain perturbation patterns. The [463] paper introduces a novel detection method termed Universal Soldier for Backdoor detection (USB), which observes that UAPs generated from backdoored models require significantly fewer perturbations to mislead the classifier compared to clean models. By reverse-engineering potential triggers through these perturbations, USB effectively identifies infected models across hundreds of architectures without prior knowledge of the trigger shape. Expanding on this, the [464] study proposes a lightweight detection mechanism that applies intentional adversarial perturbations to input images; if the model’s prediction remains consistent despite the perturbation, the input is flagged as a backdoor instance, achieving high detection rates on standard benchmarks while maintaining visual quality. Furthermore, the [465] addresses the challenge of diverse trigger sizes and transparencies by employing a global-to-local probing framework. This method, known as Adaptive Adversarial Probe (A2P), utilizes attention-guided region generation and box-to-sparsity scheduling to dynamically adjust perturbation budgets, thereby activating latent backdoors that static probes might miss.

Parallel to perturbation-based methods, interpretability-driven detection has emerged as a powerful tool for uncovering the semantic discrepancies introduced by backdoors, particularly when triggers mimic natural data distributions. These approaches argue that while backdoored models maintain high performance on clean data, their internal reasoning processes diverge significantly from benign models when exposed to triggers. The [466] framework demonstrates that classification results on adversarial or poisoned samples often lack human-perceptible justification. By establishing a bi-directional correspondence between attributes and internal neurons, this technique amplifies critical neuron activations to expose uninterpretable computation paths, successfully detecting various attacks with high accuracy. Similarly, the [467] approach employs synergistic anomaly detectors trained on clean validation data. One detector identifies anomalous features in the extraction layers, while the other flags anomalous mappings from features to outputs, effectively capturing

the dual nature of backdoor functionality. In the domain of Natural Language Processing, where stealth relies on linguistic subtleties, the [468] leverages large language models to paraphrase inputs, hypothesizing that triggers (such as infrequent words) do not alter semantic meaning. By observing prediction stability across paraphrased versions, this method effectively isolates poisoned samples, outperforming traditional baselines in precision and recall.

Forensic analysis extends beyond mere detection to the reverse engineering and categorization of attack vectors, providing critical insights required for effective mitigation. The [469] technique represents a significant advancement in this area by automatically decomposing attack samples into clean inputs and corresponding triggers. This allows for the clustering of triggers based on their properties, enabling the automatic synthesis of scanners that can generalize across different backdoor types. Addressing the challenge of false positives, the [470] method introduces a symmetric feature differencing technique. This forensic tool identifies the minimal set of features separating two classes, determining whether a trigger consists of features distinct from natural class-distinguishing characteristics, thereby drastically reducing false alarms in model scanning. Additionally, the [471] offers a scalable solution that relies on internal feature maps to detect, reverse-engineer, and identify the target class of a backdoor without access to the training dataset, proving effective against various incorporation mechanisms.

The distinction between injected backdoors and natural vulnerabilities is also crucial for robust defense. The [472] study reveals that naturally occurring backdoor vulnerabilities are widespread and can be exploited similarly to injected ones, necessitating detection frameworks that account for both inherent and malicious flaws. Moreover, the [473] paper challenges the common assumption that poison samples always form separable clusters in latent space, demonstrating adaptive attacks that bypass such defenses. This underscores the need for robust forensic tools like the [474], which detects attacks independent of the embedding mechanism by analyzing the maximum margin statistic of classifier outputs. Finally, specialized defenses such as [475] focus on extracting the backdoor functionality into a dedicated “expert” model via fine-tuning on mislabeled data, allowing for precise filtering of backdoor inputs during inference. Together, these diverse mechanisms form a comprehensive arsenal for securing foundation models against the evolving landscape of poisoning threats, providing the diagnostic foundation necessary for the unlearning and fine-tuning strategies discussed next.

7.6 Mitigation Strategies via Unlearning and Robust Fine-Tuning

7.6 Mitigation Strategies via Unlearning and Robust Fine-Tuning

Building upon the diagnostic insights and forensic capabilities detailed in the previous section, the transition from identifying poisoned models to actively remediating them requires a shift from passive detection to active intervention. As foundation models become increasingly integrated into critical infrastructure, the necessity for efficient mitigation strategies has never been more pressing. Traditional retraining from scratch is often computationally prohibitive and impractical for massive models, necessitating the development of targeted techniques such as machine unlearning and robust fine-tuning. These approaches aim to excise malicious influences—whether identified backdoors or privacy leaks revealed by forensic analysis—while preserving the model’s utility on clean data. A prominent paradigm in this domain is Shared Adversarial Unlearning, which establishes a theoretical connection between backdoor risk and adversarial risk to derive a novel upper bound for mitigation. By generating Shared Adversarial Examples (SAEs) that act as a bridge between the backdoored model and the purified model, this method unlearns specific associations to

mitigate the backdoor effect without requiring extensive clean datasets [476]. This bi-level optimization approach represents a significant advancement over simple fine-tuning, offering state-of-the-art performance in neutralizing targeted attacks while maintaining general classification accuracy.

In the realm of multimodal systems, where vulnerabilities often arise from the complex fusion of different data modalities, mitigation strategies must be equally sophisticated to address the diverse trigger types uncovered by forensic tools. Recent research highlights that multimodal contrastive learning systems are particularly susceptible to backdoor attacks due to their open nature. To address this, localized unlearning techniques have been proposed, specifically targeting the “token” level within the model architecture. The method known as Unlearning Backdoor Threats (UBT) focuses on constructing a small set of poisoned samples to strengthen backdoor shortcuts temporarily, thereby identifying suspicious parameters. Subsequently, it applies a token-based localized forgetting regime to specifically erase these backdoor associations. This approach is crucial because it avoids the degradation of clean accuracy often seen in broad fine-tuning methods and eliminates the need for constructing extensive clean training pairs [477]. Such precision is vital for large-scale vision-language models where global parameter updates could destabilize learned semantic alignments.

Beyond unlearning specific malicious patterns, robust fine-tuning and purification strategies play a critical role in defending against adaptive attacks that may evade initial detection. However, the efficacy of current purification methods is increasingly being challenged by sophisticated adversaries who can adapt their strategies to bypass defense mechanisms. For instance, while adversarial purification using generative models like diffusion networks has shown promise in removing perturbations, rigorous evaluations suggest that many existing methods fail against strong adaptive attacks tailored to the purification process itself. Studies indicate that without careful consideration of the purification model’s own robustness, these defenses can be circumvented, leading to a false sense of security [478]. Furthermore, the trade-off between robustness and standard accuracy remains a persistent challenge; methods that aggressively purify inputs may inadvertently destroy semantic information, resulting in degraded performance on benign samples [479].

To counter these limitations, advanced fine-tuning strategies such as Adversarial Training on Purification (AToP) have been developed. This pipeline combines random transforms to prevent overfitting to known attacks with fine-tuning the purifier model using adversarial loss, thereby enhancing both robustness and generalization against unseen attack vectors [480]. Similarly, in the context of federated learning, where models are trained across distributed and potentially malicious clients, mitigation requires resilient aggregation and detection mechanisms. The Metric-Cascades (MESAS) framework exemplifies this by employing multiple detection metrics simultaneously to identify poisoned updates, creating a complex optimization landscape that is difficult for adaptive attackers to circumvent [481]. Additionally, ensemble-based approaches like FLGuard utilize contrastive learning to detect and discard malicious local updates, proving particularly effective in non-IID data settings where traditional Byzantine-robust methods often fail [482].

The scope of unlearning has also expanded to address privacy regulations and the “Right to be Forgotten,” extending beyond security to include the removal of specific data influences, a transition that foreshadows the ethical complexities discussed in the following section. In Large Language Models (LLMs), unlearning is emerging as a computationally efficient alternative to Reinforcement Learning from Human Feedback (RLHF) for aligning models by removing harmful responses or copyright-protected content. This process leverages negative examples to efficiently steer model behavior, achieving alignment with a fraction of the computational cost required for full retraining [483]. However, the evaluation of these unlearning techniques reveals significant challenges;

standard Membership Inference Attacks (MIAs) often underestimate the residual information retained by models, necessitating stronger “per-example” attack formulations to truly verify privacy preservation [484]. Moreover, in reinforcement learning settings, the concept of “environment unlearning” has been introduced to revoke an agent’s knowledge of specific environments, addressing unique challenges such as preventing performance degradation in remaining tasks while ensuring the complete erasure of sensitive environmental dynamics [485].

Despite these advancements, limitations persist. Many current purification and unlearning methods struggle against adaptive adversaries who can anticipate defense mechanisms. For example, while some methods claim to remove unlearnable examples or poison, adaptive attackers can utilize auxiliary data to reverse these protections, rendering the defense ineffective [486]. Consequently, future mitigation strategies must evolve towards more dynamic, self-correcting frameworks that can continuously adapt to emerging threats without compromising the foundational capabilities of foundation models. The integration of forensic tools, such as those capable of tracing poison attacks back to their source through iterative clustering and pruning, offers a complementary path to proactive defense, enabling precise identification and removal of compromised data subsets [487]. Ultimately, while these technical strategies provide a robust arsenal for model remediation, their deployment introduces profound ethical ramifications regarding privacy, intellectual property, and societal safety, setting the stage for the critical discussion on the privacy implications and ethical risks of automated knowledge modification in the subsequent section.

7.7 Privacy Implications and Ethical Risks of Automated Knowledge Modification

7.7 Privacy Implications and Ethical Risks of Automated Knowledge Modification

Building upon the technical mitigation strategies discussed in the previous section, the deployment of automated knowledge modification techniques—encompassing model editing, fine-tuning, and unlearning—introduces profound ethical ramifications that extend far beyond simple performance metrics. As foundation models evolve into dynamic entities, the very mechanisms designed to update, correct, or erase knowledge create novel vectors for privacy breaches, intellectual property theft, and the dissemination of undetectable misinformation. A critical concern emerging from this dynamic landscape is the concept of “privacy backdoors,” which fundamentally alters the risk profile for models deployed in sensitive environments. Recent research has uncovered a vulnerability where attackers tamper with pre-trained model weights to install these backdoors; when a victim subsequently fine-tunes such a corrupted model on proprietary or sensitive data, the backdoor activates, allowing the adversary to reconstruct individual training samples with guaranteed success [488]. This threat is particularly insidious as it amplifies membership inference risks, demonstrating that even models trained with differential privacy guarantees can be compromised if the underlying pre-trained checkpoint is untrusted [489].

The intersection of model editing and membership inference attacks further complicates the ethical landscape, revealing that defensive mechanisms can be subverted. While machine unlearning is often advocated to comply with regulations like the “Right to be Forgotten,” the process itself is susceptible to exploitation. Adversaries have demonstrated that malicious unlearning requests can be leveraged to implant backdoors rather than remove data, effectively transforming a protective measure into an offensive tool [490]. Moreover, the relationship between unlearning and privacy is non-linear; removing specific data points can trigger the “Privacy Onion Effect,” where eliminating the most vulnerable outlier samples exposes a new layer of previously safe data points to membership inference attacks [491]. This phenomenon suggests that naive approaches to automated knowledge

removal may inadvertently degrade the privacy of remaining users in the dataset, challenging the assumption that deletion always equates to enhanced security.

Beyond privacy, intellectual property (IP) theft represents a significant ethical challenge exacerbated by the ease of automated knowledge modification. The ability to hijack models or extract proprietary information through subtle parameter updates threatens the viability of commercial AI deployment. Although techniques such as backdoor-based watermarking have been developed to protect model ownership, they remain vulnerable to sophisticated “capsulation attacks” where adversaries encapsulate the model to block ownership verification without affecting functionality [492]. Conversely, attackers can repurpose backdoor techniques to perform membership inference, enabling malicious actors to determine if specific copyrighted or private data was used in training a model, thereby facilitating IP theft or privacy violations [493]. The dual-use nature of these technologies means that methods designed for protecting IP can be inverted to steal it, fostering a complex arms race in model security.

The societal impact of these vulnerabilities is further amplified by the potential for foundation models to be edited to harbor malicious capabilities, leading to undetectable misinformation. The ease with which large language models can be manipulated to provide instructions for creating biological, cyber, or chemical weapons underscores the urgent need for robust unlearning and editing frameworks capable of reliably removing hazardous knowledge without degrading general utility [395]. However, the persistence of such knowledge presents a double-edged sword: while retention poses safety risks, incomplete removal during editing processes can leave models vulnerable to exploitation. Furthermore, ethical implications extend to fairness and bias; automated modifications that fail to account for societal biases can perpetuate discrimination and exclusion, contributing to a broad spectrum of risks ranging from toxic language generation to the erosion of trust in shared information [494].

In interactive settings, these risks are compounded by the model’s frequent inability to reason about contextual privacy. Instruction-tuned models often fail to maintain contextual integrity, revealing private information in scenarios where human judgment would dictate secrecy [495]. This failure becomes critical when models are edited to adapt to new domains without rigorous safety alignment, potentially leading to the leakage of Personally Identifiable Information (PII) even in models that have undergone scrubbing procedures [496]. Consequently, the ethical deployment of AI requires a holistic framework that considers not only the technical efficacy of editing and unlearning but also the broader sociotechnical vulnerabilities they introduce. As noted in comprehensive reviews of AI ethics, addressing these challenges demands a multidisciplinary approach that integrates technical safeguards with regulatory standards and value-sensitive design principles [497]. Ultimately, the future of safe and trustworthy AI depends on developing automated knowledge modification techniques that are transparent, auditable, and resilient against the evolving landscape of privacy and security threats [498]. Without such advancements, the promise of lifelong intelligent systems may be overshadowed by the risks of pervasive privacy erosion and unchecked misinformation.

8 Theoretical Limits, Open Challenges, and Future Research Directions

8.1 Theoretical Bounds on Memory, Information Flow, and Computational Complexity

8.1 Theoretical Bounds on Memory, Information Flow, and Computational Complexity

The pursuit of artificial systems capable of lifelong adaptation is fundamentally constrained by the physical and mathematical limits of computation. As foundation models scale toward trillion-parameter architectures, the theoretical underpinnings of Continual Learning (CL) must evolve from heuristic empirical observations to rigorous bounds on memory, information flow, and thermodynamic cost. These theoretical constraints provide the essential framework for understanding why the brute-force scaling strategies discussed in the following section are approaching a wall; without addressing these fundamental limits, simply adding parameters yields diminishing returns. A primary constraint in this domain is the relationship between system memory and computational tractability. The Space-Bounded Church-Turing Thesis (SBCT) posits that memory acts as a limiting factor in computation, independent of time constraints, suggesting that predicting the long-term behavior of bounded-memory systems is computationally tractable only within specific entropy bounds [499]. This thesis implies that for a continual learner to maintain stability over an unbounded stream of tasks, its memory capacity must scale appropriately with the complexity of the task sequence. Indeed, complexity-theoretic studies within the PAC framework have established that any continual learner requires memory that grows linearly with the number of tasks k , strongly suggesting that the problem is inherently intractable without improper learning strategies or relaxed constraints [57].

Beyond static memory bounds, the dynamics of information flow through neural network layers dictate the feasibility of retaining old knowledge while acquiring new representations. Theoretical analyses reveal that optimizing information preservation between layers is critical for managing task-specific knowledge passing throughout the network. Without explicit mechanisms to regulate this flow, the interference between sequential tasks leads to unavoidable performance degradation. Research has established a probabilistic framework to analyze how information adaptation within layers impacts learning performance, demonstrating that maximizing information preservation is a necessary condition for alleviating catastrophic forgetting [59]. Furthermore, the phenomenon of “distal interference,” where training on a subset of the domain induces non-local changes in distant input regions, imposes severe architectural constraints. It has been theoretically shown that uniformly trainable models devoid of distal interference must be exponentially large, implying that polynomial-complexity models inevitably require data augmentation or algorithmic interventions to function effectively in continual settings [500].

These information-theoretic challenges are inextricably linked to the thermodynamic costs of computation, which introduce fundamental limits on the speed and fidelity of knowledge updates. Landauer’s Principle dictates a minimum energy cost for information erasure, proportional to the change in Shannon entropy. However, in the context of rapid, online continual learning, systems operate far from equilibrium. The cost of “fast thought”—performing computations in finite time—incurs additional dissipated work beyond the Landauer bound. Studies on shortcuts to thermodynamic computing demonstrate that counterdiabatic potentials are required to guide systems rapidly along desired computational paths, and this dissipated work scales with the computation rate and the square of the system’s length scale [501]. This creates a tangible trade-off: achieving high plasticity (fast adaptation) in a continual learner comes at an increasing energetic cost, which

becomes prohibitive as model sizes grow. Moreover, the topology of the computational circuit itself restricts physical coupling, thereby affecting the minimal entropy production required to run the system. In multipartite processes, constraints on rate matrix dependencies derive a strictly nonzero lower bound on entropy production, linking the “learning rate” of the system directly to its thermodynamic inefficiency [502] [503].

These thermodynamic and information-theoretic constraints converge to define the hard limits of sustainable intelligence, foreshadowing the “Energy Wall” facing trillion-parameter systems. For instance, the read-once machine limitation, where a system scans input only once, is associated with unavoidable thermodynamic costs even with infinite memory, highlighting the intrinsic difficulty of single-pass continual learning scenarios [504]. Similarly, the analysis of Turing Machines through the lens of stochastic thermodynamics reveals a fundamental trade-off between heat generation, the Kolmogorov complexity of the heat function, and the complexity of the input-output map, suggesting that universal continual learners face infinite expected heat generation under certain distributions [505].

Consequently, future research must address these hard limits by designing algorithms that operate efficiently within space-bounded and energy-constrained regimes, rather than relying solely on scaling compute. Theoretical investigations into variational principles and mean-field potentials in statistical mechanics offer a promising direction, translating continual learning into a Franz-Parisi thermodynamic potential framework where previous knowledge acts as a prior [506]. Additionally, exploring analog computing systems reveals that infinite precision is forbidden by fundamental physical laws, as the entropy production per degree of freedom during erasure is governed by the logarithm of the configurational volume [507]. Understanding these bounds is not merely an academic exercise but a prerequisite for developing scalable, energy-efficient, and truly lifelong intelligent systems that can navigate the stability-plasticity dilemma without violating the laws of physics. These theoretical insights set the stage for examining the practical manifestation of these limits: the scaling challenges and the impending energy crisis in next-generation foundation models.

8.2 Scaling Challenges and the Energy Wall in Trillion-Parameter Systems

8.2 Scaling Challenges and the Energy Wall in Trillion-Parameter Systems

The theoretical constraints on memory, information flow, and thermodynamic costs outlined in the previous section are not merely abstract limits; they manifest concretely as the “Energy Wall” currently confronting the development of trillion-parameter foundation models. While the relentless pursuit of larger architectures promises unprecedented capabilities in reasoning and multimodal understanding, this trajectory is colliding with a convergence of thermodynamic ceilings, economic unsustainability, and hardware stagnation. As models expand, the computational resources required for training and inference have grown at an exponential rate, doubling approximately every few months—a trend that is neither ecologically sustainable nor economically scalable in the long term [508]. The current paradigm, reliant on the brute-force scaling of parameter counts and dataset sizes discussed in theoretical bounds, faces a critical juncture where marginal gains in model performance are increasingly outweighed by the disproportionate surge in energy consumption and carbon emissions.

This unsustainability is rooted in the breakdown of historical hardware efficiency trends that once

underpinned the scaling era. For decades, the industry relied on Moore’s Law and Dennard scaling to deliver automatic performance improvements and energy efficiency; however, these foundational principles have effectively stalled [509]. The end of Dennard scaling has elevated power consumption to a primary design constraint, equal in importance to raw performance, forcing a reevaluation of computational architecture [510]. In the context of trillion-parameter systems, the energy cost of moving data between memory and processing units—the “memory wall”—has become a dominant factor, often exceeding the energy cost of the computation itself [511]. Modern computing systems, designed to move data to computation, are fundamentally misaligned with the data-intensive nature of large-scale AI, creating severe bottlenecks in bandwidth, latency, and energy efficiency [512]. Consequently, simply adding more GPUs to a cluster yields diminishing returns due to communication overheads and thermal constraints, rather than the linear scaling of capability once expected [513].

Furthermore, the empirical reality of scaling laws suggests we are approaching a point of saturation where increasing model size no longer guarantees proportional improvements in utility without a corresponding explosion in resource requirements. While early empirical scaling laws successfully predicted performance gains based on compute, data, and parameter counts, recent analyses indicate that parameter scaling is losing steam for certain architectures, and data scaling may soon hit the limits of available high-quality tokens [514]. The assumption that “bigger is better” is being challenged by the finding that scaling laws cannot be deceived by spurious parameters; test loss depends strictly on the number of trainable parameters, meaning that inefficient architectural choices or frozen parameters do not contribute to capability but still incur significant energy costs [515]. Moreover, the communication costs associated with distributed training of massive models are projected to constitute a significant portion, potentially up to 75%, of the total runtime in future hardware scenarios, further exacerbating the energy crisis [516].

Addressing these challenges necessitates a paradigm shift from pure algorithmic scaling to a rigorous hardware-algorithm co-design strategy, directly responding to the thermodynamic and space-bounded limits identified earlier. The solution lies not merely in optimizing software for existing hardware or vice versa, but in developing unified approaches where algorithms are tailored to the physical constraints of emerging computing substrates, and hardware is designed to exploit the specific statistical properties of AI workloads [517]. This co-design philosophy must embrace heterogeneity, leveraging specialized accelerators, near-threshold voltage computing, and novel memory technologies to break the efficiency deadlock [518]. For instance, integrating processing-in-memory (PIM) architectures can drastically reduce data movement energy by performing computations directly where data is stored, offering a promising pathway to overcome the von Neumann bottleneck inherent in current architectures [519]. Additionally, exploring inexact computing and reduced precision arithmetic allows for trading solution quality for significant energy savings, a strategy that is particularly viable for the stochastic nature of deep learning gradients [509].

The future of trillion-parameter systems also depends on rethinking the very nature of computation through thermodynamic and biological lenses, bridging the gap between theoretical bounds and practical implementation. Concepts such as “Thermodynamic Computing” propose harnessing non-equilibrium, self-organizing systems to utilize nature’s innate computational capacity, potentially bypassing the energy inefficiencies of traditional digital logic [520]. Similarly, adopting brain-inspired architectures that utilize sparse, event-driven processing, such as Spiking Neural Networks (SNNs), could offer a route to energy-proportional computing that mirrors the efficiency of the human brain [521]. The integration of such novel paradigms requires a holistic view that considers the entire stack, from the physical device level to the algorithmic abstraction, ensuring that

energy efficiency is treated as a first-order metric alongside accuracy [522]. Without this radical restructuring of the AI development lifecycle, the ambition of deploying trillion-parameter models risks hitting an impassable energy wall, rendering the next generation of foundation models physically impossible to train or deploy at scale. Therefore, the research community must prioritize the development of sustainable scaling laws that explicitly account for energy complexity, moving beyond FLOPs-centric metrics to encompass the full thermodynamic cost of intelligence [523]. Only through such a concerted effort in co-design and architectural innovation can the field hope to sustain the trajectory of AI advancement in the face of finite global energy resources, thereby enabling the transition to the task-agnostic, open-ended systems explored in the following section.

8.3 Bridging the Gap to Task-Agnosticism and Open-Ended Development

8.3 Bridging the Gap to Task-Agnosticism and Open-Ended Development

While the previous section highlighted the thermodynamic and hardware barriers preventing the indefinite scaling of static, brute-force models, the path forward requires not only architectural efficiency but a fundamental shift in learning paradigms. The transition from static, task-specific artificial intelligence to truly dynamic, open-ended systems represents one of the most significant frontiers in the evolution of foundation models. Current paradigms predominantly rely on predefined objective functions and curated datasets, creating a disconnect between how machines learn and how biological agents acquire competence through lifelong interaction with non-stationary environments. Bridging this gap necessitates moving away from offline processing of massive data collections toward “collectionless,” online environmental interactions where agents autonomously generate their own curricula driven by intrinsic motivation.

A primary barrier to task-agnosticism is the reliance on explicit task definitions and reward signals. In real-world scenarios, agents must often operate without clear goals, necessitating mechanisms for self-directed exploration and skill acquisition. The concept of “collectionless” artificial intelligence proposes a paradigm where data acquired from the environment is processed immediately to update internal representations without storing the temporal stream, thereby promoting self-organized memorization and reducing dependency on centralized data repositories [524]. This approach aligns with the energy-constrained future outlined previously, offering a privacy-preserving and computationally efficient alternative to massive dataset storage. Furthermore, true task-agnostic learning involves acquiring fragmented knowledge from diverse interactions that can be recombined to solve novel tasks, a capability demonstrated by methods that organize experience into knowledge graphs to facilitate action proposal in unseen scenarios [525].

Central to achieving open-ended development is the mechanism of automatic curriculum generation. Instead of relying on human engineers to design progression paths, advanced agents must be capable of self-generating goals and sequencing them based on their current competence. Intrinsically Motivated Goal Exploration Processes (IMGEP) provide a foundational framework for this, enabling agents to self-generate, select, and order learning goals to discover diverse skill repertoires [526]. Recent advancements have extended this by leveraging large language models to augment autotelic agents, allowing them to imagine abstract goals, decompose them into subgoals, and generate relevant reward functions without hand-coded specifications [527]. This integration of linguistic priors facilitates the cultural transmission of goals and supports the exploration of high-level, human-relevant concepts. Moreover, sophisticated curriculum learning algorithms, such as

those utilizing unsupervised task representation learning, have shown the ability to decouple task manifolds from policy optimization, overcoming non-stationarity issues and significantly improving zero-shot generalization compared to traditional methods [528].

The challenge of task-agnosticism is further complicated by the interdependencies between tasks and the non-stationary nature of real-world environments. Agents must not only learn individual skills but also understand the hierarchical relationships and preconditions necessary for complex behavior. Research into autonomous open-ended learning has highlighted the necessity of treating task selection as a Markov Decision Process, where agents maximize competence across interdependent tasks by dynamically adjusting their learning focus [529]. This is particularly critical in scenarios where task relationships evolve over time, requiring architectures like H-GRAIL that can store and modify autonomously acquired task sequences to adapt to non-stationary interdependencies [530]. Additionally, the ability to transfer knowledge across heterogeneous domains and morphologies is essential for robust generalization. Meta-Automatic Curriculum Learning (Meta-ACL) addresses this by formalizing curriculum generation for unknown distributions of learners, enabling the transfer of curriculum strategies across agents with varying capabilities [531].

Intrinsic motivation serves as the driving force behind these open-ended systems, providing the necessary signal for exploration in the absence of extrinsic rewards. Effective intrinsic motivation strategies must go beyond simple surprise or novelty; they should encourage synergistic behaviors and the discovery of causal structures in the environment. For instance, incentivizing actions that produce effects unpredicted by individual agent dynamics has proven effective in multi-agent synergistic tasks [532]. Similarly, structured world models that incorporate relational inductive biases allow agents to plan for future novelty, leading to zero-shot generalization in object manipulation tasks after a phase of task-agnostic exploration [533]. The combination of curiosity and empowerment drives offers a unified strategy to quantify information flow, enhancing agent autonomy by balancing the relevance of actions with the potential to control future states [534].

Despite these advances, significant gaps remain in scaling these principles to foundation models. Current approaches often struggle with the computational complexity of maintaining open-ended repertoires in high-dimensional spaces and ensuring stability while continuously acquiring new skills. The vision of “collectionless” AI challenges the prevailing data-centric scaling laws, suggesting a future where models learn dynamically from streaming interactions rather than static corpora [524]. Achieving this will require novel architectural innovations that explicitly support the cognitive mechanisms discussed in the following section, including dual-process reasoning, working memory dynamics, and the seamless integration of perception, policy, and meta-learning. Future research must focus on developing benchmarks that evaluate not just task performance, but the quality of the learned curriculum, the diversity of discovered skills, and the agent’s ability to generalize to entirely novel domains without retraining. By synthesizing insights from developmental robotics, intrinsic motivation, and automated curriculum generation, the field can move closer to realizing artificial general intelligence capable of lifelong, self-directed evolution.

8.4 Cognitive-Inspired Architectures for Dual-Process Reasoning and Memory

8.4 Cognitive-Inspired Architectures for Dual-Process Reasoning and Memory

While Section 8.3 established the necessity of “collectionless,” intrinsically motivated agents capable of open-ended development in non-stationary environments, the realization of such systems

demands more than just adaptive curricula; it requires a fundamental restructuring of the underlying computational architecture. Current Foundation Models, despite their linguistic fluency, often falter when faced with the complex, multi-step reasoning and coherent long-term context retention required for true lifelong interaction. To bridge the gap between the dynamic learning paradigms outlined previously and the robust execution mechanisms needed for AGI, future research must integrate insights from cognitive science. Specifically, architectures must evolve to embody dual-process theories, sophisticated working memory dynamics, and global workspace coordination. These biological principles provide the structural blueprint for systems that can iteratively reason, distinctly manage episodic and semantic knowledge, and sustain the task-agnosticism envisioned in prior sections.

Central to this architectural evolution is the explicit implementation of dual-process reasoning, mirroring the cognitive distinction between fast, intuitive processing (System 1) and slow, deliberate reasoning (System 2). While large language models effectively simulate System 1 through rapid token generation, they frequently lack the metacognitive oversight characteristic of System 2 operations. Future architectures must incorporate a dedicated “slow” reasoning module capable of intervening when intuitive responses are insufficient or error-prone. Research indicates that integrating a dual-process architecture allows agents to decompose complex queries into manageable cognitive steps, utilizing an intuitive system for rapid hypothesis generation and a reflective system for verification and scoring [535]. This separation enables a critical balance between computational efficiency and logical rigor, mitigating the hallucination errors common in purely probabilistic approaches. Furthermore, recognizing the thermodynamic implications of these distinct modes suggests that future systems may require differentiated energy budgets, where fast thought operates as an associative machine and slow thought necessitates voluntary, logical-deductive processing [536].

Parallel to advanced reasoning mechanisms, the development of sophisticated, differentiated memory systems is paramount for supporting the lifelong intelligence described in Section 8.3. Human cognition relies on a dynamic interplay between short-term working memory, episodic memory for specific experiences, and semantic memory for generalized knowledge. In contrast, current AI systems often conflate these distinct memory types, leading to catastrophic forgetting or an inability to generalize from single instances—a significant barrier to the “collectionless” ideal. A promising direction involves constructing explicit, hybrid memory structures where working memory acts as a centralized hub for iterative updates, while episodic and semantic stores serve as specialized long-term repositories. Architectures that simulate human working memory through sustained neural activity and synaptic potentiation can maintain mental continuity, allowing successive states of thought to overlap and blend incrementally [537]. Such systems enable the iterative updating of thought processes, where each state preserves a proportion of co-active representations from the preceding state, fostering a chain of associatively linked intermediate states capable of advancing toward complex goals [538].

The separation of episodic and semantic memory is particularly vital for agents operating in the non-stationary environments highlighted previously. Theoretical frameworks suggest that episodic memory should capture fine-grained spatiotemporal representations of specific events, while semantic memory extracts compact, generalized categories from these experiences. Models employing dual-memory self-organizing architectures have demonstrated superior performance in lifelong learning scenarios by allowing episodic networks to learn unsupervised instances while semantic networks regulate structural plasticity [539]. Moreover, integrating spatial awareness into episodic memory—moving beyond simple temporal ordering to include place-centric contexts—can significantly enhance an agent’s ability to plan and reason about its environment [540]. By adopting

a subjective timescale for learning transition dynamics, agents can vary the temporal extent of their predictions, incentivizing exploration and improving decision-making in complex, open-world settings [541].

To unify these disparate modules—fast and slow reasoning, working memory, and dual long-term stores—future architectures should leverage Global Workspace Theory (GWT). GWT posits a large-scale system that integrates information among specialized modules to create higher-level cognition and awareness, a mechanism essential for coordinating the autonomous goal generation discussed in Section 8.3. Implementing a Global Latent Workspace (GLW) via unsupervised neural translation between multiple latent spaces can create a unique, amodal workspace where diverse sensory inputs and task-specific representations converge [542]. This approach facilitates the broadcasting of relevant information to all specialized modules, enabling conscious access to data and coordinated problem-solving. Recent proposals suggest that such a workspace can be realized through task-attention modules that capture task-specific information from a common representation space, effectively reducing task interference and mitigating catastrophic forgetting [543].

Ultimately, the convergence of dual-process reasoning, structured memory systems, and global workspace integration represents the structural prerequisite for the operational intelligence discussed in the following section. These cognitive-inspired architectures promise to endow machines with the ability to perform complex knowledge work, engage in mental time travel, and adapt continuously to new challenges without erasing past lessons. As research progresses, the focus must shift from merely scaling parameter counts to designing algorithms that emulate the functional dynamics of the human mind, ensuring that future AI systems are not only powerful but also reliable, interpretable, and capable of true generalization [544]. By grounding these developments in established cognitive theories and validating them through rigorous benchmarking against human behavioral data, the field can construct the necessary “blueprint” for a mind, setting the stage for the “engine” of self-correcting, neuro-symbolic, and compositional intelligence detailed in Section 8.5 [545].

8.5 Towards Self-Correcting, Neuro-Symbolic, and Compositional Intelligence

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Building upon the cognitive-inspired architectures detailed in the previous section, which establish the structural necessity for dual-process reasoning and distinct memory systems, the next critical frontier lies in operationalizing these mechanisms through self-correcting, neuro-symbolic, and compositional frameworks. While Section 8.4 outlines the “blueprint” for a mind capable of System 1 and System 2 thinking, this section addresses the “engine” required to execute such thinking: the deep integration of hybrid neuro-symbolic approaches that combine the perceptual fluency of neural networks with the rigorous, structured reasoning of symbolic logic. This synthesis is essential to overcome the inherent limitations of static, monolithic foundation models, which frequently suffer from hallucinations, logical inconsistencies, and an inability to systematically generalize to novel scenarios outside their training distribution. By moving beyond mere pattern matching, these hybrid systems offer a pathway toward agents that can not only learn from data but also refine their own knowledge bases, correct errors through logical deduction, and compose complex behaviors from reusable conceptual primitives.

A primary avenue for achieving the self-correcting intelligence alluded to in dual-process theories is

the development of frameworks where symbolic reasoning acts as an explicit supervisory signal to refine neural outputs iteratively. Traditional neural approaches often lack the mechanisms to verify the logical validity of their predictions, leading to the accumulation of errors—a vulnerability that is particularly acute in continual learning settings where stability is paramount. Recent advancements demonstrate that integrating external solvers with large language models can significantly enhance reliability by mimicking the “slow” verification step of human cognition. For instance, abductive-deductive frameworks have been proposed to refine step-wise natural language explanations, jointly verifying their correctness to reduce incompleteness and minimize redundancy [546]. By leveraging backward-chaining solvers, such systems can generate formal proofs that support the model’s reasoning, thereby enabling a form of introspective revision where the model identifies and corrects its own logical fallacies. This capability is crucial for domain-specific applications where factual accuracy and ethical alignment are paramount, suggesting that future foundation models must incorporate similar feedback loops to maintain consistency over time, effectively bridging the gap between the intuitive generation of System 1 and the reflective oversight of System 2.

Furthermore, achieving the robust generalization required for lifelong intelligence necessitates moving beyond mere statistical correlation to true compositional understanding. Human intelligence is characterized by the ability to decompose complex problems into reusable sub-tasks and recombine known concepts to solve unseen challenges, a trait that aligns closely with the modular memory structures discussed previously. Neural-symbolic systems are uniquely positioned to emulate this capability by factorizing inference processes into modular components. The concept of dynamic modularized reasoning, where inference is broken down into functional units that can be dynamically selected and routed, has shown superior performance in compositional generalization tasks compared to standard baselines [547]. By treating reasoning modules as interchangeable building blocks, these architectures allow models to adapt to new task structures without catastrophic forgetting, a critical requirement for continual learning that complements the episodic-semantic separation strategies. This modularity facilitates the emergence of grounded symbol-like representations, addressing the binding problem in artificial neural networks by enabling the flexible combination of distributed information [548].

The challenge of symbol grounding—connecting abstract symbolic concepts to raw perceptual data—remains a central hurdle in building effective neuro-symbolic systems that can interact with the non-stationary environments described in prior sections. Successful frameworks must bridge the gap between sub-symbolic sensory input and high-level logical reasoning without relying on brittle, pre-defined rules. Innovations in this space include softened symbol grounding processes that model symbol solution states as Boltzmann distributions, allowing for efficient sampling and mutually beneficial interactions between network training and symbolic reasoning [549]. Additionally, differentiable logic formalisms, such as differentiable fuzzy description logics, provide a unified semantic framework where neural models can be infused with knowledge bases while maintaining consistency with logical theories [550]. These approaches ensure that the representations learned by neural components are not only statistically valid but also semantically consistent with prior knowledge, enabling the system to revise its grounding rationally when faced with conflicting evidence, thus supporting the adaptive memory dynamics essential for AGI.

Moreover, the integration of generative modeling with symbolic structures offers a promising direction for learning rich, task-general representations that can sustain the “mental time travel” and planning capabilities envisioned in cognitive architectures. Generative neuro-symbolic models have demonstrated the ability to capture causal and compositional structures directly from raw data, allowing for one-shot learning and generalization across multiple tasks [551]. By coupling the control

flow of probabilistic programs with neural subroutines, these systems can express nonparametric statistical relationships while adhering to symbolic constraints. This duality is essential for creating agents that can plan and reason in open-world environments, where they must continuously update their internal world models based on new experiences. The combination of generative capabilities with logical constraints also aids in mitigating reasoning shortcuts, a common failure mode where models satisfy logical constraints vacuously without truly understanding the underlying concepts [552].

Looking forward, the realization of self-correcting, compositional intelligence will depend on the development of unified frameworks that seamlessly blend learning and reasoning, serving as the computational backbone for the cognitive architectures proposed earlier. Approaches that unify connectionist structures for learning with probabilistic program semantics for compositionality represent a significant step toward this goal [553]. Such frameworks must support both bottom-up feature extraction and top-down logical refinement, enabling models to induce logic theories from raw data while simultaneously optimizing neural parameters [554]. However, as we move toward trillion-parameter systems, the energy efficiency and computational tractability of these hybrid architectures become increasingly critical, foreshadowing the sustainability challenges discussed in the following section. Future research must focus on scalable algorithms that can perform real-time symbolic inference alongside neural processing, ensuring that the benefits of structured reasoning do not come at the cost of prohibitive latency or excessive carbon footprints. Ultimately, the convergence of neuro-symbolic methods and compositional learning structures promises to deliver AI systems that are not only more accurate and robust but also capable of the kind of flexible, self-improving intelligence that defines human cognition, while laying the groundwork for the ethical and sustainable deployment of such powerful technologies.

8.6 Ethical Reconciliation, The Right to be Forgotten, and Sustainable AI

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While Section 8.5 established the neuro-symbolic and compositional architectures necessary for self-correcting intelligence, and foreshadowed the critical challenges of energy efficiency and computational tractability, this section addresses the profound ethical and environmental imperatives that must govern these advanced systems. The rapid evolution of foundation models from static artifacts into dynamic, lifelong entities creates a palpable tension with established legal frameworks, most notably the “Right to be Forgotten” (RTBF). As models continuously ingest and integrate new information, the traditional notion of data deletion becomes increasingly obsolete. The core ethical challenge lies in reconciling the permanent, distributed nature of knowledge storage within neural weights with the legal mandate for complete and verifiable erasure. Fundamental research suggests that deep learning models, due to their complex structure and sheer scale, function more like mechanical brains than tabular databases, making the guarantee of total forgetting theoretically and practically difficult to achieve [555]. This structural irreconcilability necessitates a paradigm shift from simple data removal to sophisticated, reversible model editing and accountable unlearning mechanisms, directly linking the theoretical capabilities discussed in the previous section to practical governance requirements.

The implementation of machine unlearning to satisfy RTBF requests introduces a complex trade-off between privacy compliance and model fairness, a nuance critical for the trustworthy deployment of the compositional agents envisioned earlier. While unlearning algorithms aim to efficiently erase specific data points without full retraining, the process itself can inadvertently introduce bias. Stud-

ies have revealed that under non-uniform data deletion scenarios, certain unlearning methods may compromise the fairness of the resulting model compared to full retraining baselines, highlighting a critical gap where privacy protections might conflict with equity goals [556]. Furthermore, the act of unlearning can paradoxically degrade privacy rather than enhance it. Research indicates that the differential behavior between an original model and its unlearned counterpart can be exploited by adversaries to perform membership inference attacks, potentially revealing more about the deleted data than the original model would have [557]. Consequently, ethical reconciliation requires robust verification frameworks. Novel approaches utilizing backdoor-based mechanisms have been proposed to allow users to quantitatively verify data deletion with high confidence, ensuring that service providers genuinely comply with erasure requests in Machine Learning as a Service (MLaaS) environments [558].

Beyond the binary of forgetting versus remembering, the intersection of RTBF with other user rights, such as the right to actionable explanation or algorithmic recourse, presents further ethical dilemmas that complicate the design of self-correcting systems. Theory and empirical evidence suggest that fulfilling a data deletion request can invalidate previously generated recourses, meaning that a user’s ability to contest an unfavorable decision may be erased along with their data [559]. This interdependence implies that ethical AI systems must adopt holistic governance frameworks that consider the cascading effects of editing operations on multiple user rights simultaneously. To address these complexities, future research must focus on developing “deletion-as-control” formalisms that allow data to be used freely prior to deletion while imposing strict, verifiable constraints post-deletion, moving beyond narrow definitions of confidentiality [560].

Parallel to the ethical imperative of forgetting is the urgent need for sustainable AI, a theme explicitly anticipated in the discussion of scalable neuro-symbolic frameworks. The computational cost of continually updating foundation models to incorporate new knowledge or excise old information poses a significant environmental threat. The “Energy Wall” looms large as models scale to trillion parameters, where the carbon footprint of frequent retraining or inefficient editing becomes untenable. Tools designed to track and predict the energy and carbon footprint of training deep learning models, such as Carbontracker, emphasize the necessity of reporting environmental metrics alongside performance benchmarks to promote responsible computing [561]. This concern is exacerbated in edge and IoT deployments, where the end-to-end carbon footprint includes both operational energy and embodied hardware costs, requiring precise modeling tools to ensure sustainability [562].

Sustainable lifelong intelligence demands algorithms that minimize computational overhead while maximizing retention and editing efficiency, thereby enabling the real-time symbolic inference and neural processing required for next-generation agents. Methods that overcome catastrophic forgetting with fewer network updates directly contribute to reducing the carbon footprint associated with model maintenance [563]. Moreover, the development of low-carbon lifelong systems involves not just algorithmic efficiency but also architectural innovations that enable localized updates, such as parameter-efficient fine-tuning and sparse editing, which avoid the energy-intensive process of full-model retraining. The integration of these efficiency metrics into a broader Responsible AI framework is essential. Recent efforts to create comprehensive metrics catalogues for AI accountability underscore the need to evaluate AI systems not only on their accuracy but also on their procedural integrity, resource consumption, and societal impact [564].

Looking forward, the path toward ethical reconciliation involves the co-design of reversible editing architectures and sustainable learning protocols, laying the essential groundwork for the embodied and distributed paradigms explored in the subsequent section. This includes developing frame-

works for “concept correction and disentanglement” that allow for the precise removal of harmful or private concepts from generative models without impairing general capabilities, as seen in recent advancements for text-to-image diffusion models [565]. Ultimately, achieving low-carbon, lifelong intelligence requires a systemic approach that bridges the gap between legal mandates, ethical fairness, and environmental sustainability. By resolving these tensions, we ensure that the evolution of AI remains aligned with human values and planetary boundaries, creating the responsible foundation necessary for the emergence of embodied AGI and metaverse-calibrated intelligence.

8.7 Emerging Paradigms: Embodied AGI, Metaverse Calibration, and Distributed Flows

8.7 Emerging Paradigms: Embodied AGI, Metaverse Calibration, and Distributed Flows

Building upon the imperative for sustainable and ethically reconciled lifelong intelligence, the trajectory of artificial intelligence is increasingly pointing toward a synthesis of embodied cognition, immersive virtual environments, and decentralized governance structures. As foundational models approach scaling limits in static text and image domains, the next frontier for achieving Artificial General Intelligence (AGI) lies in embodied agents that operate within dynamic, interactive worlds. In this context, the metaverse emerges not merely as a platform for social interaction or entertainment, but as a critical calibration environment where embodied AGI can develop, test, and refine its cognitive architectures through grounded perception-action cycles. This paradigm shift necessitates a re-evaluation of how intelligence is defined, moving from passive pattern recognition to active engagement with complex, multi-modal realities, thereby extending the concept of “lifelong learning” from data streams to physical and virtual experiences.

Central to this emerging vision is the concept of embodied AGI, which posits that true general intelligence requires a physical or virtual body to interact with an environment, thereby grounding abstract reasoning in sensory experience. Recent theoretical frameworks suggest that the metaverse plays a pivotal role in facilitating the relationship between embodied agents and human consciousness, serving as a symbolic interface where alternate inner states of being can be simulated and explored [566]. By leveraging the metaverse as a “digital twin” of reality, researchers can create ecologically valid conditions for bootstrapping increasingly complex cognitive skills, effectively initiating a cognitive arms race between embodied agents that drives evolutionary progress [567]. This approach directly addresses the limitations of static learning paradigms discussed in previous sections by emphasizing the integration of perception, action, memory, and learning as essential, continuous components of an intelligent agent [568].

The calibration of these embodied agents within metaverse environments relies heavily on the seamless fusion of physical and virtual data streams, a process that demands the efficient model editing and updating mechanisms highlighted in the pursuit of sustainable AI. The metaverse offers a unique opportunity to bridge the gap between digital avatars and physical entities through real-time synchronization, enabling agents to learn from both simulated and real-world interactions. For instance, generative AI technologies are being employed to ensure effective physical-virtual synchronization in vehicular metaverses, where digital twins of autonomous vehicles are continuously updated based on real-time sensor data [569]. This synchronization is crucial for overcoming state space paradoxes, where the complexity of the environment exceeds the computational capacity of the agent. By utilizing distributed radiance fields and edge computing, systems can achieve

high-fidelity reconstruction of 3D scenes with significant data compression, allowing for real-time updates of digital twins in autonomous driving scenarios [570], thus aligning with the low-carbon goals of sustainable lifelong systems.

Furthermore, the development of embodied AGI within the metaverse is intrinsically linked to the concept of distributed flows and decentralized governance, offering a structural solution to the ethical and accountability challenges previously outlined. The sheer scale and heterogeneity of metaverse environments necessitate a shift away from centralized control mechanisms toward decentralized, self-sovereign architectures. Blockchain technology and self-sovereign identity (SSI) protocols are emerging as foundational elements for ensuring trust, interoperability, and privacy in these distributed ecosystems [571]. Such decentralized frameworks enable the creation of robust, fault-tolerant networks where multiple agents can collaborate, share knowledge, and coordinate actions without relying on a single point of failure. This is particularly evident in social metaverse applications, where hierarchical federated learning frameworks leverage social ties among users to optimize privacy-utility trade-offs while preventing model theft and collusion [572], thereby providing a scalable mechanism for managing the “Right to be Forgotten” and data provenance in complex multi-agent systems.

The interplay between embodied agents and the metaverse also gives rise to novel forms of multi-agent cooperation and collective intelligence, extending the scope of continual learning to collaborative settings. Research into compositional world models demonstrates how decentralized agents can cooperate effectively despite having only partial egocentric views of the world, utilizing vision-language models to infer the actions of others and plan jointly [573]. These capabilities are further enhanced by the integration of large language models as robotic brains, which unify egocentric memory and control through natural language dialogues, enabling agents to perform complex tasks such as active exploration and embodied question answering [574]. The metaverse serves as an ideal testbed for these multi-agent systems, providing a rich, interactive environment where agents can practice coordination, negotiation, and collaborative problem-solving, effectively functioning as a distributed continual learning laboratory.

Looking ahead, the realization of embodied AGI calibrated within metaverse environments will require addressing significant challenges related to scalability, energy efficiency, and ethical governance, echoing the sustainability concerns raised earlier. The computational demands of rendering immersive 3D worlds and running data-hungry AI models necessitate innovative solutions in mobile edge computing and green networking technologies [575; 576]. Moreover, the deployment of such powerful systems raises profound ethical questions regarding agency, accountability, and the potential impact on human society. Proposals for dual governance frameworks, which combine centralized regulation with crowdsourced safety mechanisms, offer a promising path forward for ensuring the responsible development and deployment of generative AI and embodied agents [577], ensuring that the autonomy of these systems remains aligned with human values.

Ultimately, the convergence of embodied AGI, metaverse calibration, and distributed flows represents a transformative paradigm for the future of artificial intelligence. By grounding intelligence in embodied experience, leveraging the metaverse as a dynamic calibration space, and adopting decentralized governance structures, we can overcome the limitations of current AI systems and move closer to achieving true general intelligence. This holistic approach not only addresses technical challenges but also aligns with broader societal goals of sustainability, inclusivity, and human-centric design, paving the way for a future where intelligent agents and humans coexist and collaborate in shared physical and virtual realities [578]. As we continue to explore this frontier, the insights gained from embodied AI research will undoubtedly shape the next generation of intelligent systems,

redefining our understanding of cognition, agency, and the nature of reality itself.

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